

Introduction to Astrophysics and Supernova Phenomenology

– Supplementary Material –

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27 June 2026

- This supplementary material is for the talk
Introduction to Astrophysics and Supernova Phenomenology
at JCP HEP School
- For convenience, no explicit citations are provided for the images
in this supplementary material included in the lecture note;
Please refer to the lecture note for the original sources of each figure
- The original material includes video contents
- If you would like to access the original version,
please contact the author via the email below

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Landscape of Stellar Evolution



CRAB NEBULA



ORION



EYES



CAT'S PAW



GALACTIC CENTER



M16 PILLARS



BAT SHADOW



NGC 7469



VIRGO CLUSTER



WR 124



G21.5-0.9



CENTAURUS A



CASSIOPEIA A



NGC 3532



NGC 6872



HB 5



ABELL 2125



NGC 3324



NGC 1365



MSH 15-52



ARP 220



JUPITER



NGC 1850



MACS J0035



SN 1987A

25 New Images for
Chandra's 25th
Anniversary,
NASA/SAO/CXC, 2024

Measurements

- Luminosity

$$L = 4\pi r^2 \sigma T^4$$

- Temperature

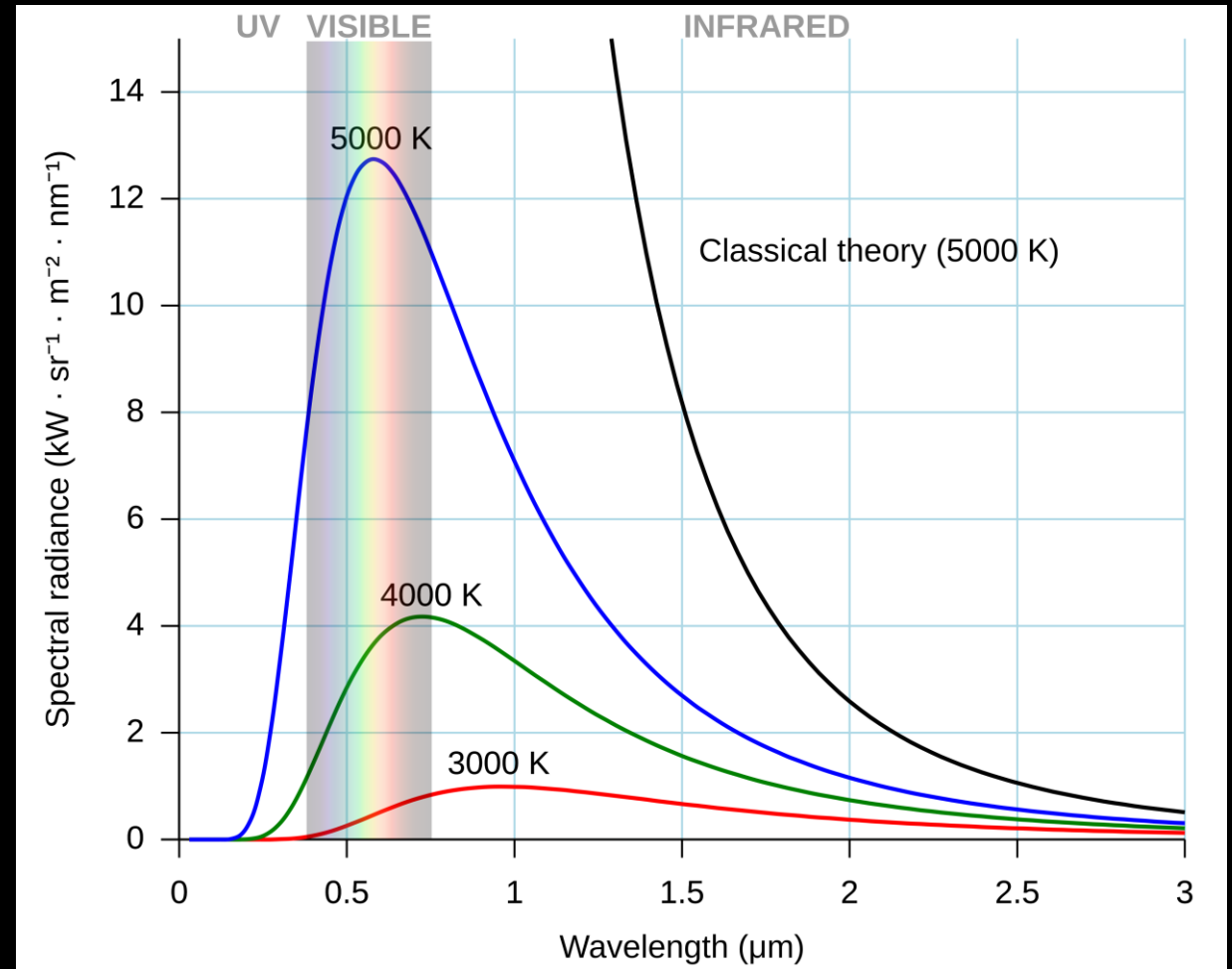
Can be measured by Planck's law

$$U = \frac{8\pi h}{c^3} \frac{\nu^3}{e^{h\nu/kT} - 1}$$

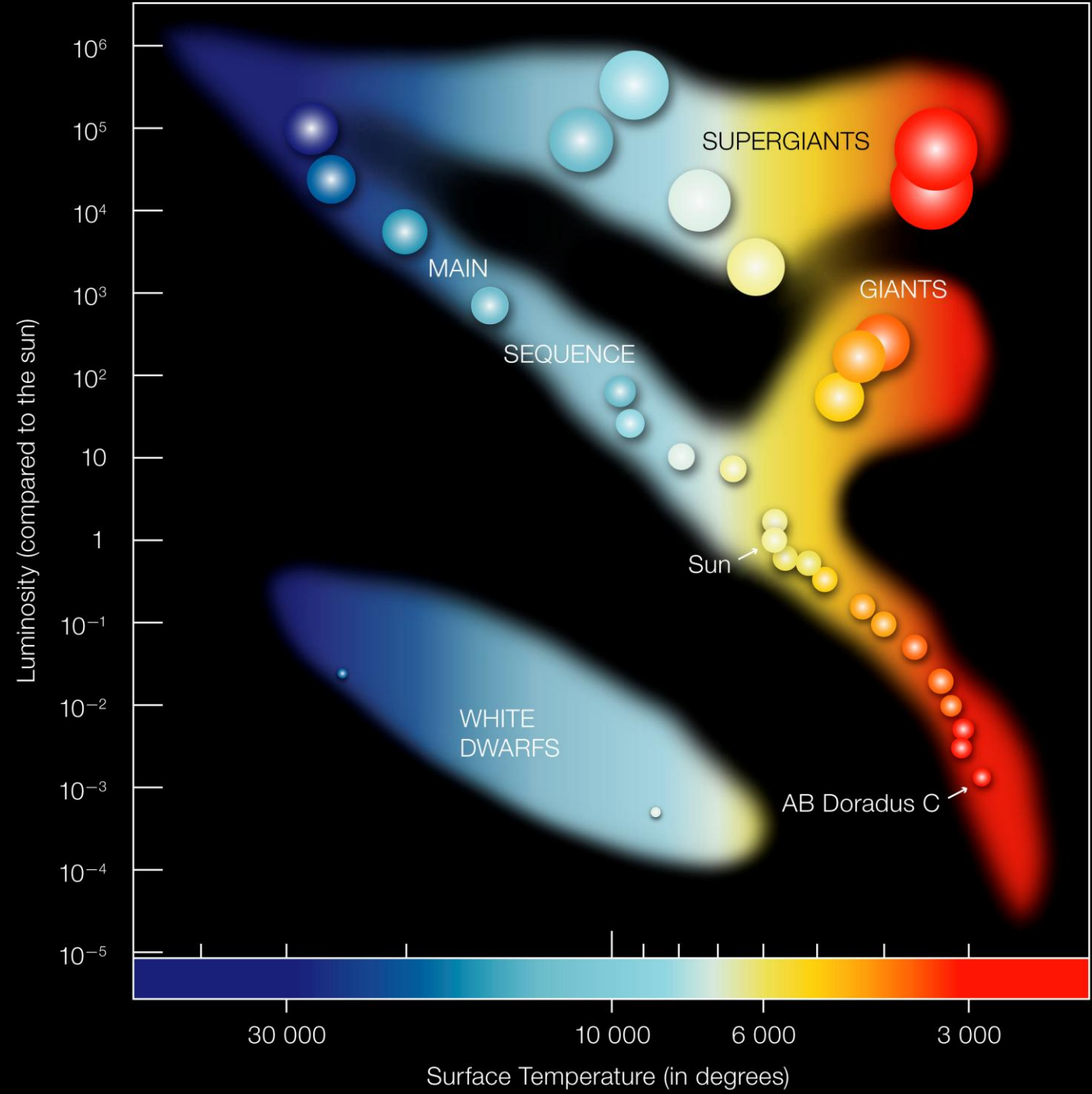
- Magnitude

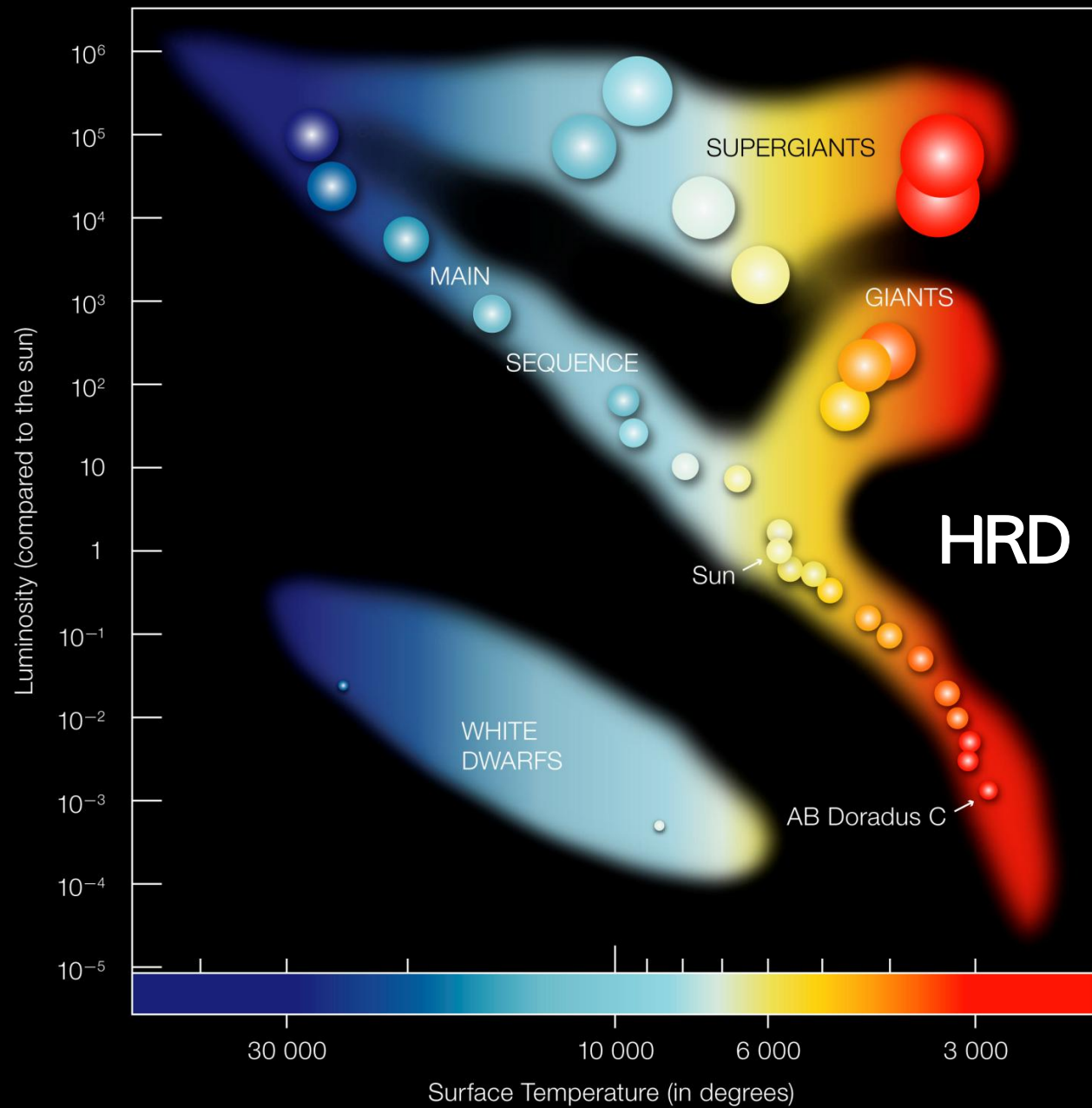
$$m = -2.5 \log\left(\frac{F}{F_0}\right)$$

$$F = \frac{L}{4\pi d^2}$$

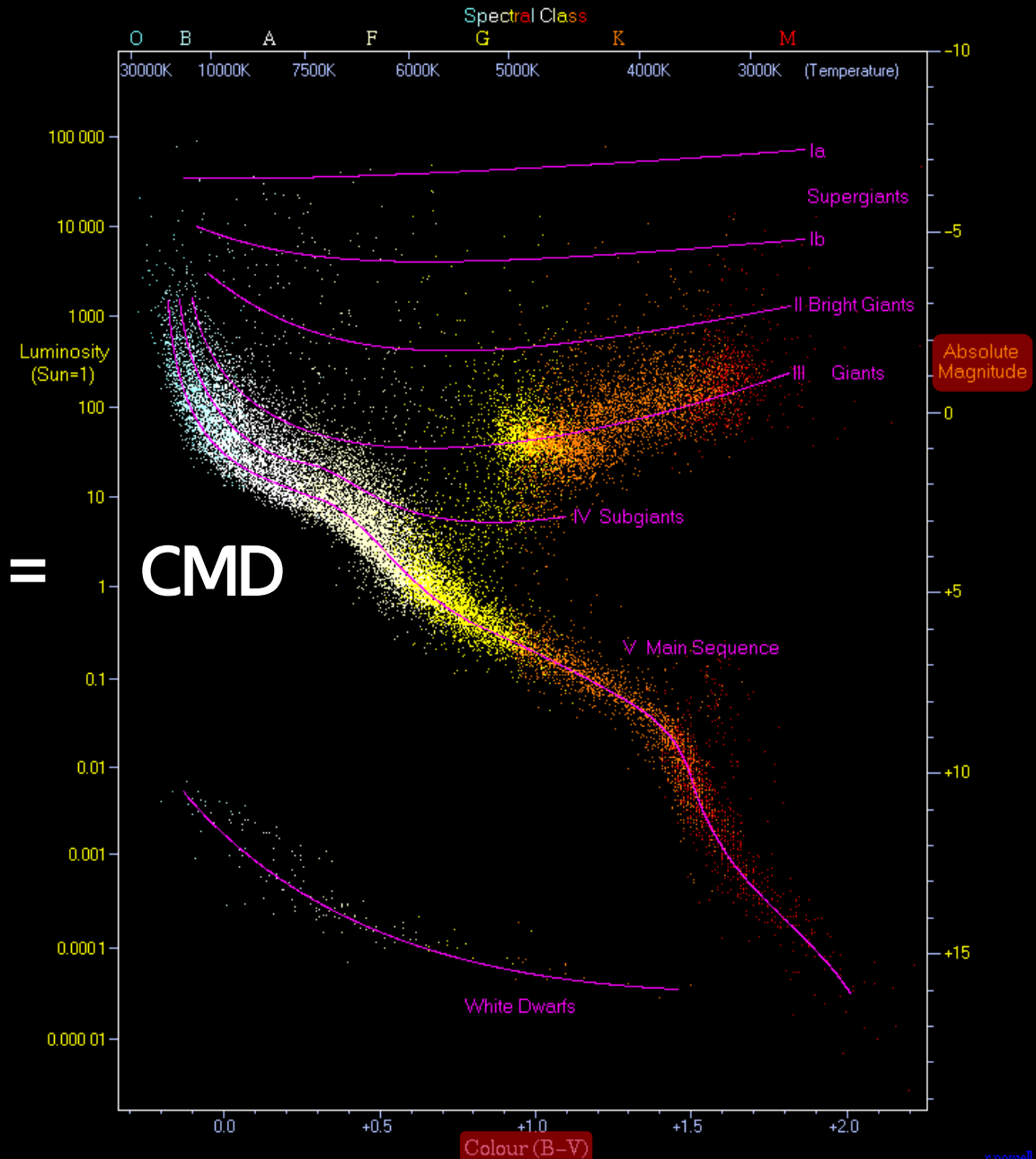


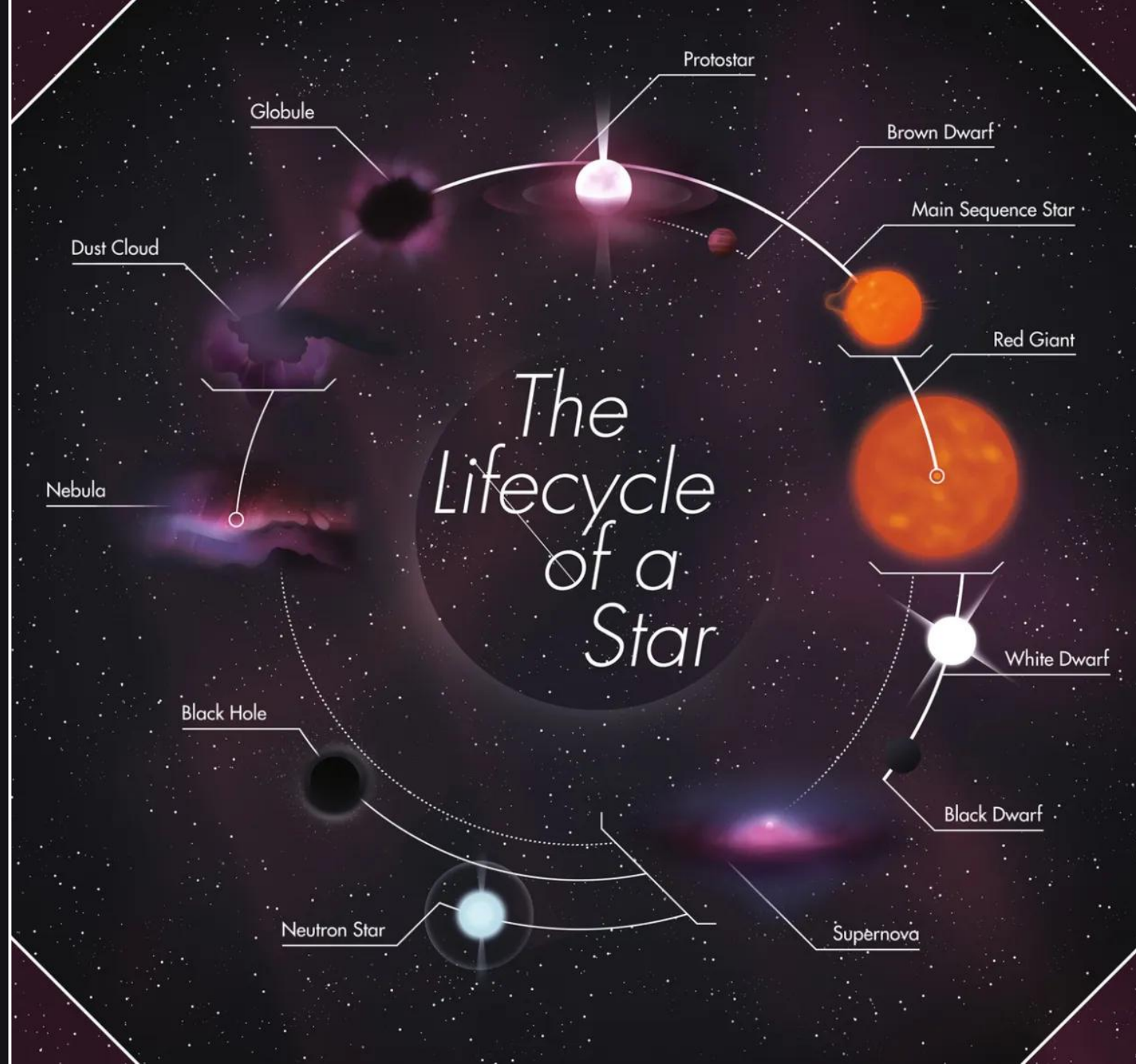
Hertzprung–Russell
Diagram (HRD)





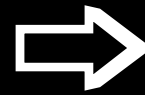
= CMD





Electron degenerate

Adiabatic exponent $\Gamma_1 \leq \frac{4}{3}$

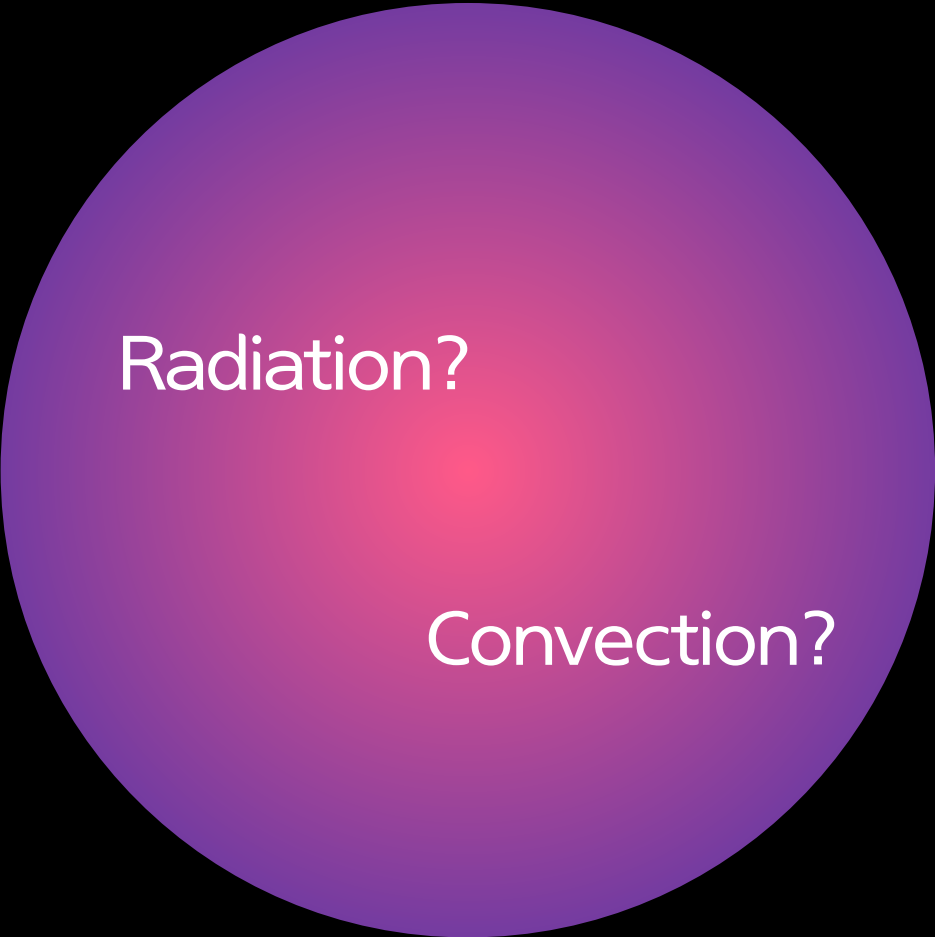


Pre-Main-Sequence (PMS)

$$M_J = \left(\frac{375k^3T^3}{4\pi G^3\mu^3m_p^3} \right)^{1/2}$$

Pressure

Gravity



Radiation?

Convection?

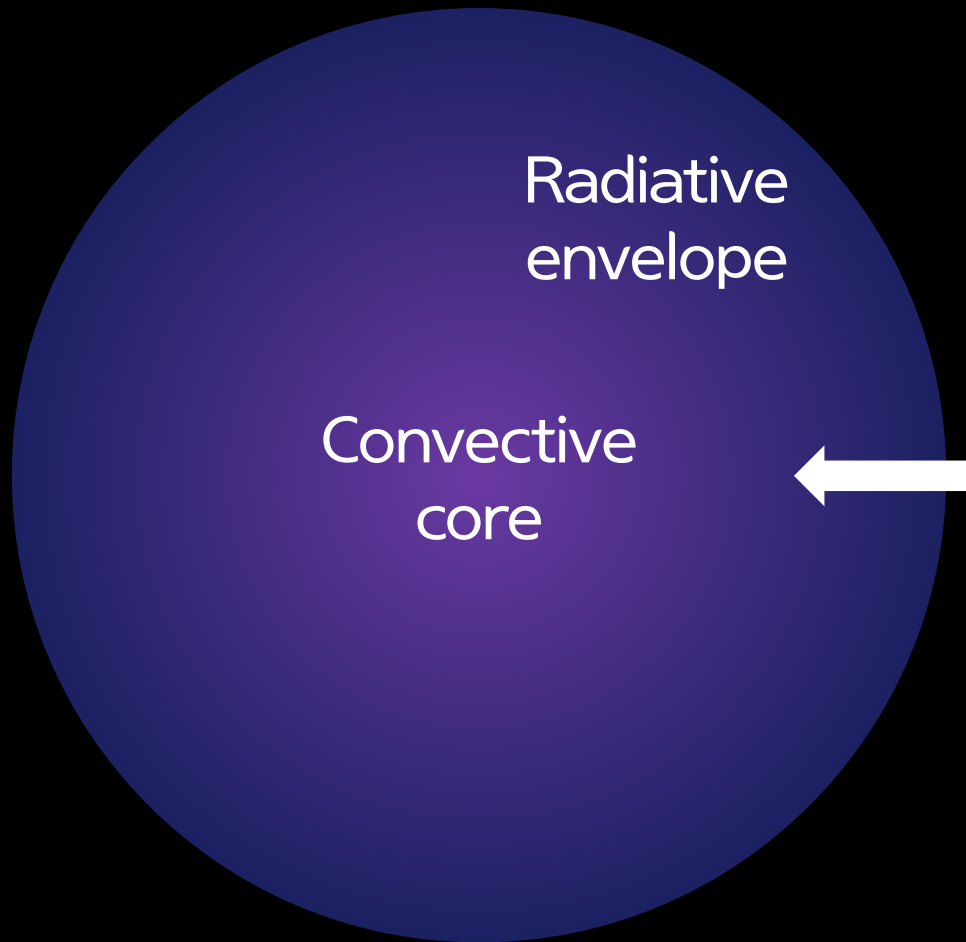
Onset of convection

$$\nabla_{\text{rad}} > \nabla_{\text{ad}}$$

$$\nabla_{\text{ad}} \equiv \left(\frac{d \ln T}{d \ln P} \right)_{\text{ad}} \propto \frac{P}{\rho T c_P}$$

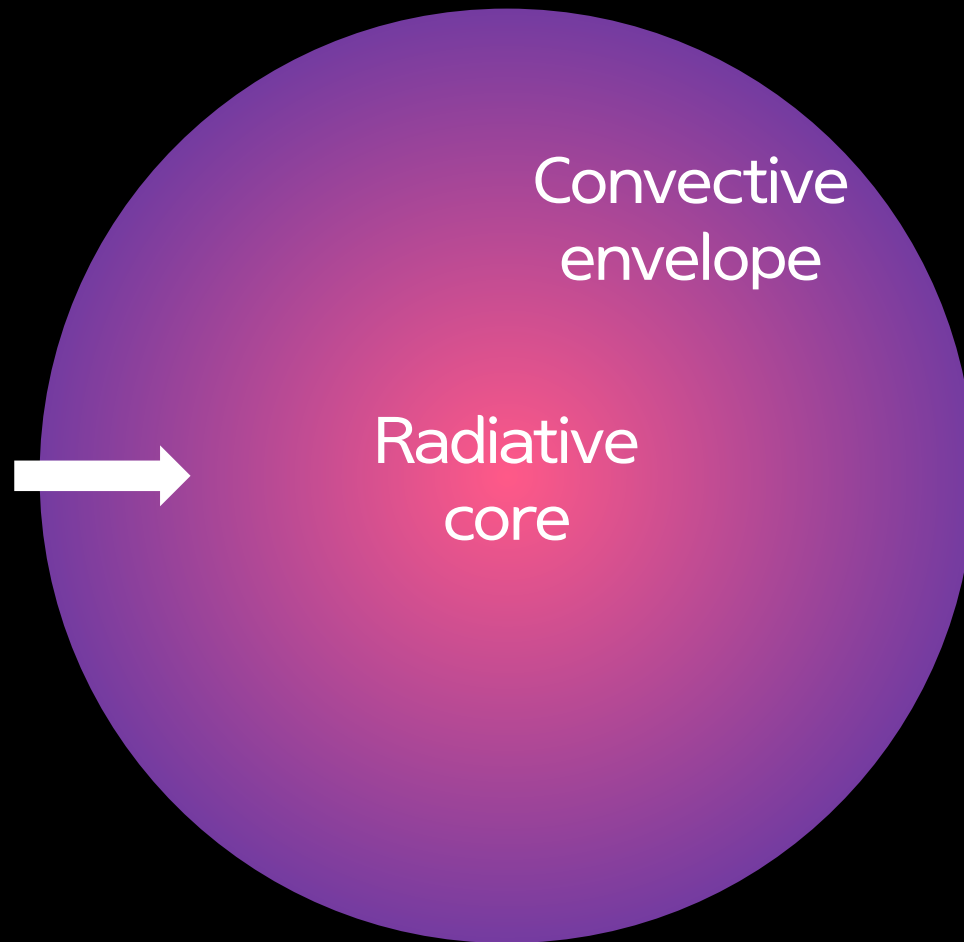
$$\nabla_{\text{rad}} \equiv \left(\frac{d \ln T}{d \ln P} \right)_{\text{rad}} \propto \frac{\kappa L P}{M T^4}$$

Low Mass



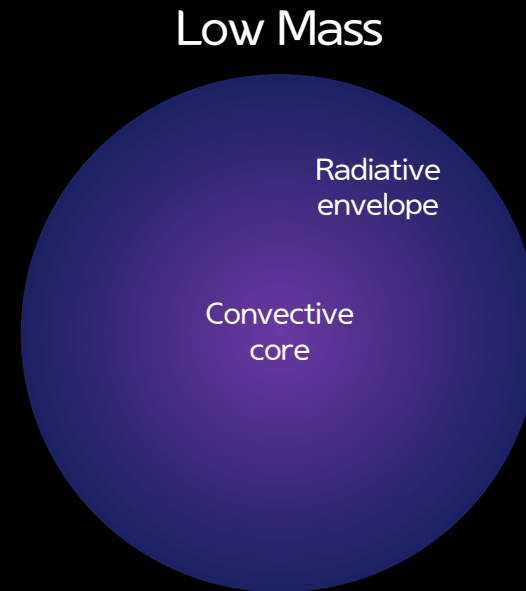
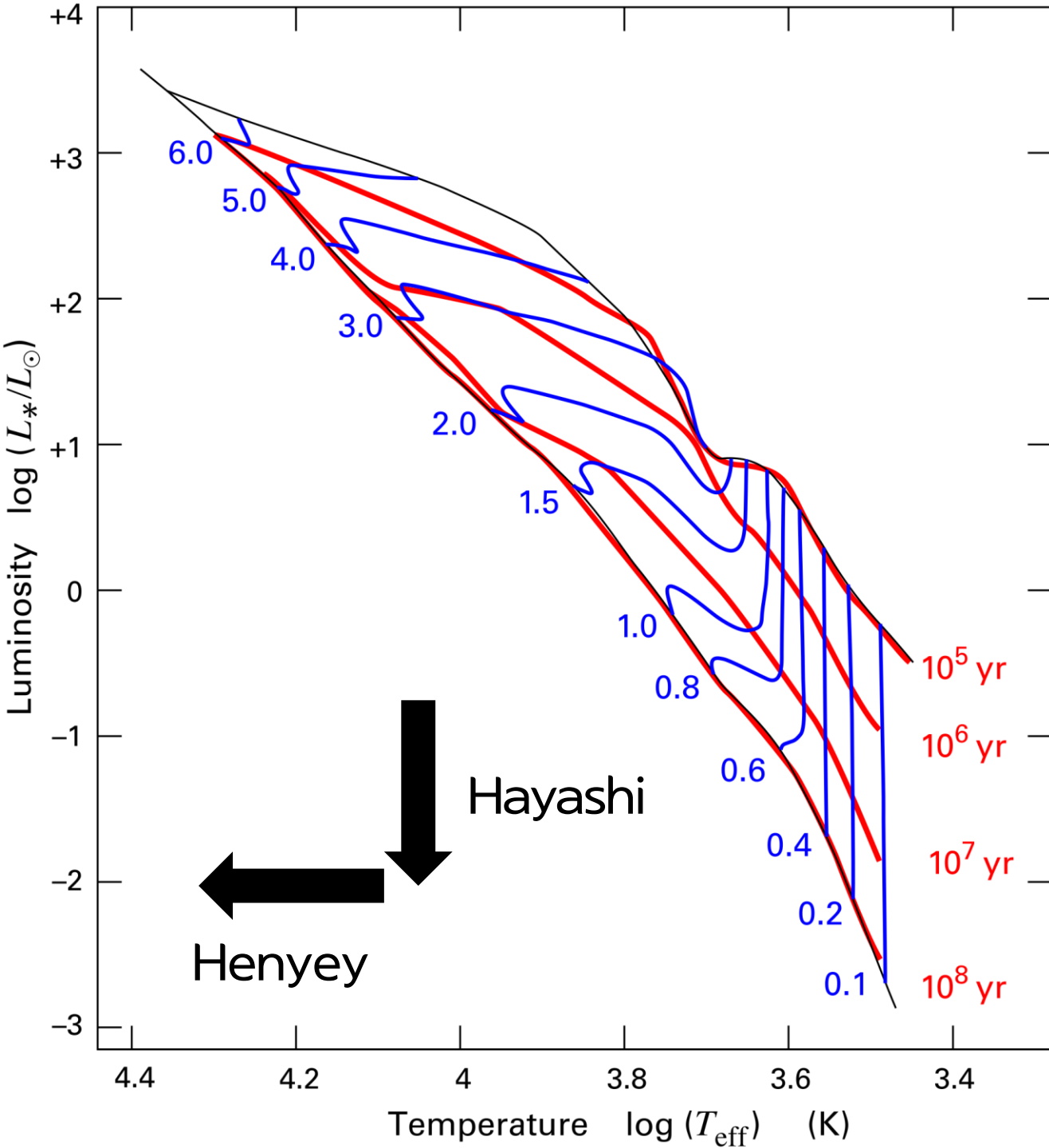
$T \sim \text{const.}, L \downarrow$

Massive

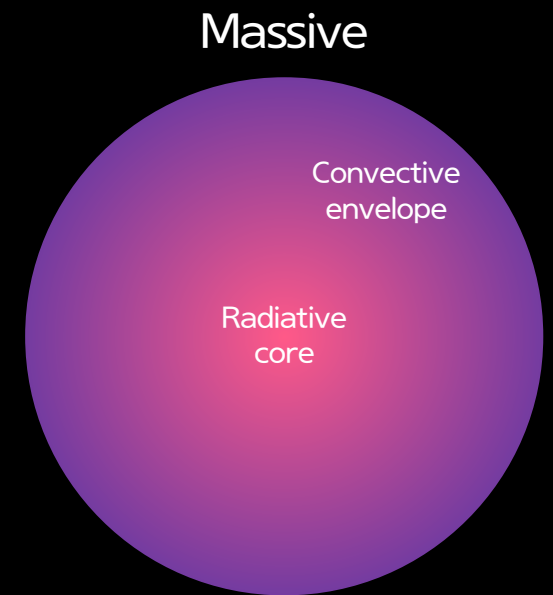


$L \sim \text{const.}, T \uparrow$

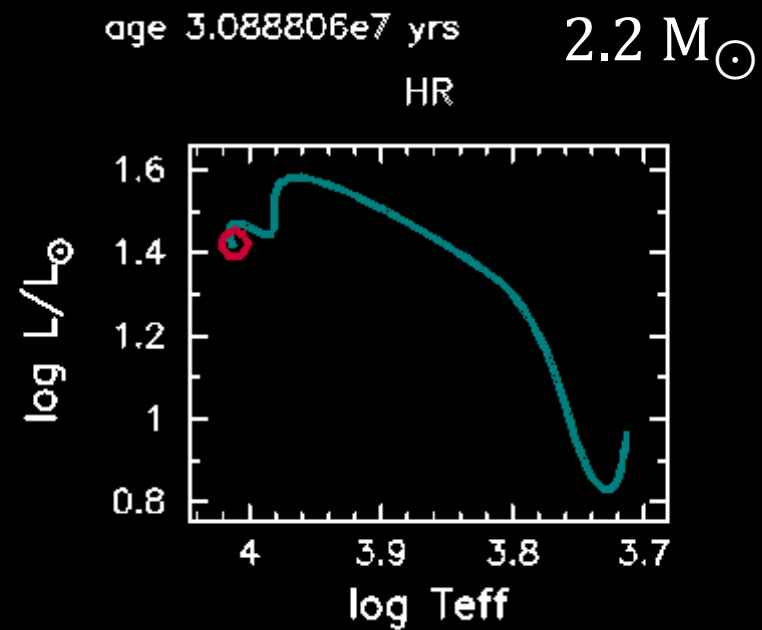
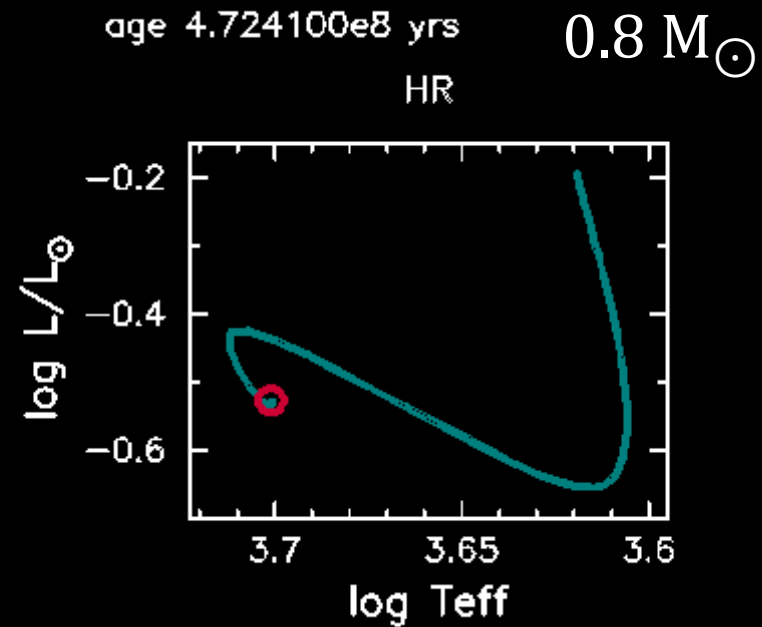
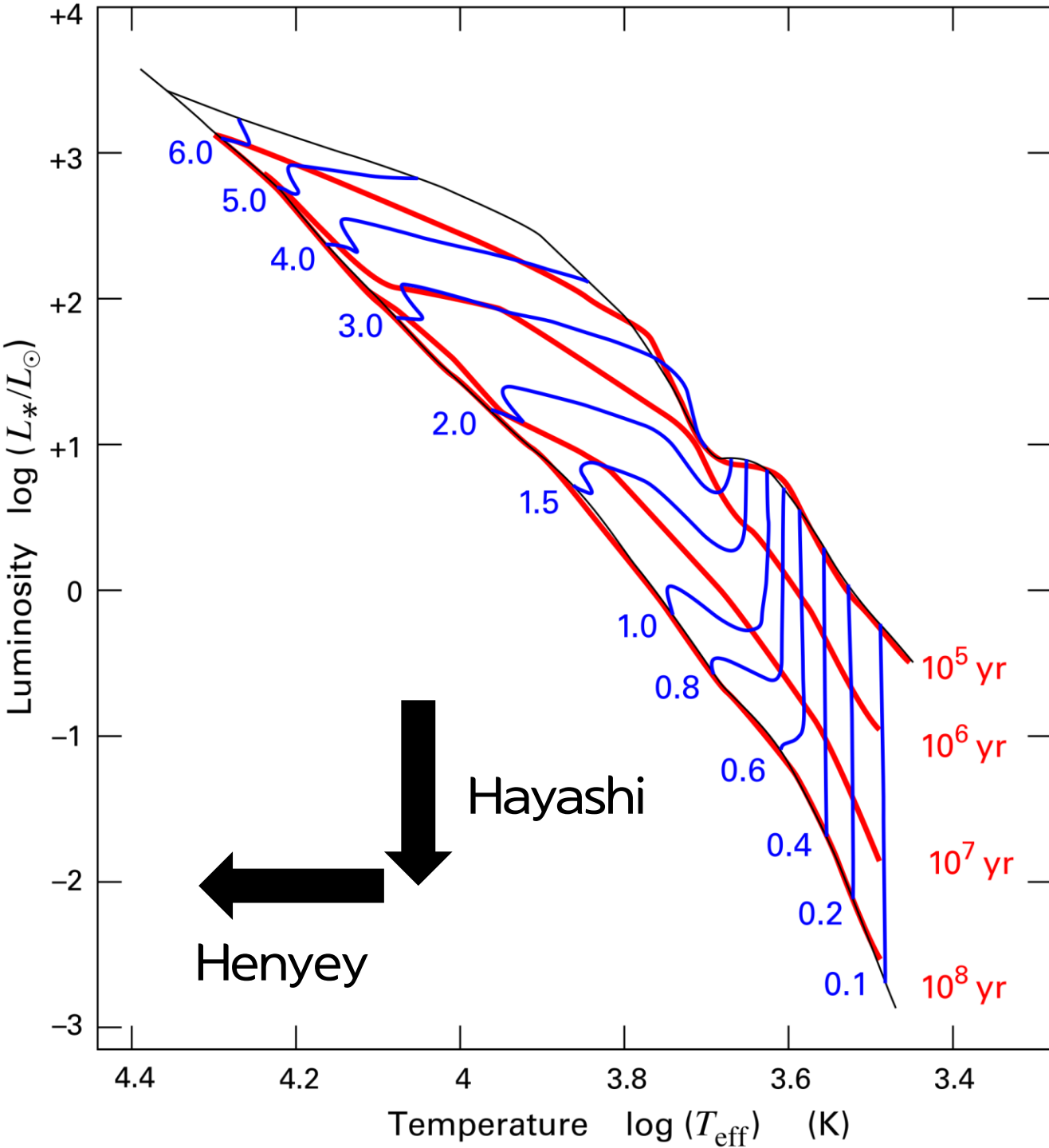
Depends
on the T_c



$T \sim \text{const.}, L \downarrow$

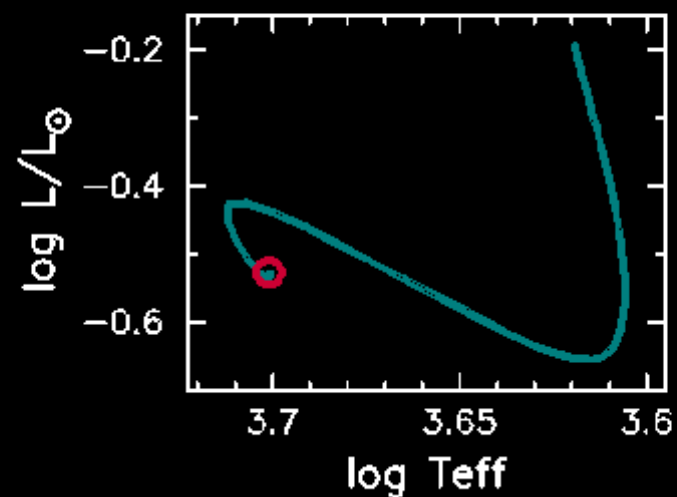


$L \sim \text{const.}, T \downarrow$



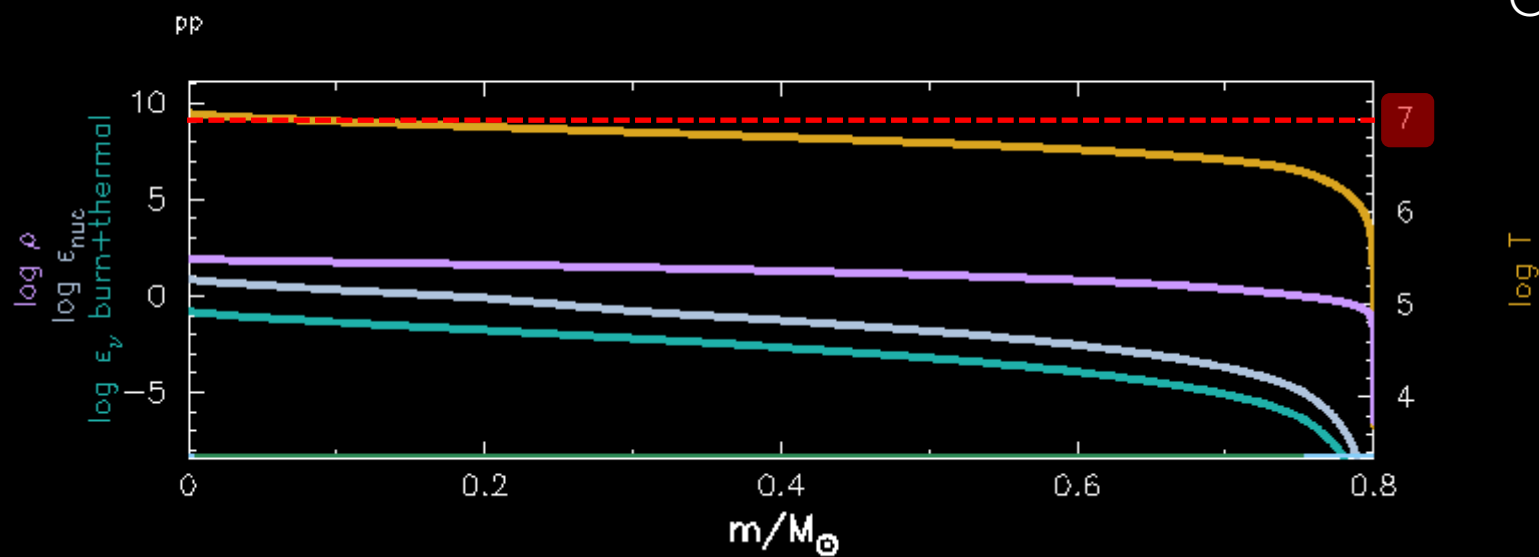
age 4.724100e8 yrs

HR



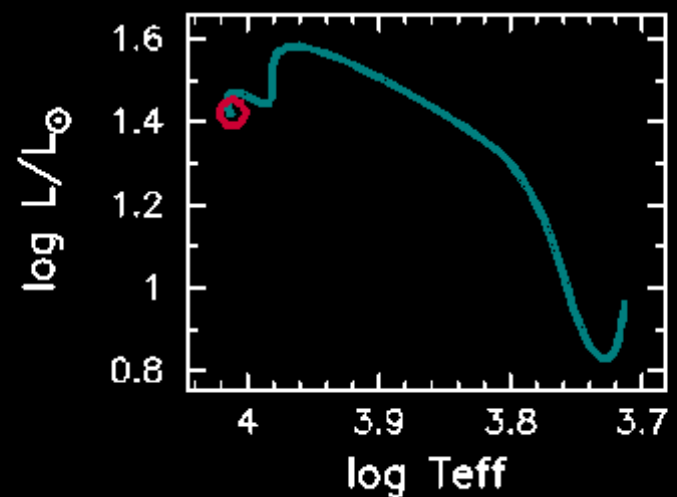
Summary_Burn

0.8 $M_\odot$



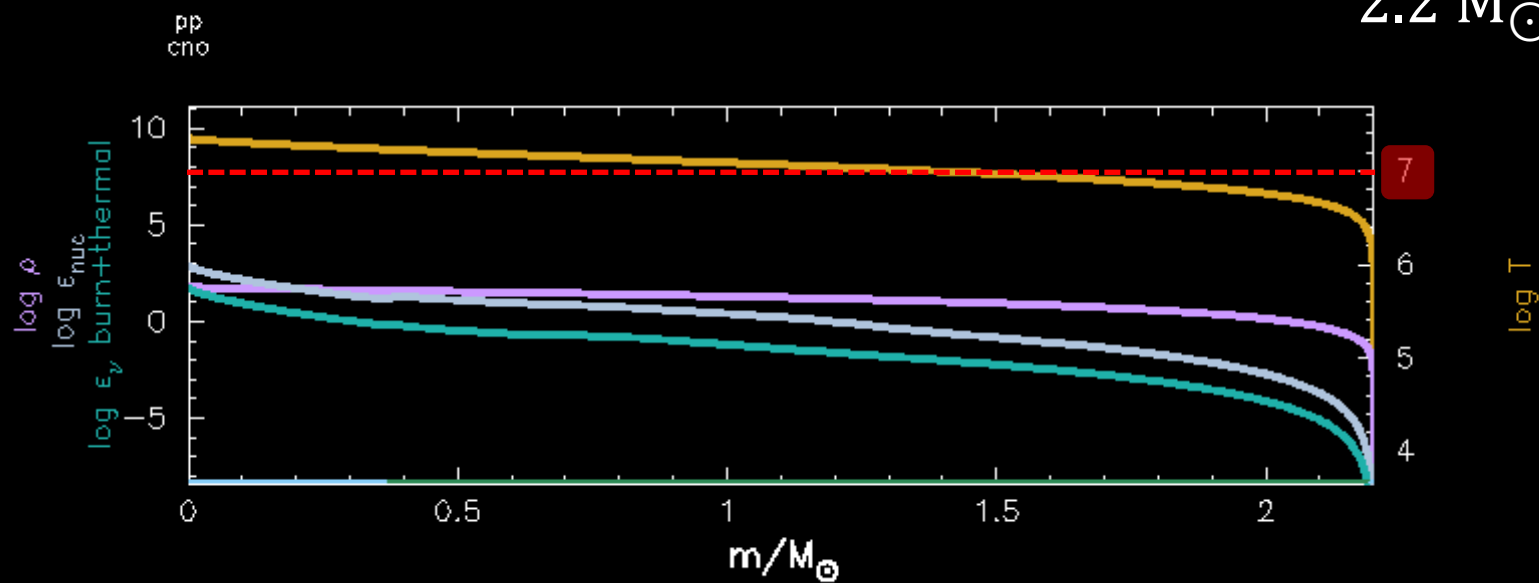
age 3.088806e7 yrs

HR

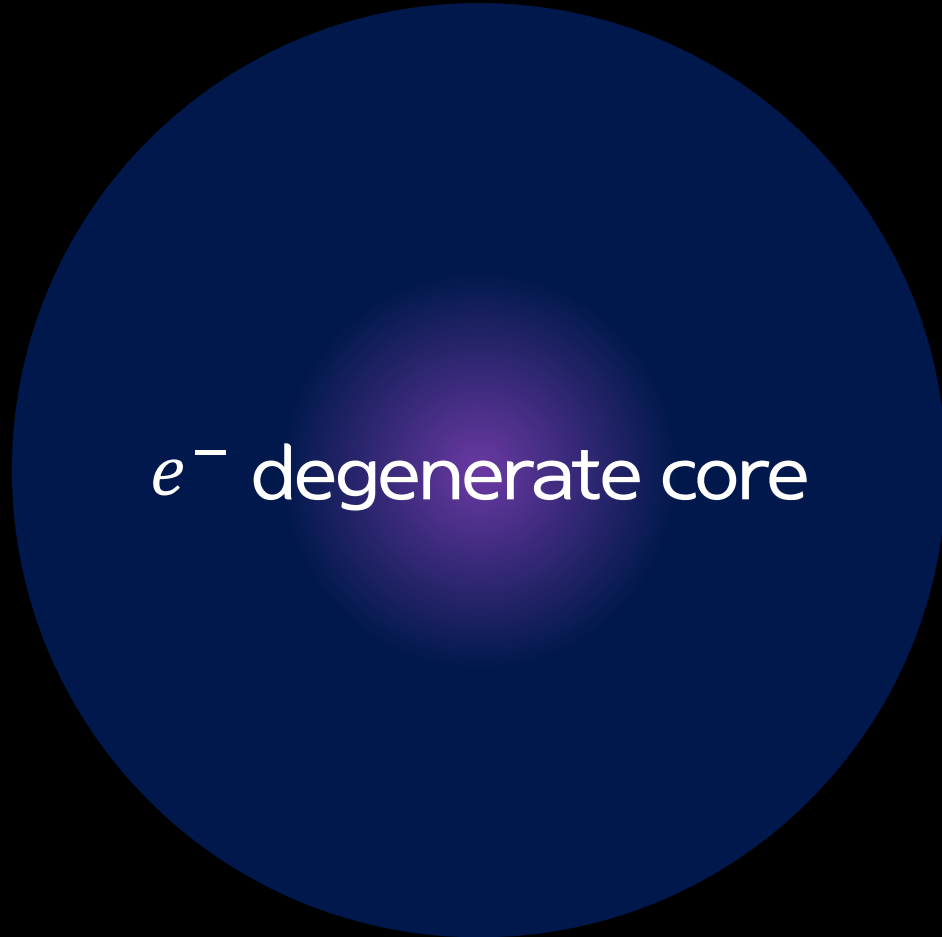


Summary_Burn

2.2 $M_\odot$



Extraordinarily Low Mass



$$M \lesssim 0.08 M_{\odot}$$

e⁻ degenerate core forms
before $T_c \sim 10^7$ K reached

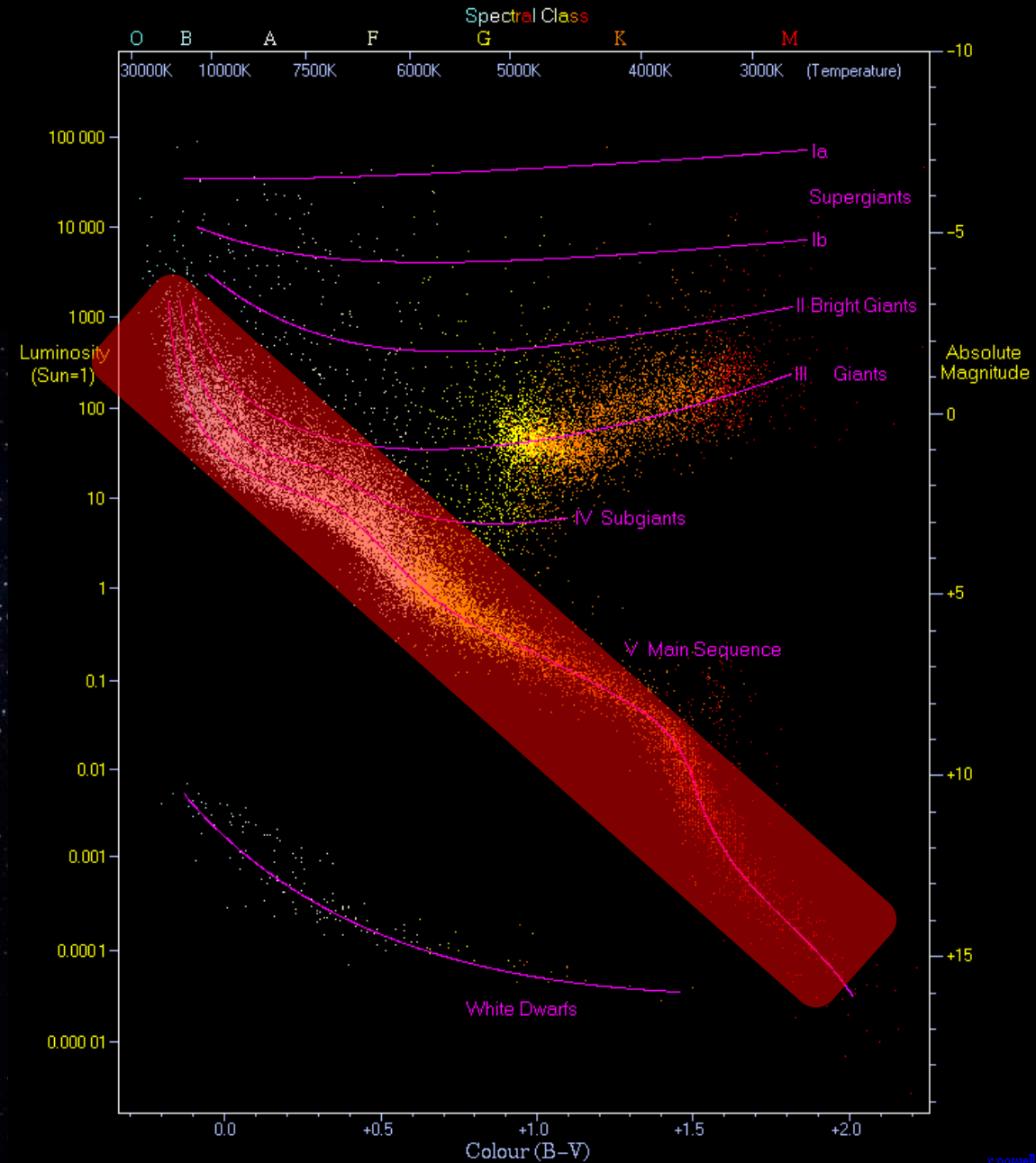
Contraction $\rightarrow E_i \uparrow \rightarrow p_F \uparrow$

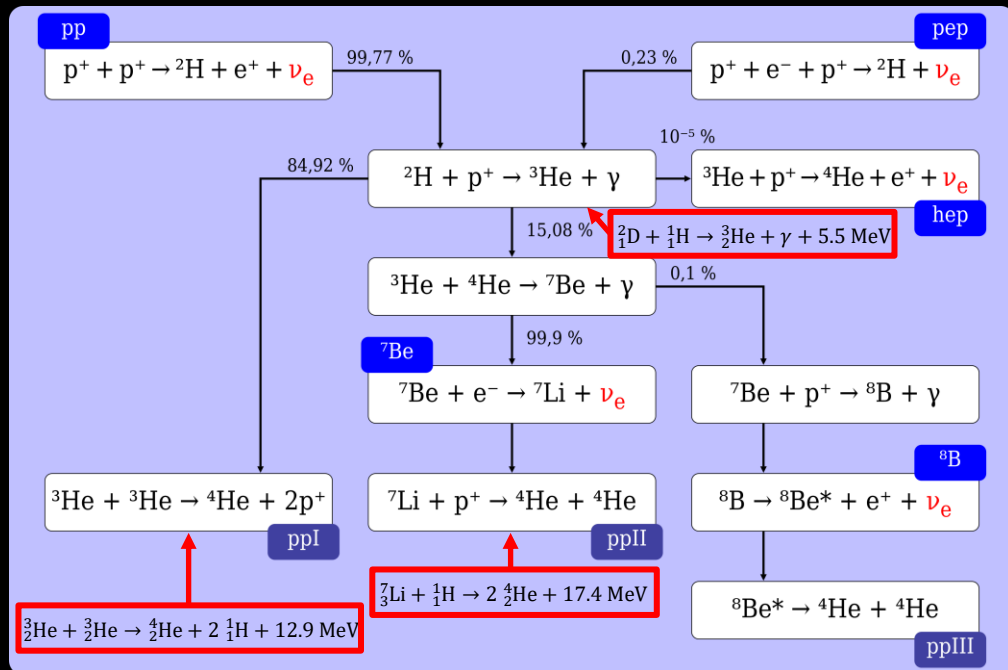
No H ignition

No main sequence entry

Main Sequence (MS)

- H burning occurs at the core



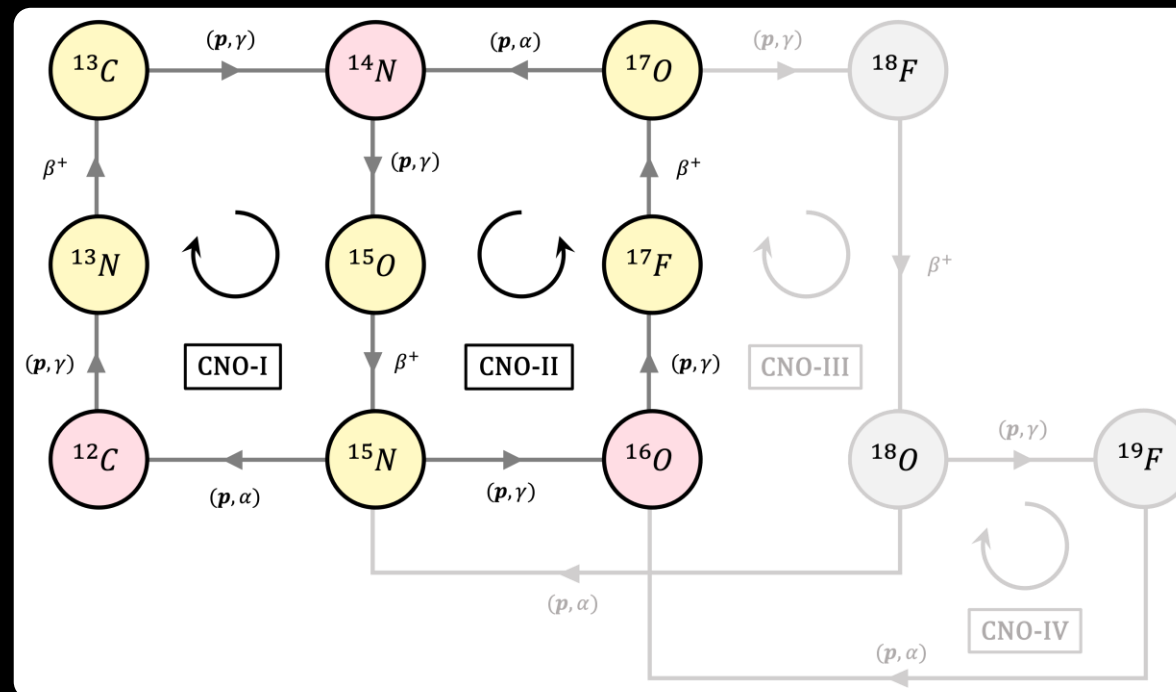


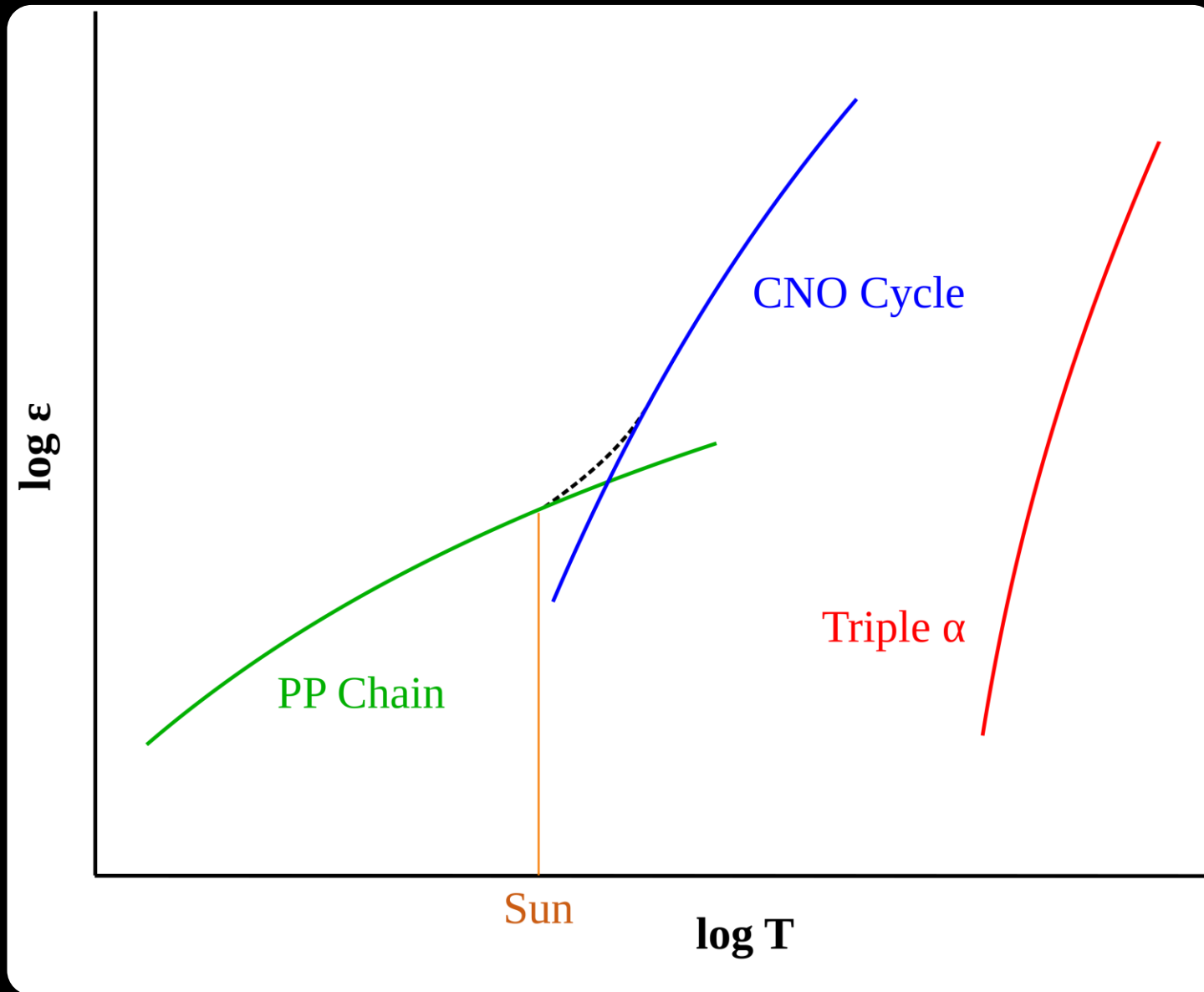
Proton-Proton Chain

$$\varepsilon_{pp} \propto \rho X^2 T_6^4$$

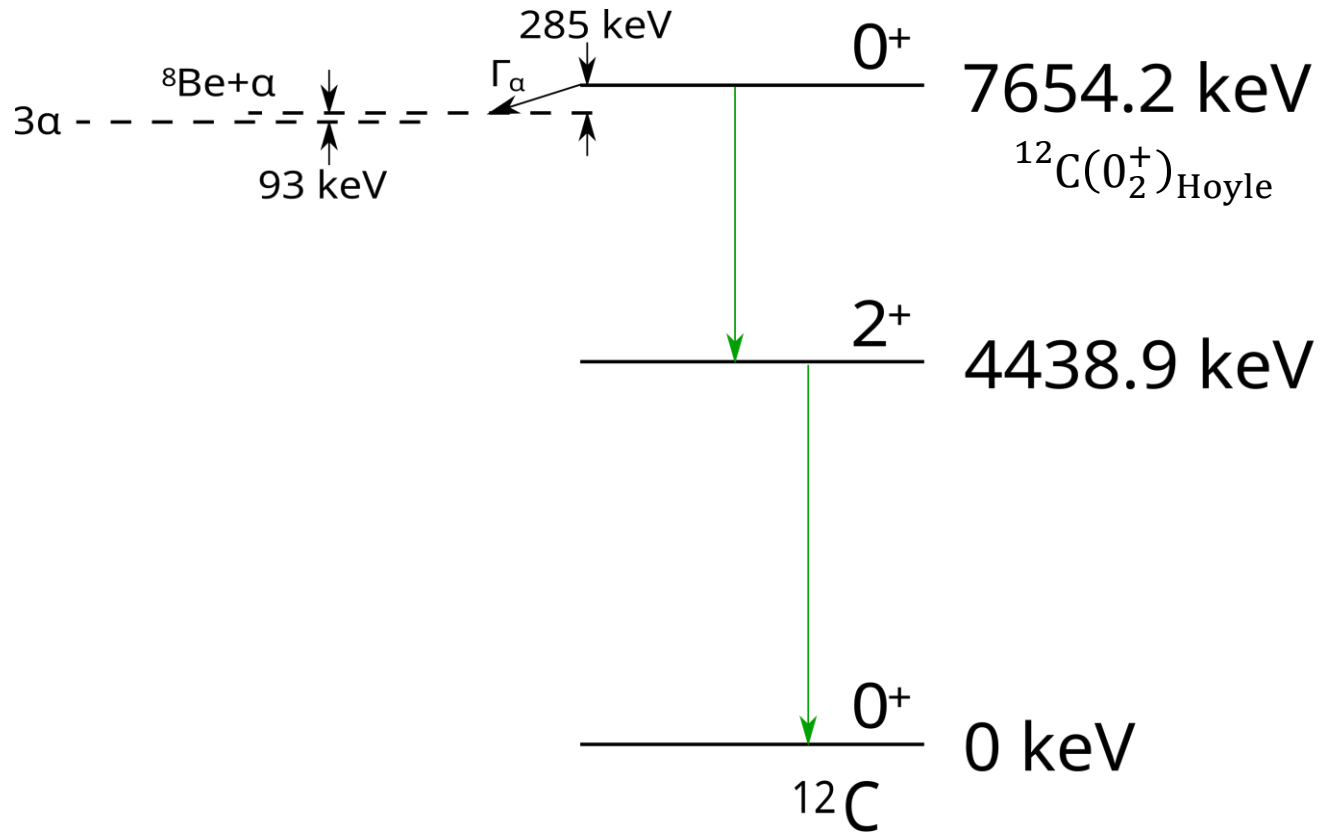
CNO Cycle

$$\varepsilon_{\text{CNO}} \propto \rho X X_{\text{CNO}} T_6^{19.9}$$



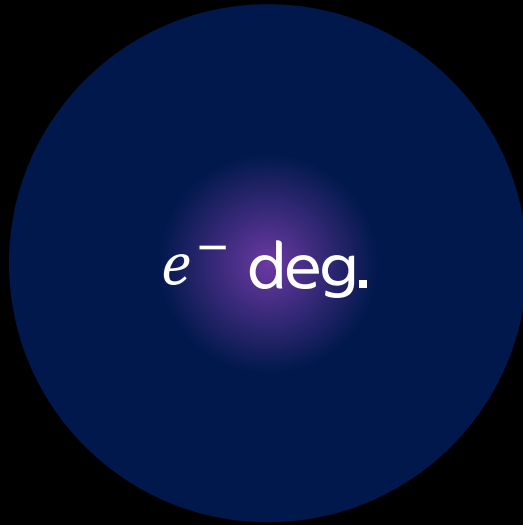


Helium Burning and Final Destiny



Triple-Alpha

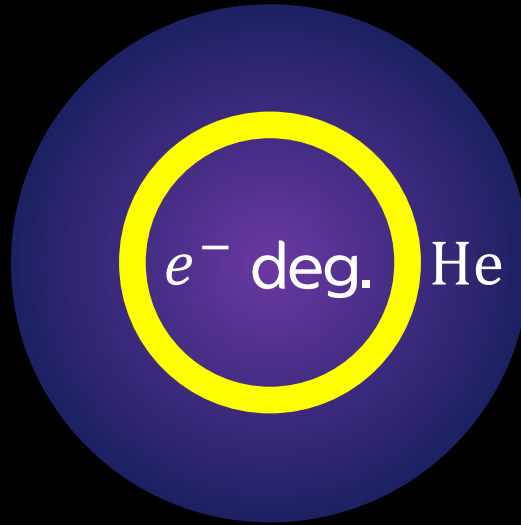
$$\varepsilon_{3\alpha} \propto \rho^2 Y^3 T_8^{41}$$



No He burning

Extremely Low

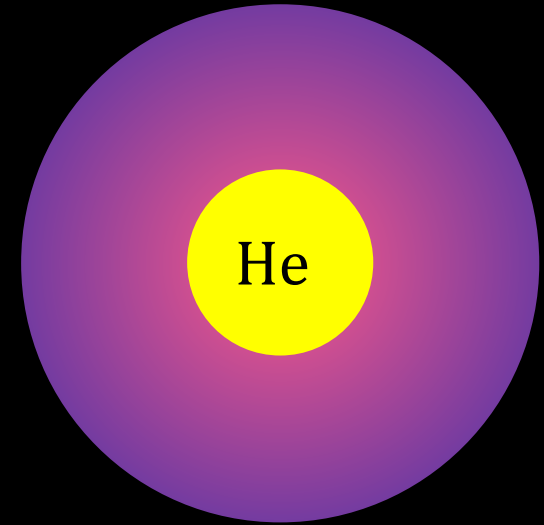
$0.08 - 0.5 M_{\odot}$



Shell He ignition

Low

$0.5 - 2.0 M_{\odot}$



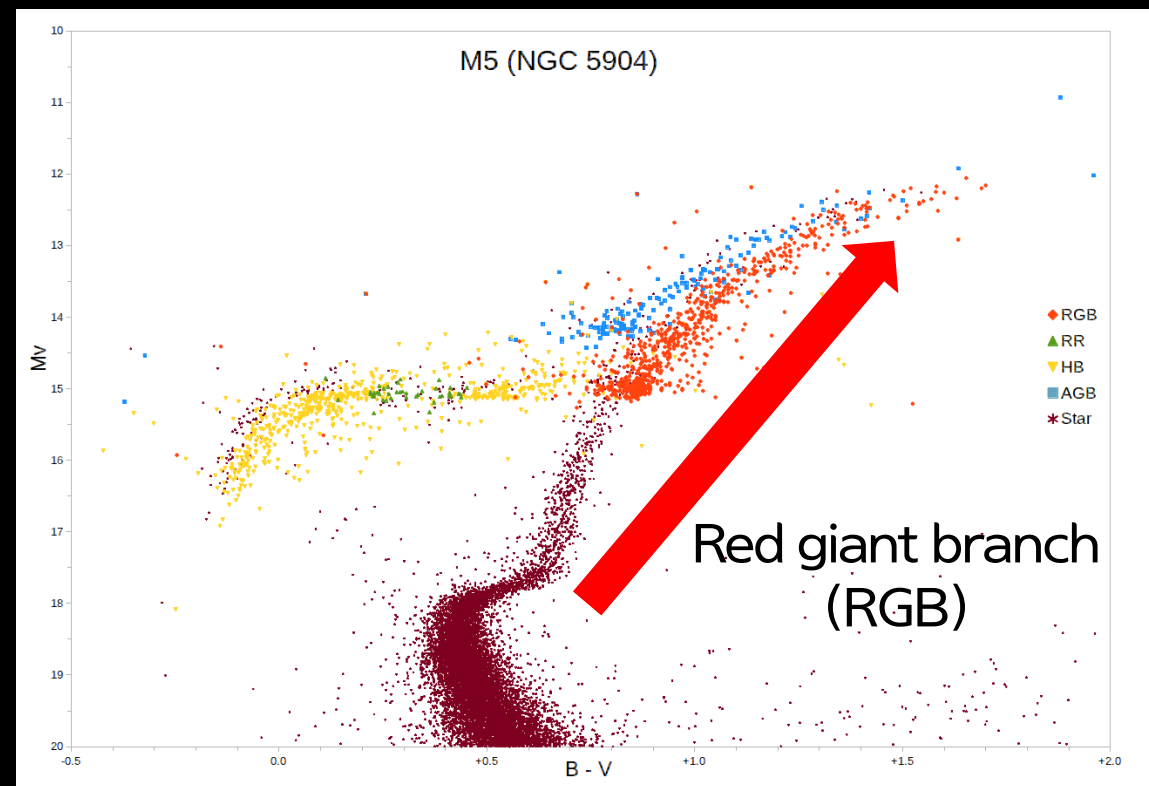
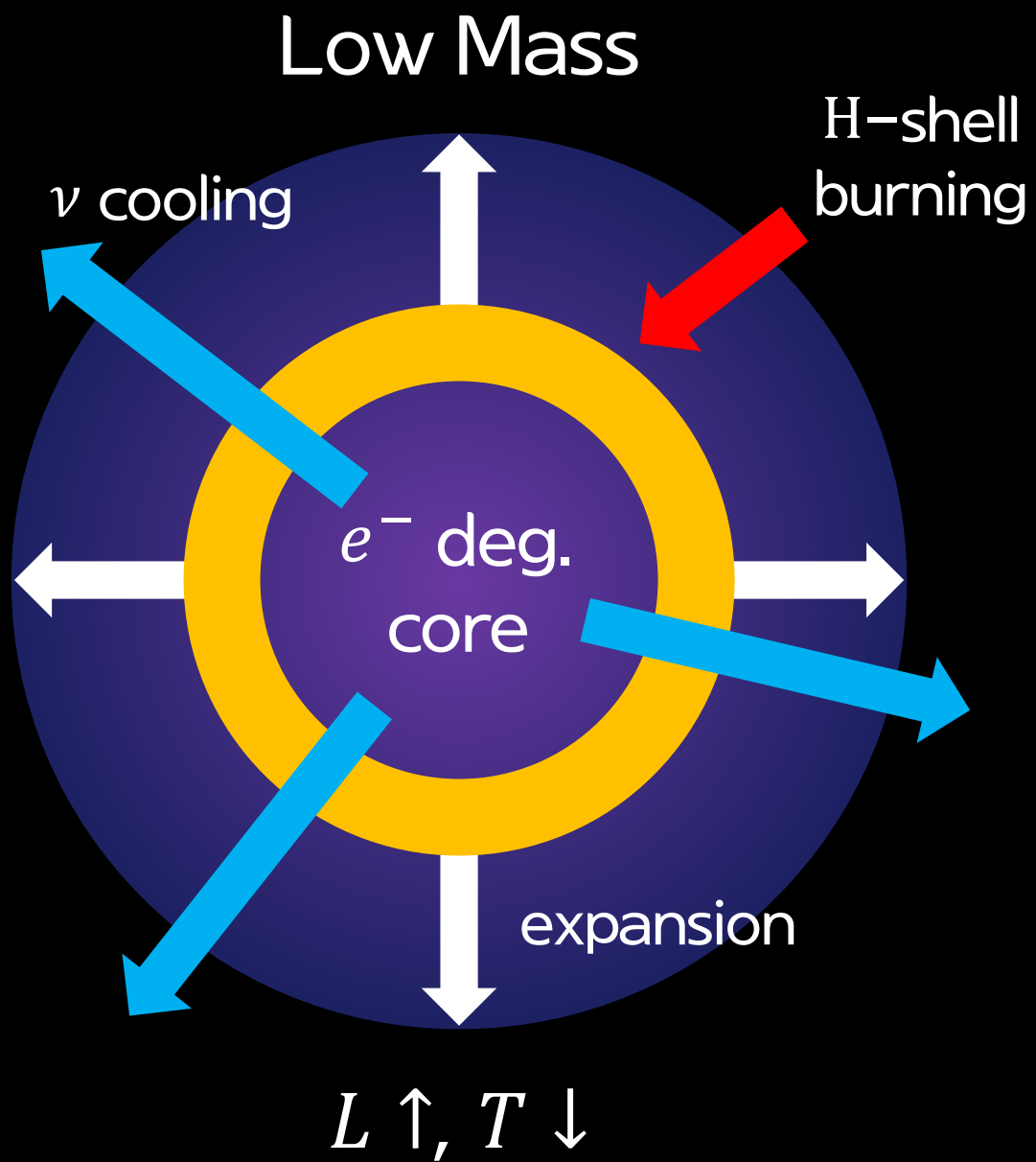
Core He ignition

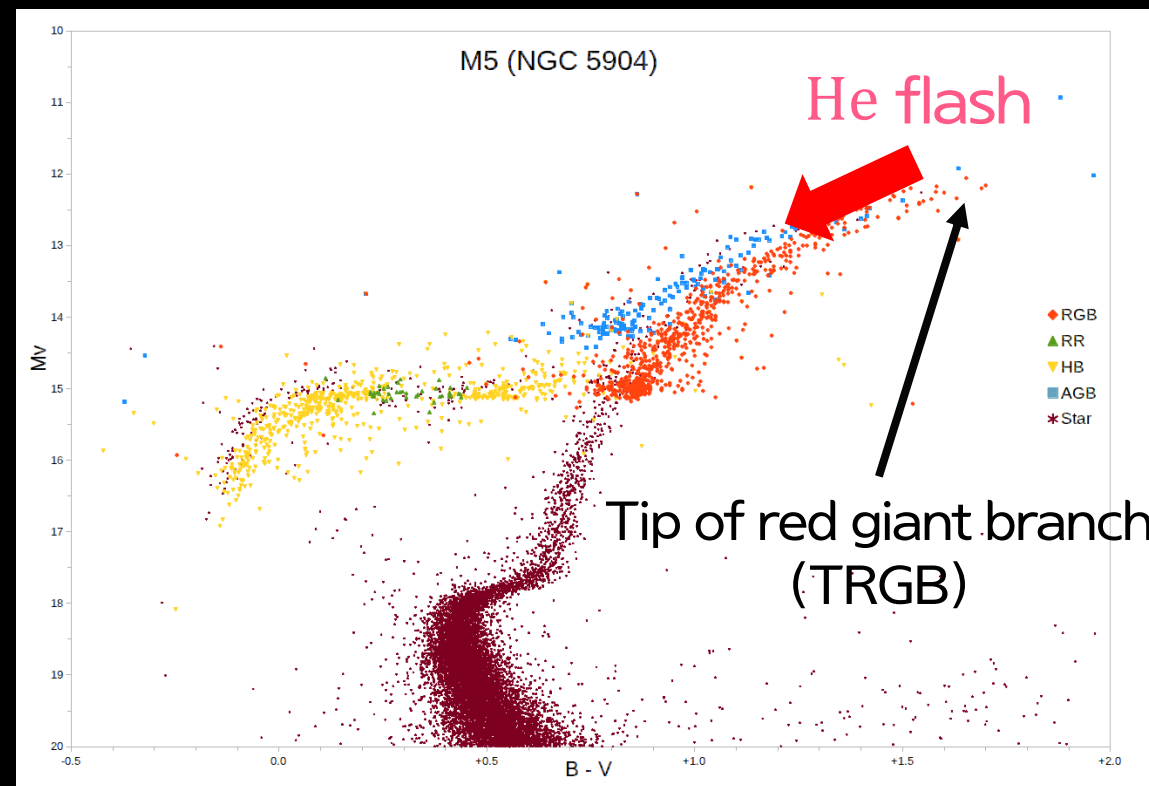
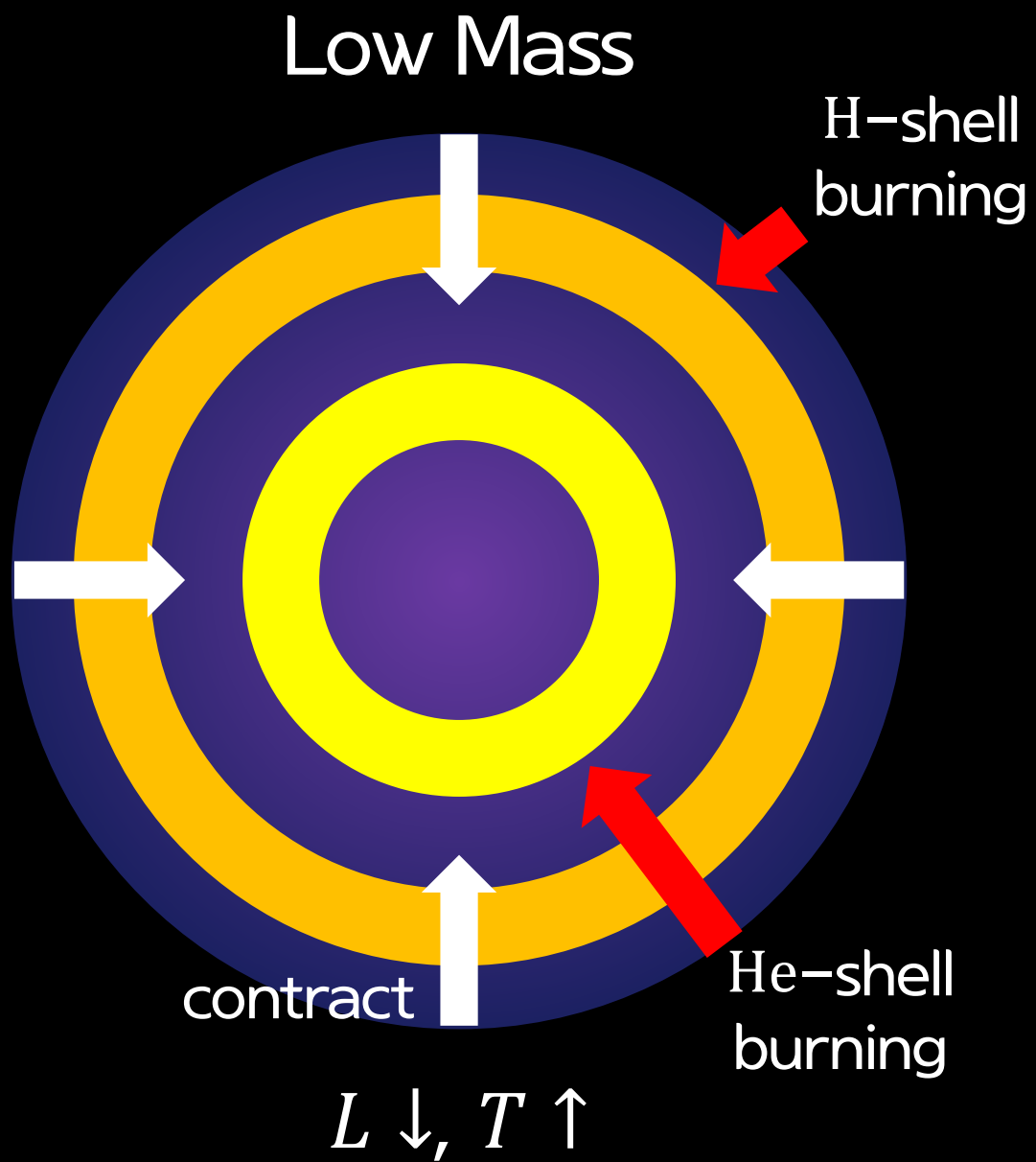
Intermediate

Massive

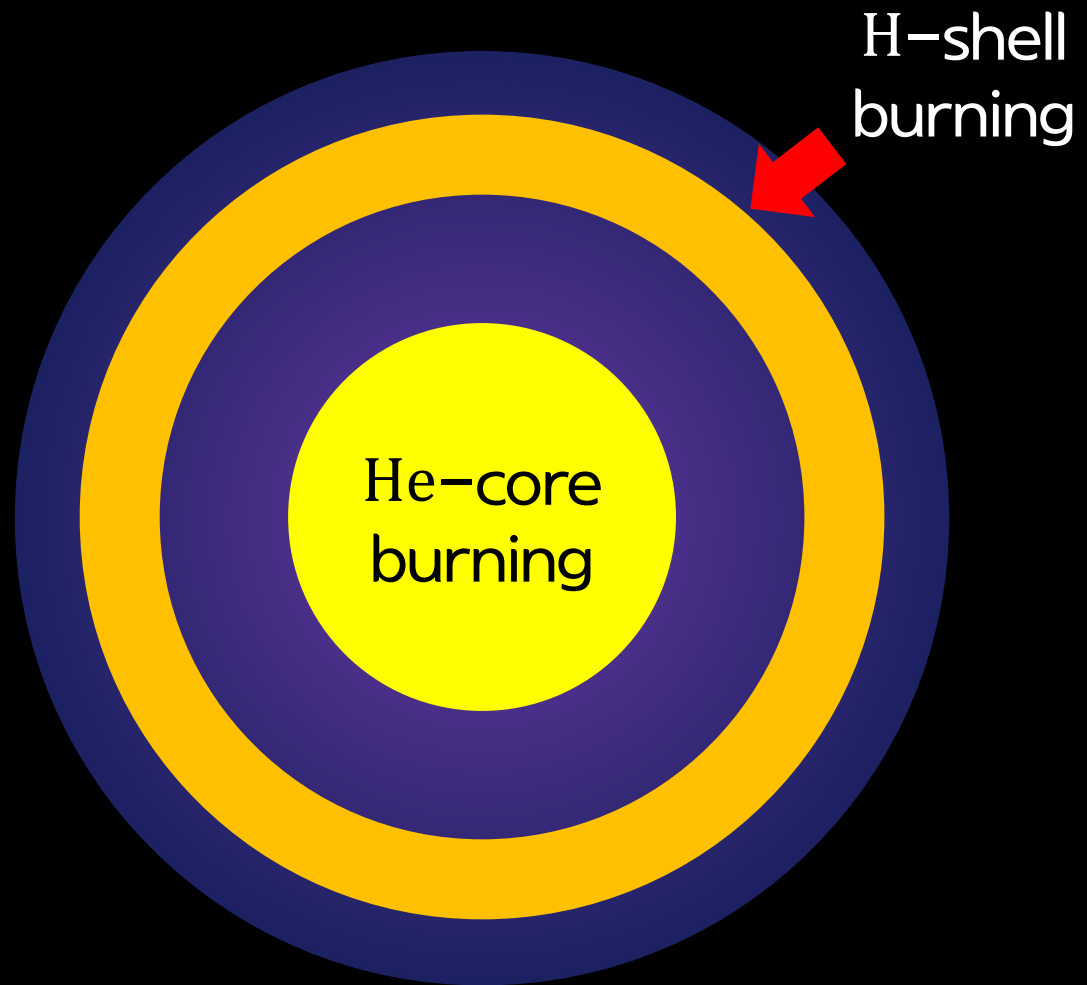
$2.0 - 8.0 M_{\odot}$

$8.0 M_{\odot} -$

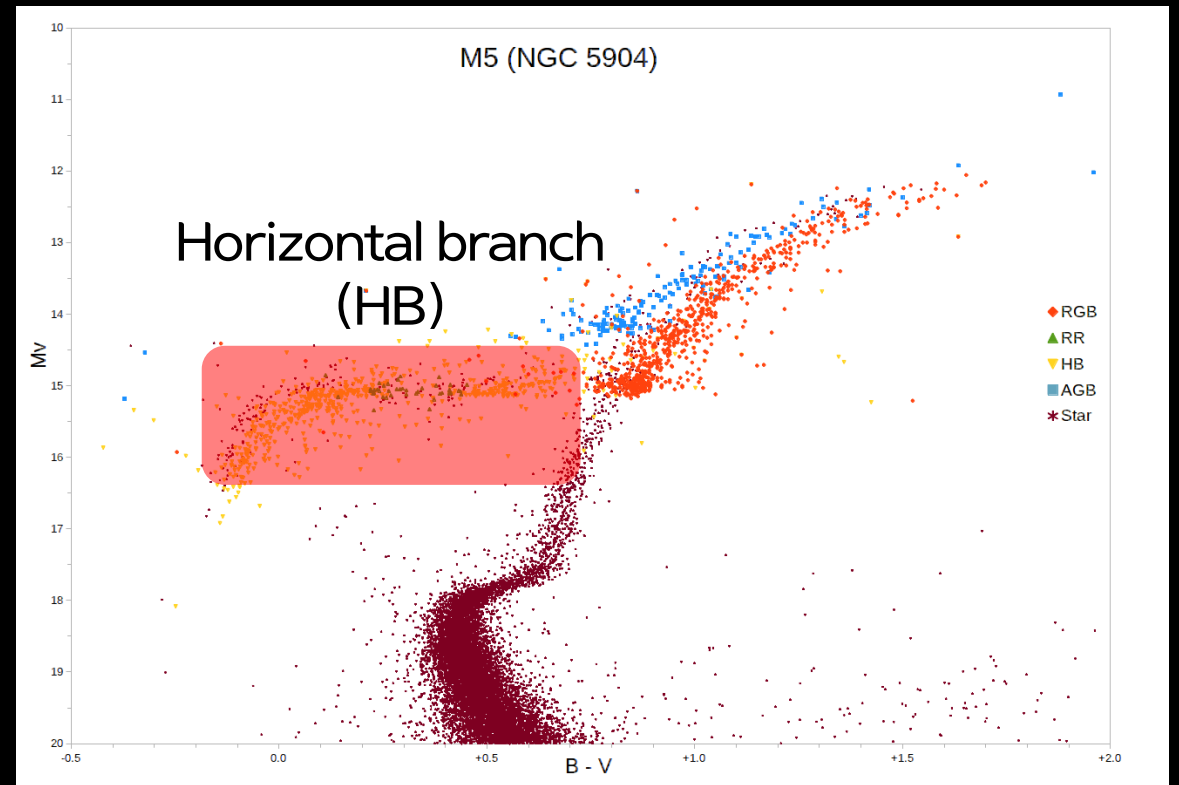




Low Mass

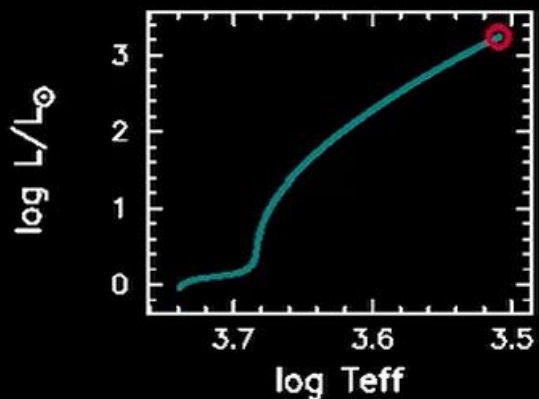


$L \uparrow, T \downarrow$

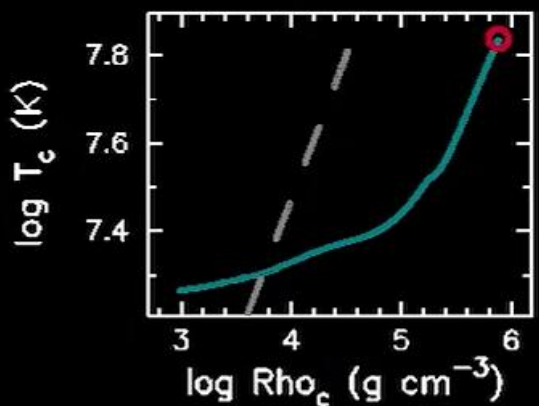
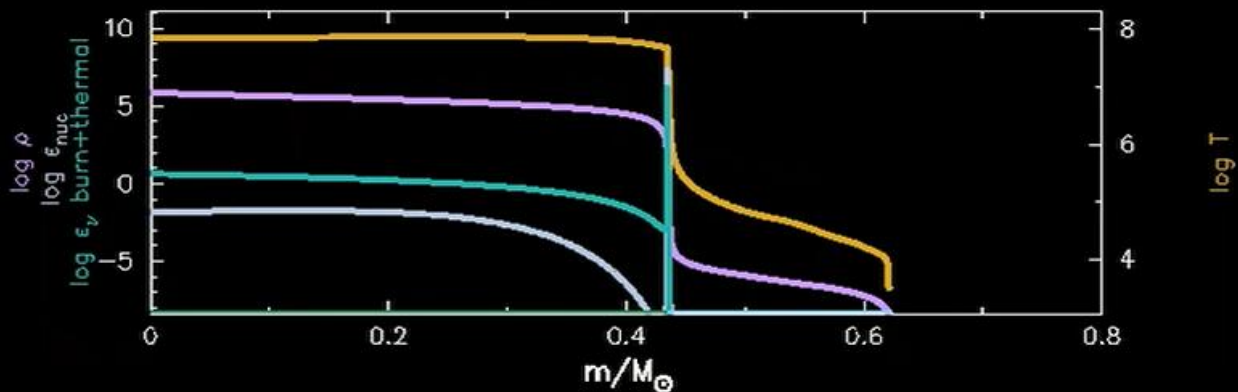


age 2.452811e10 yrs

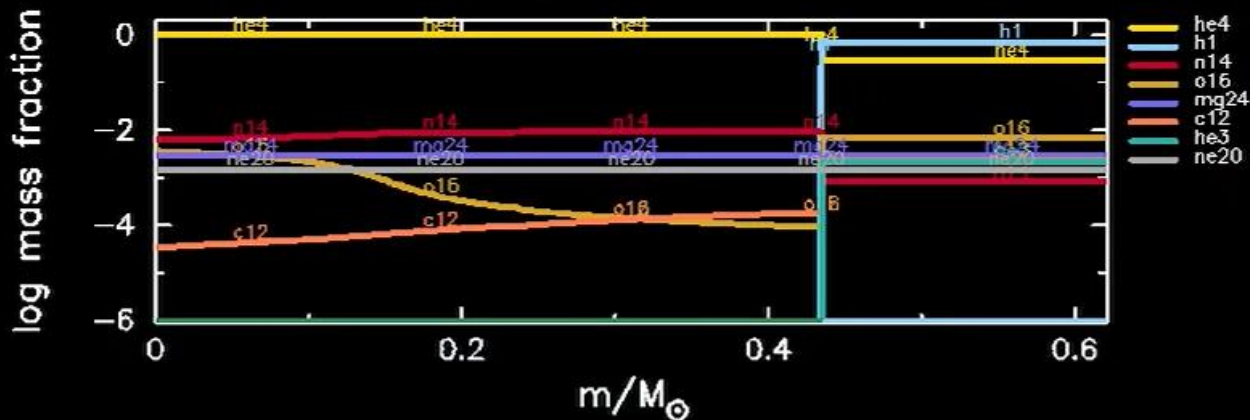
HR



TRho

pp
cno

Abundance



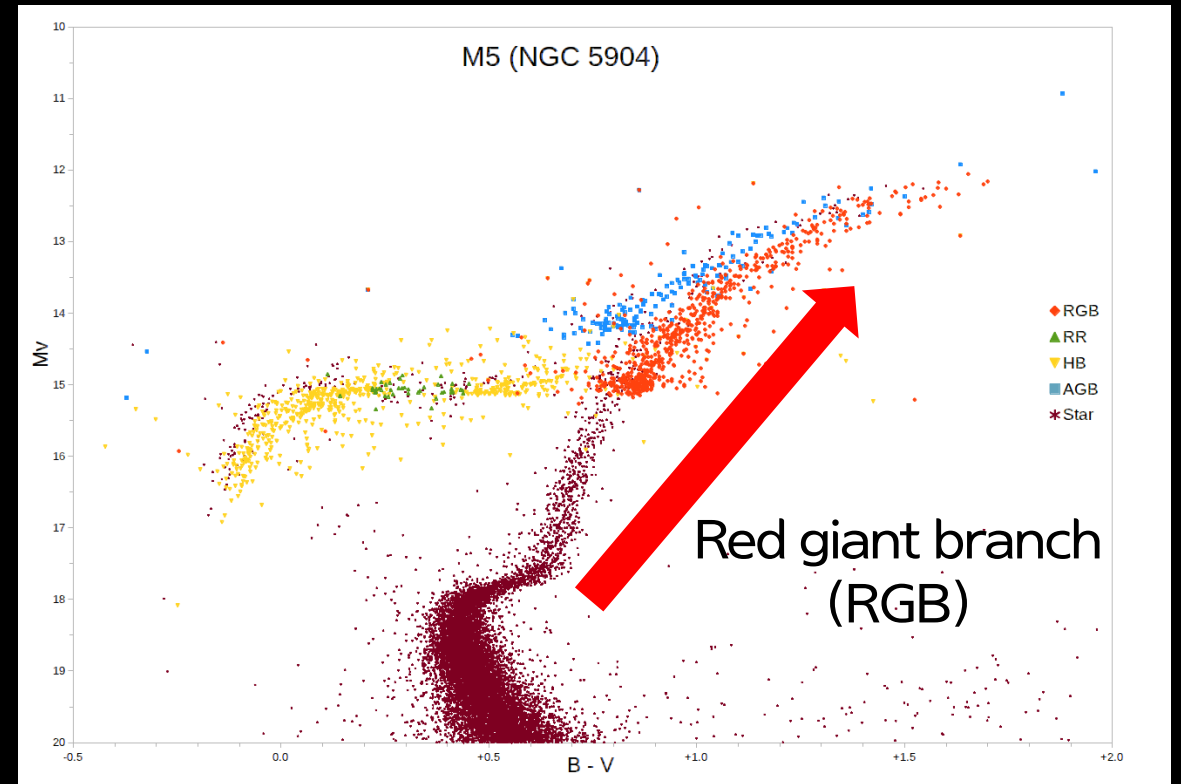
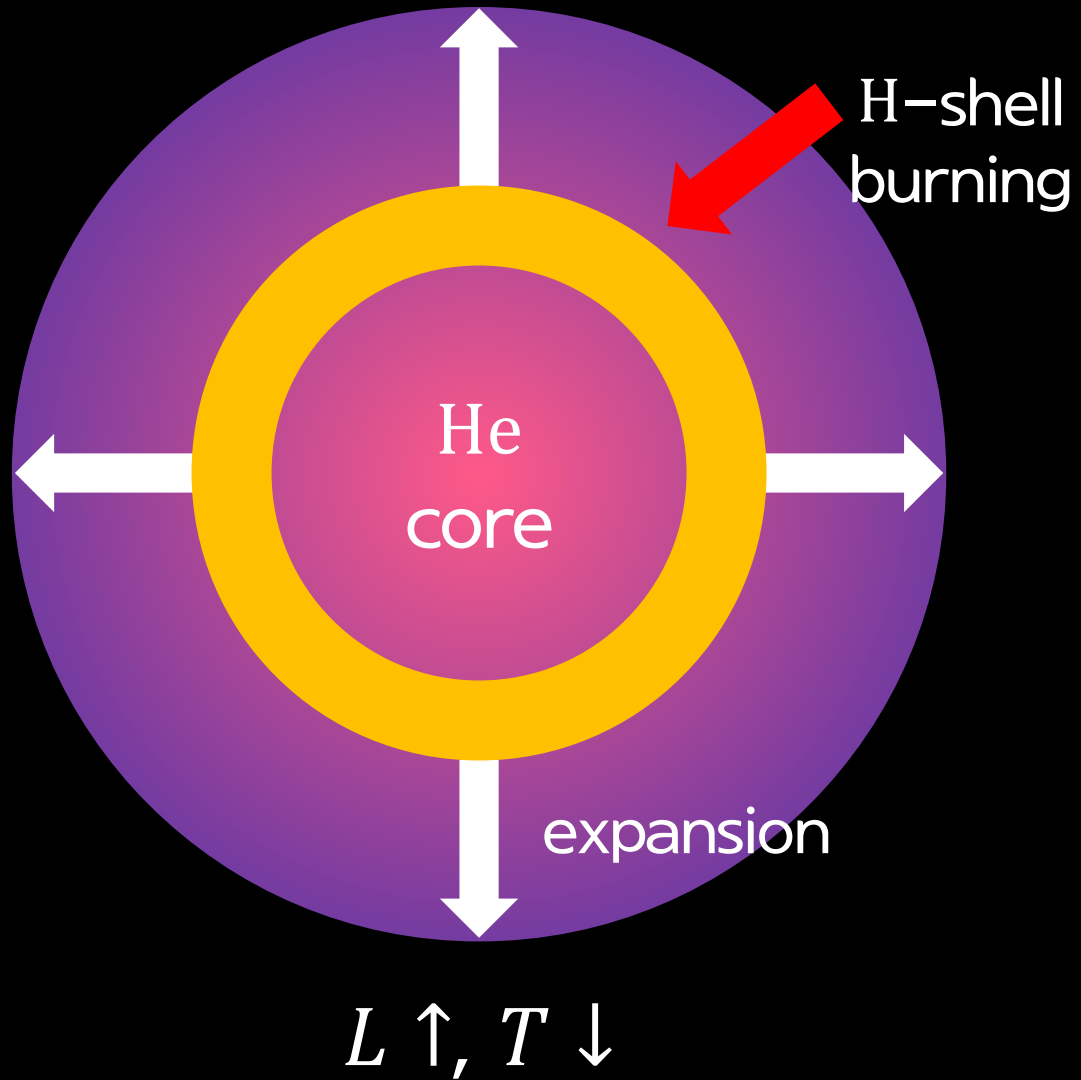
model_number 7500
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log_dt 2.7666187
log_L 3.2437355
log_Teff 3.5089980
log_R 2.1265249
log_g -0.0219140
log_surf_cell_P 2.8696419

star_mass 0.6209665
log_abs_mdott -7.2187408
he_core_mass 0.4346009
co_core_mass 0
cz_bot_mass 0.4376983
cz_top_mass 0.6207199
cz_bot_radius 1.5257401
cz_top_radius 132.3378518

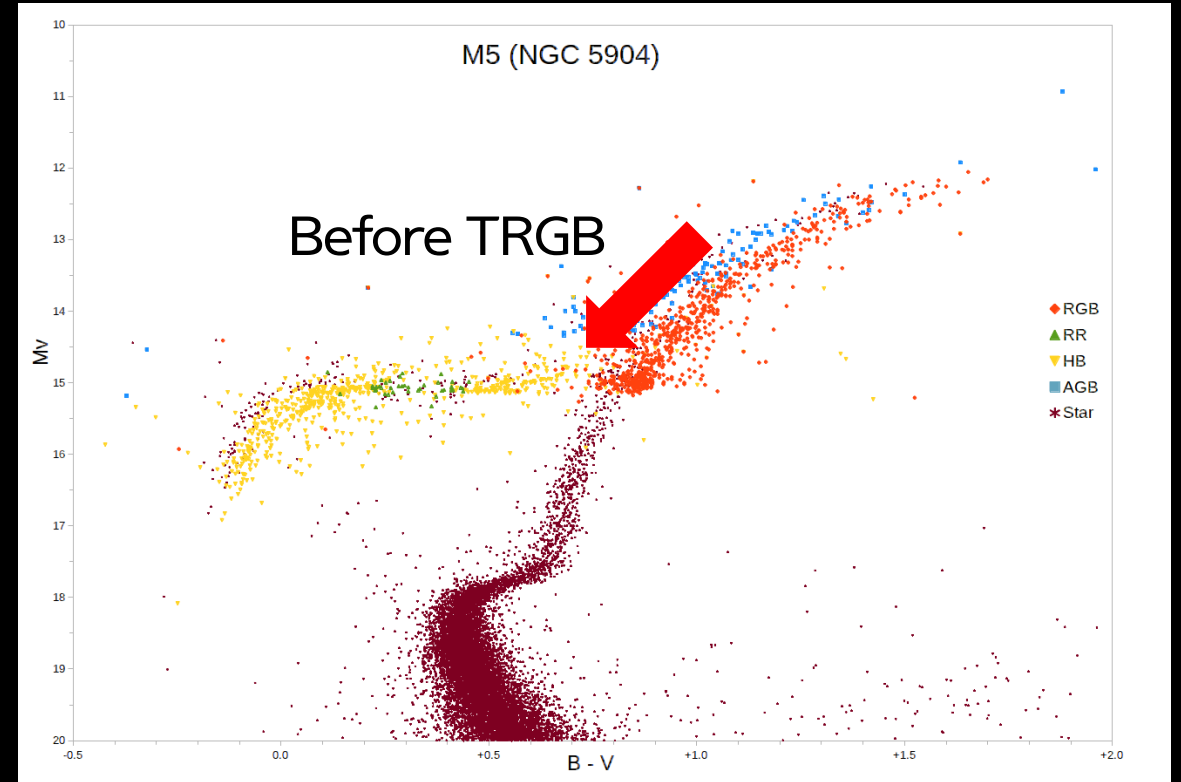
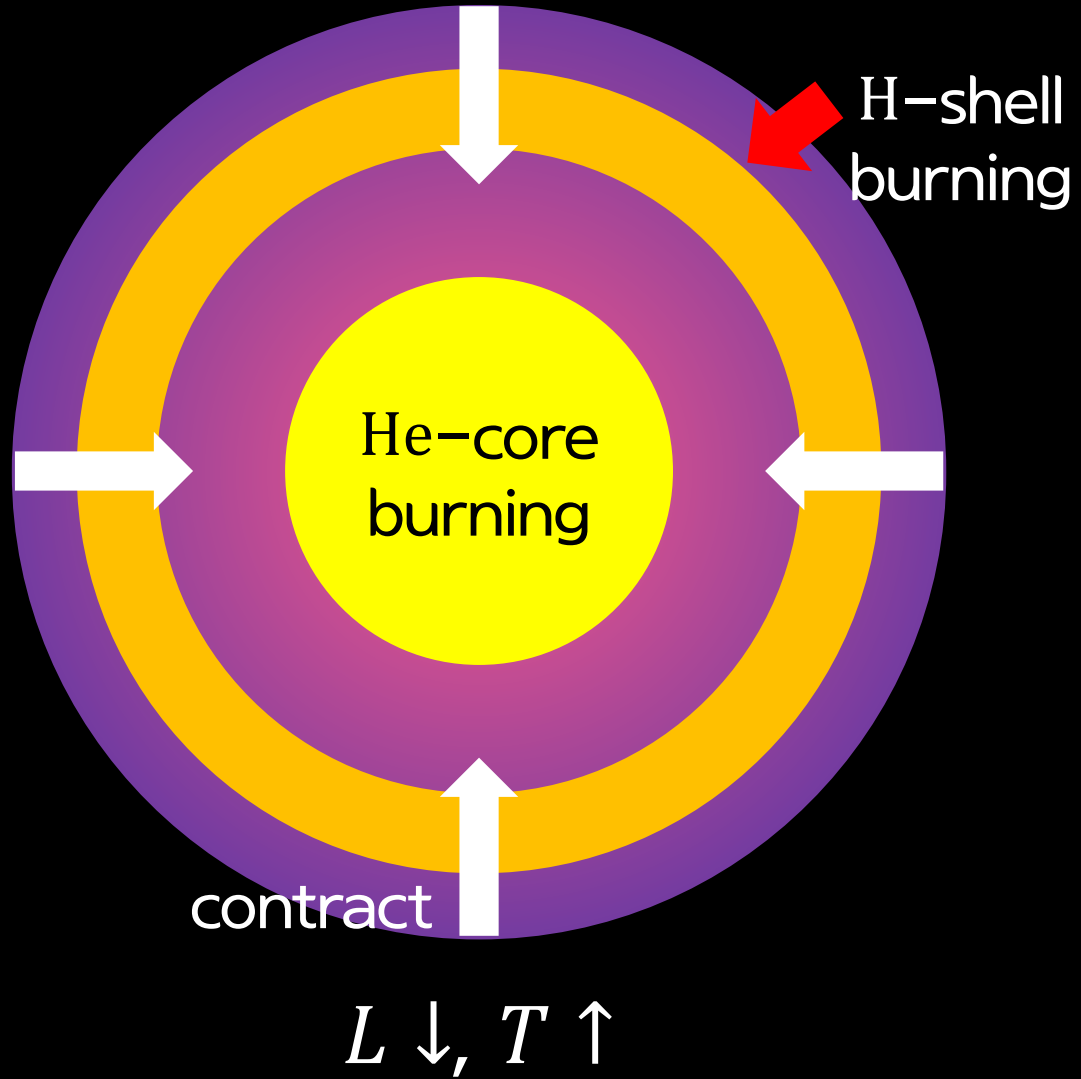
log_cntr_T 7.8380641
log_cntr_Rho 5.8909679
log_center_P 22.2725790
center h1 0
center he4 0.9859763
center c12 3.345E-05
center n14 0.0060954
center o16 0.0035813

log_Lnuc 3.2391598
log_Lneu 2.0732538
log_LH 3.2391593
log_LHe -2.6576303
log_LZ -4.7261059
num_zones 1663
num_retries 0

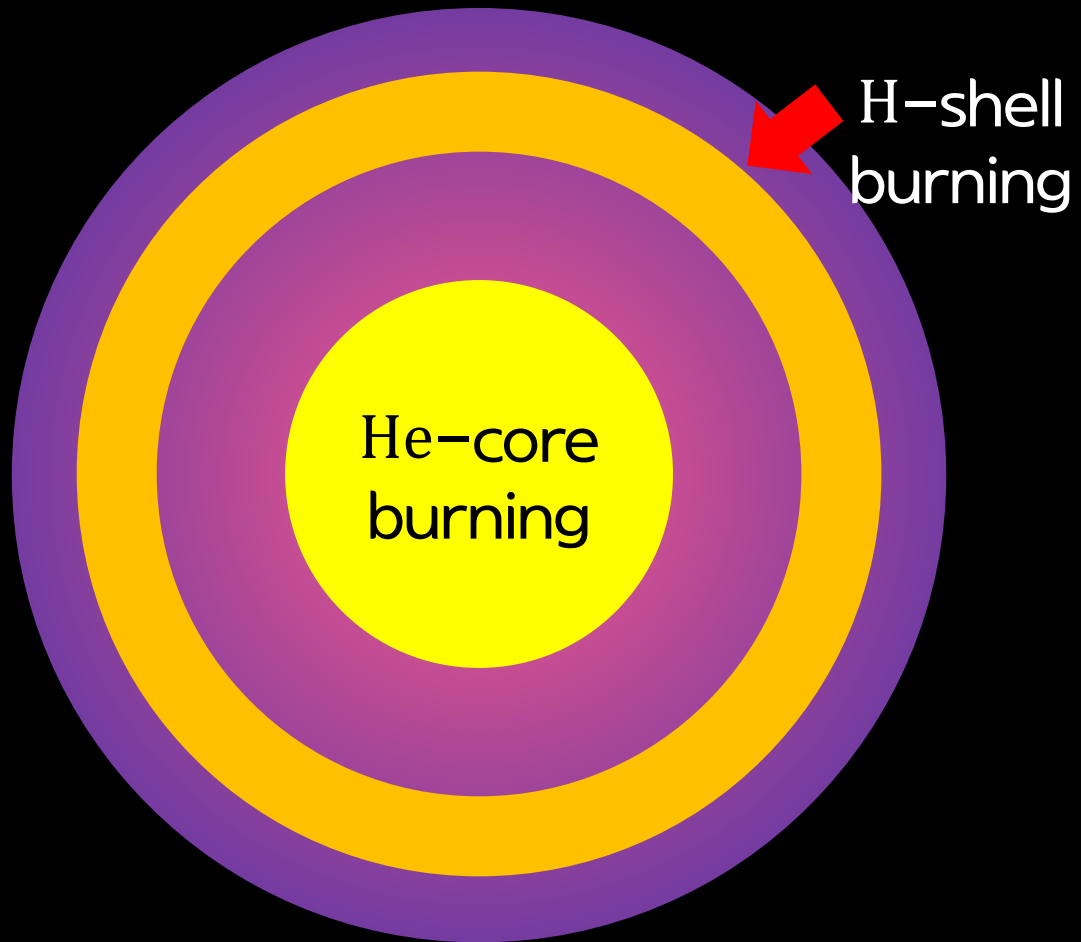
Intermediate Mass



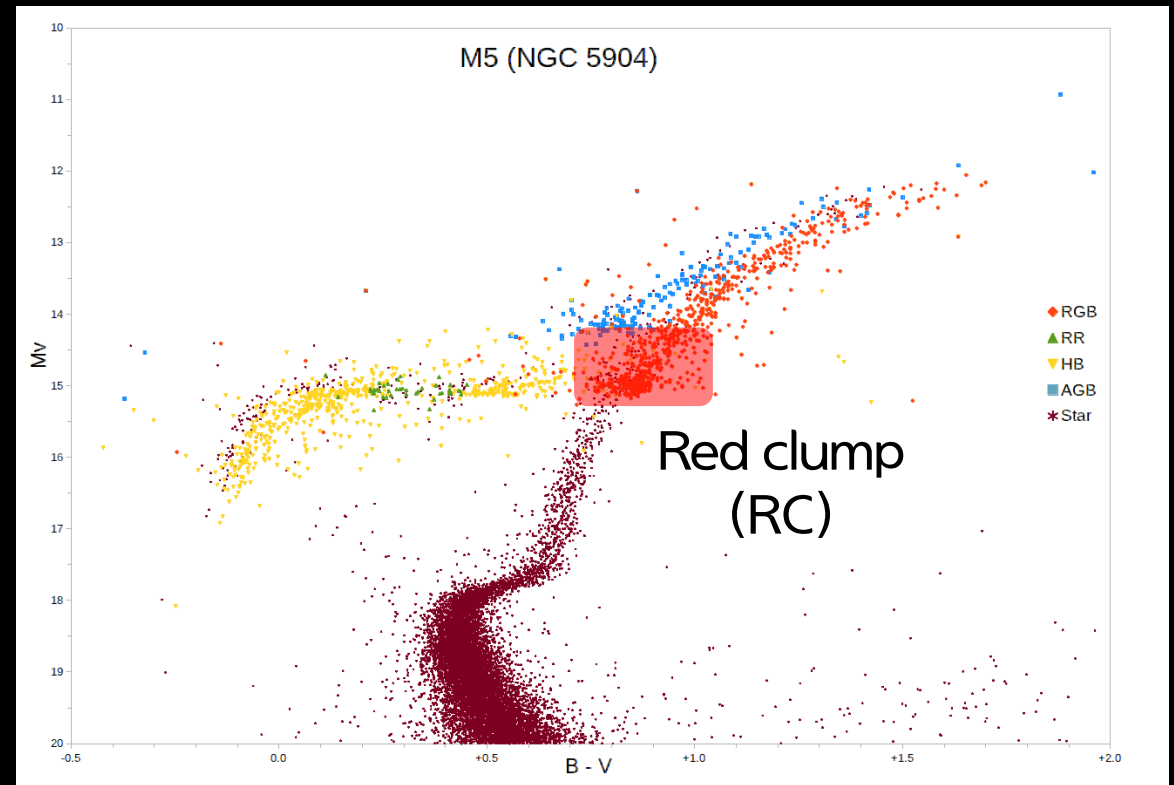
Intermediate Mass



$$2.0 M_{\odot} \lesssim M \lesssim 2.5 M_{\odot}$$

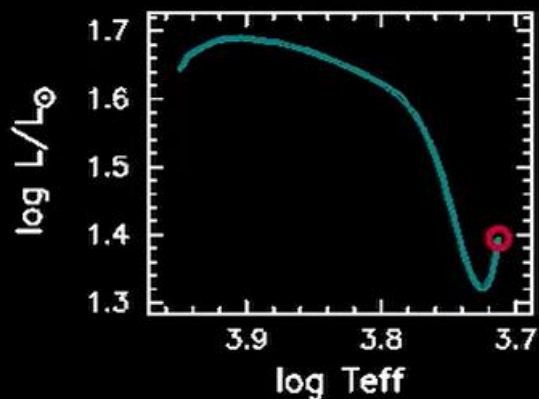


$$L \sim \text{const.}, T \sim \text{const.}$$

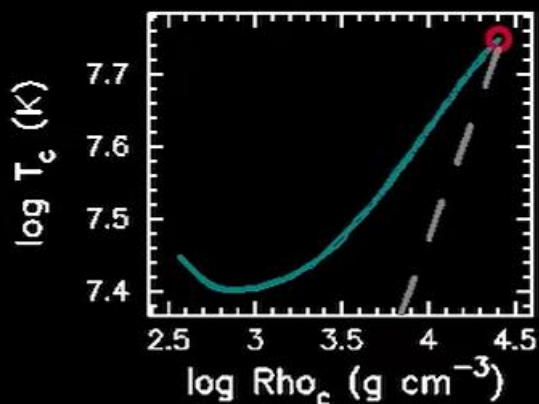
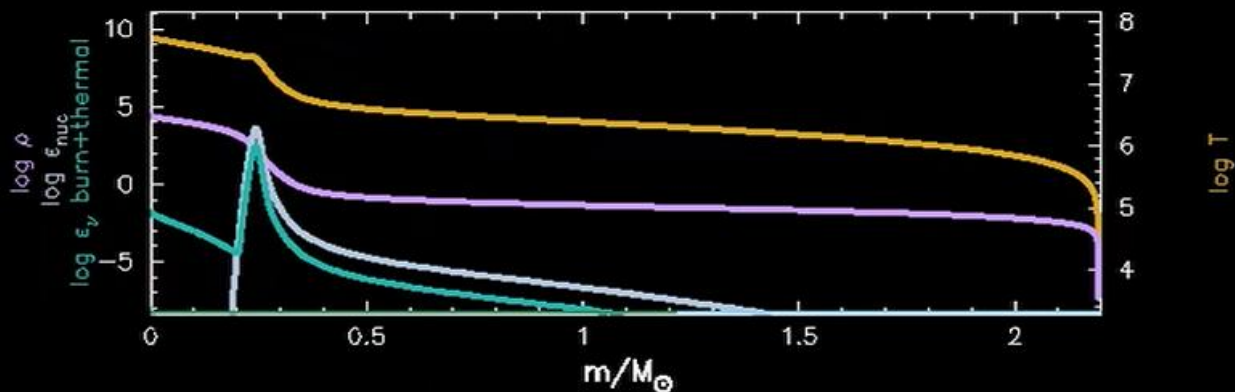


age 7.049143e8 yrs

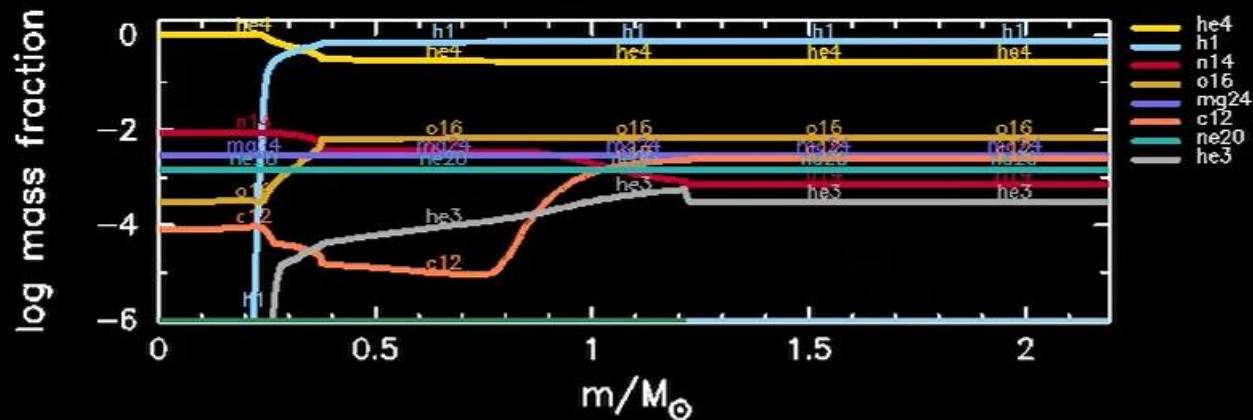
HR



TRho

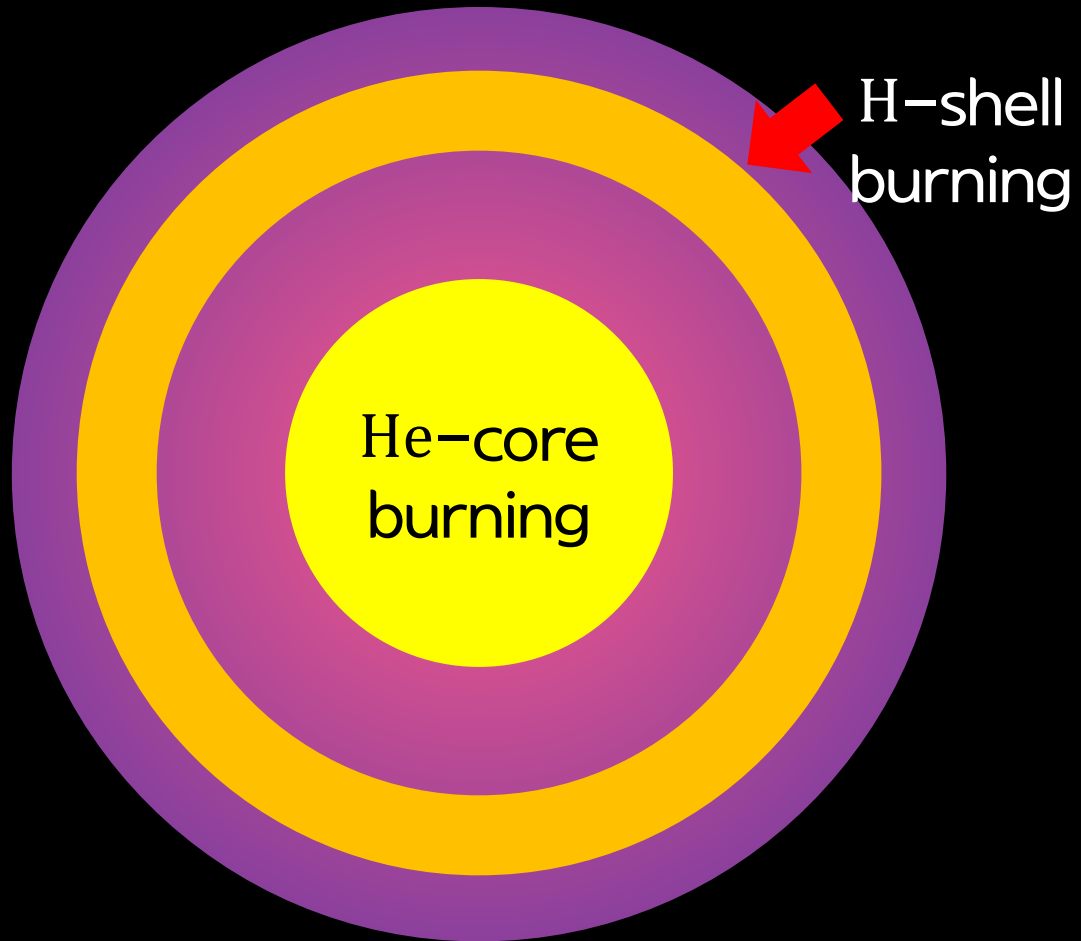
pp
cno

Abundance

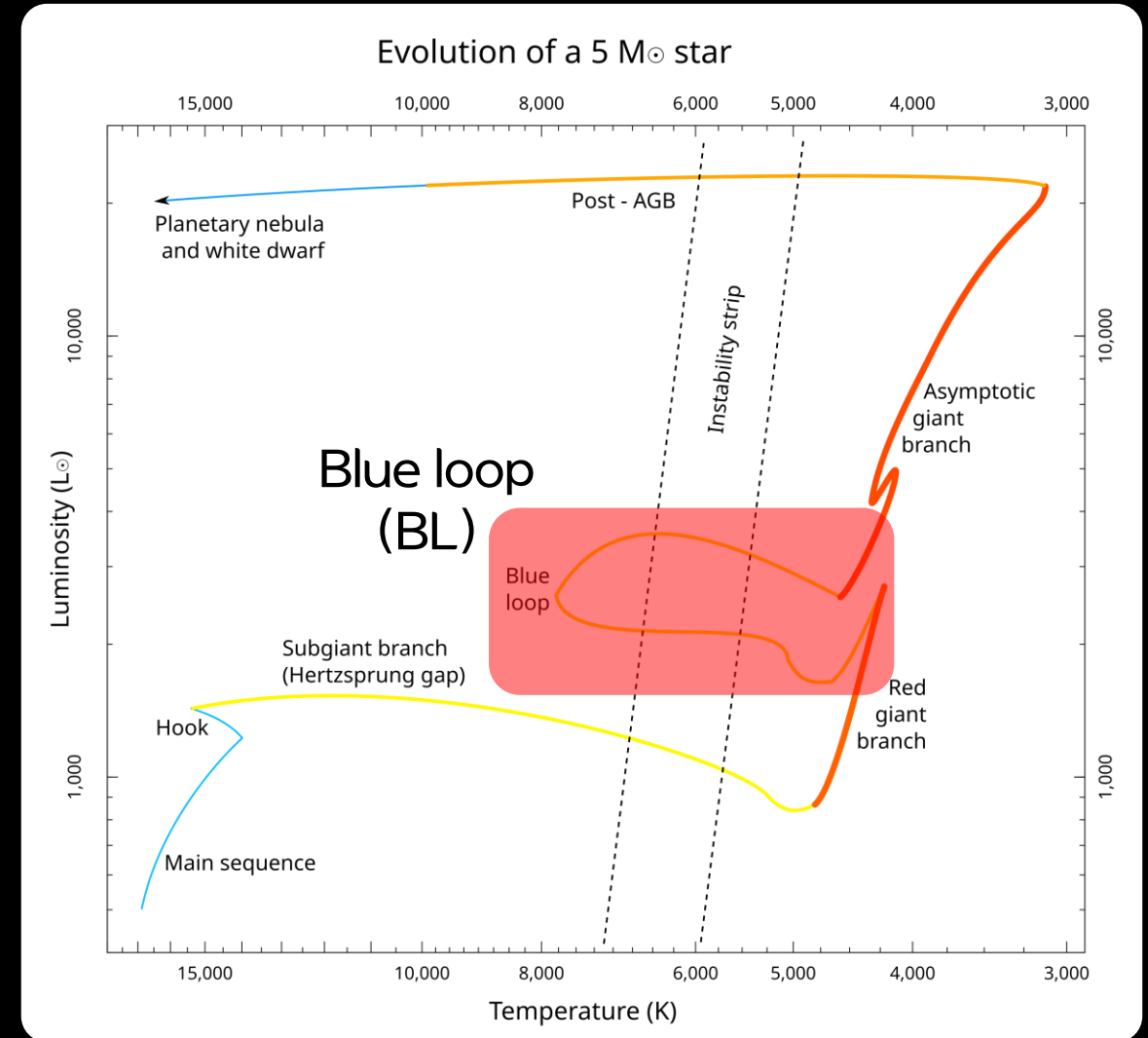


model_number	380	star_mass	2.1929911	log_cntr_T	7.7488493	log_Lnuc	1.4162880
log_star_age	8.8481363	log_abs_mdott	-10.9917241	log_cntr_Rho	4.4107872	log_Lneu	0.2490545
log_dt	5.6672621	he_core_mass	0.2470332	log_center_P	20.0932010	log_LH	1.4162880
log_L	1.3949586	co_core_mass	0	center h1	0	log_LHe	-11.4231118
log_Teff	3.7120044	cz_bot_mass	1.2252807	center he4	0.9863942	log_LZ	-10.7028834
log_R	0.7961237	cz_top_mass	2.1929910	center c12	8.379E-05	num_zones	1254
log_g	3.1868570	cz_bot_radius	2.7175030	center n14	0.0089007	num_retries	0
log_surf_cell_P	4.2205555	cz_top_radius	6.2519243	center o16	3.078E-04		

$$2.5 M_{\odot} \lesssim M \lesssim 8.0 M_{\odot}$$



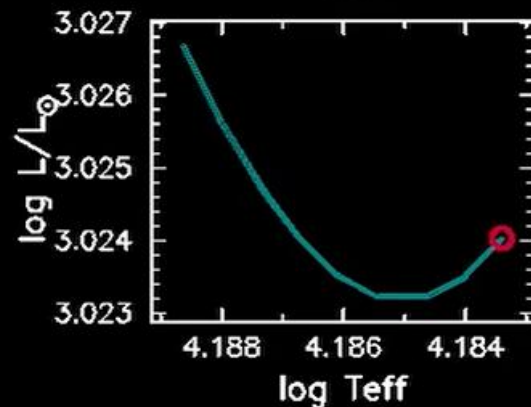
$$L \uparrow, T \uparrow \sim \downarrow$$



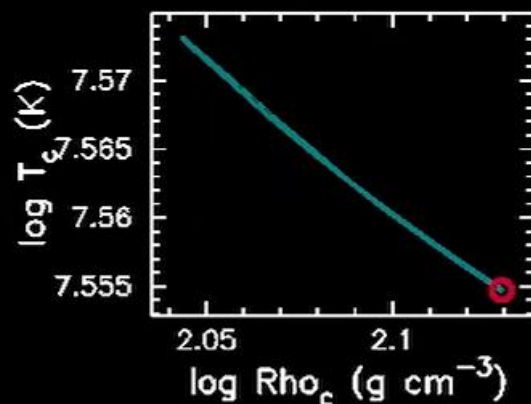
Lithopsian, Evolutionary track 5m, Wikipedia 2016

age 8.306641e7 yrs

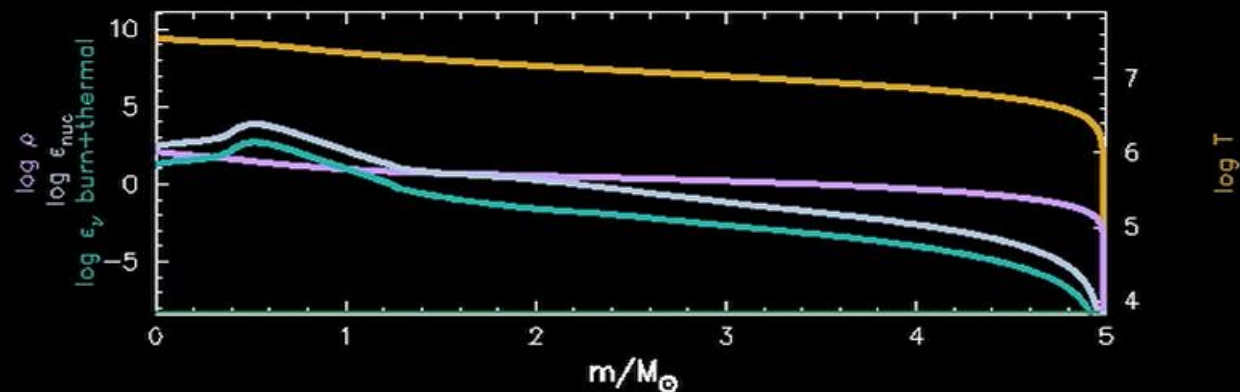
HR



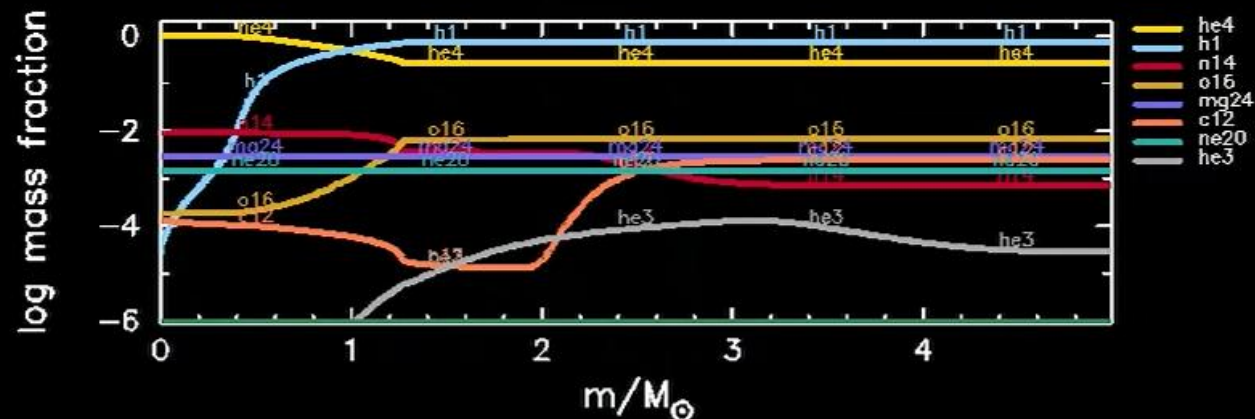
TRho



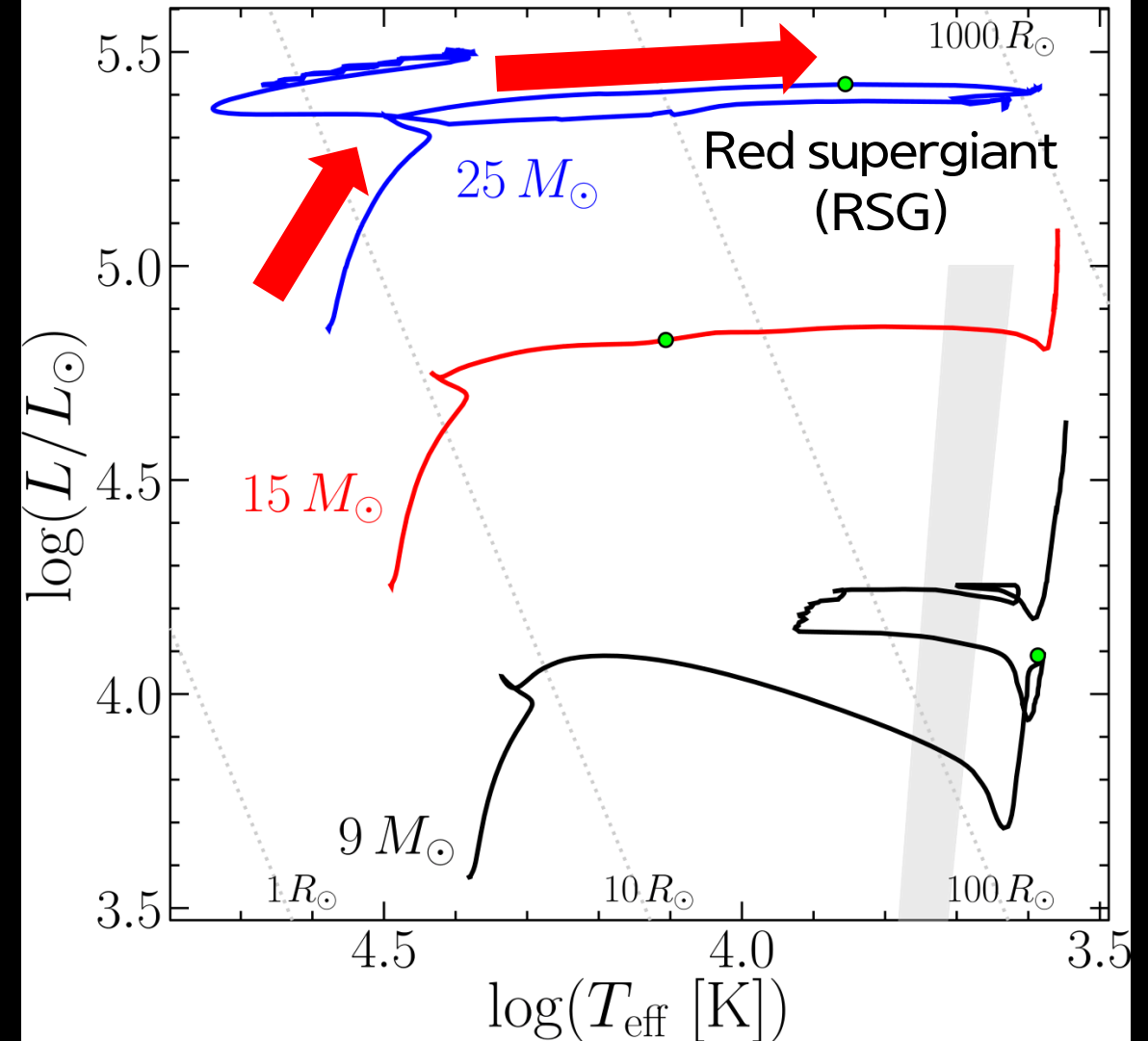
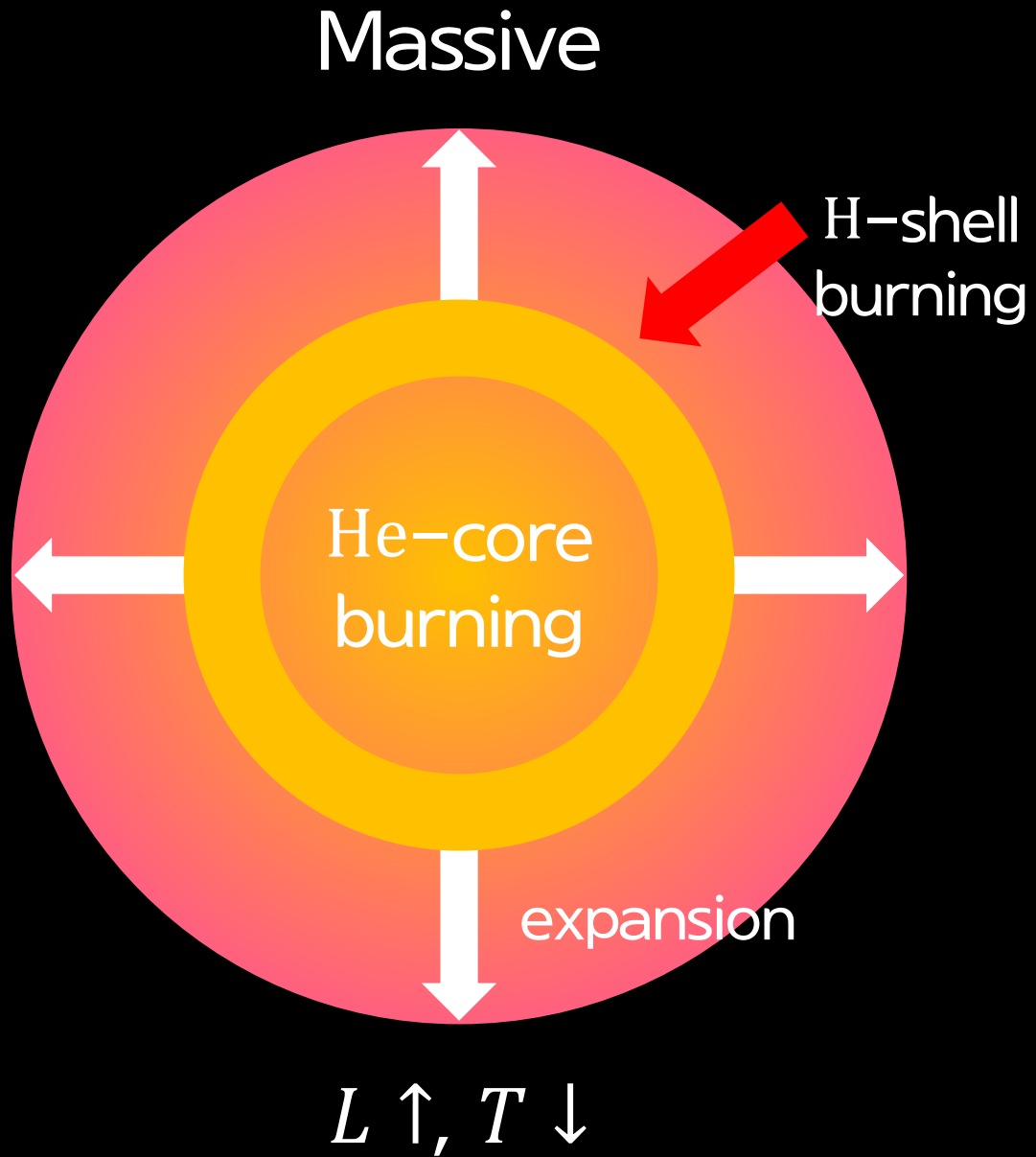
cno pp



Abundance



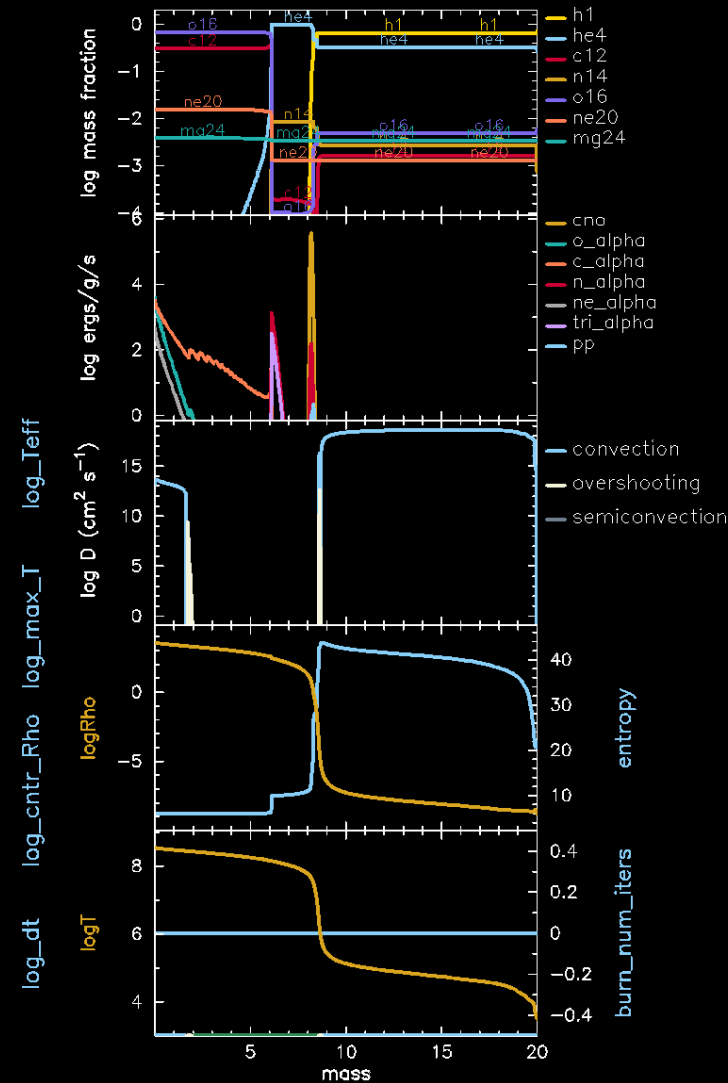
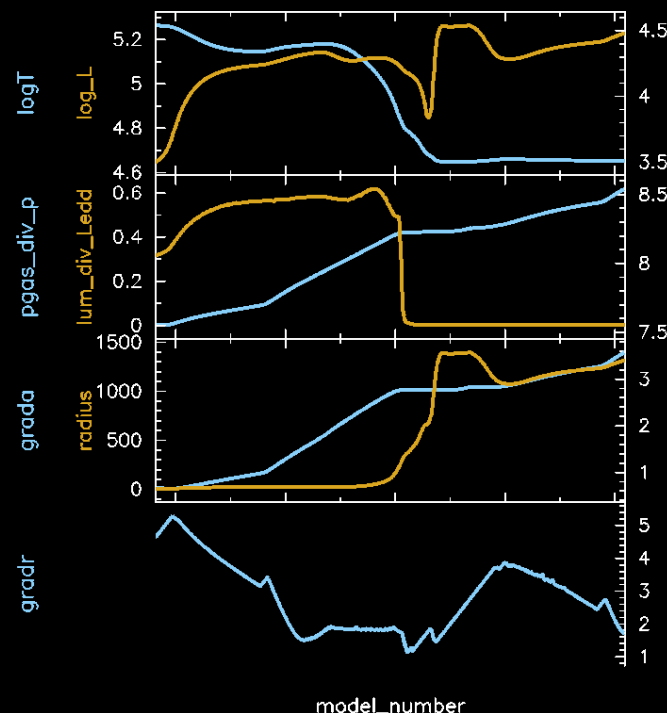
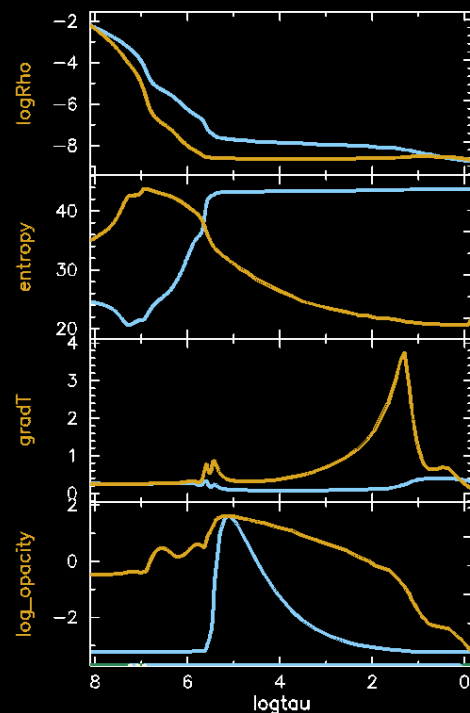
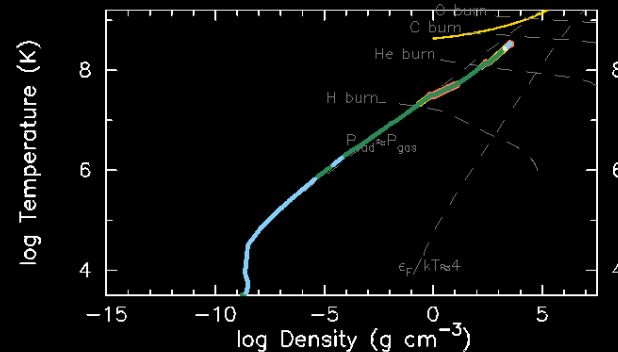
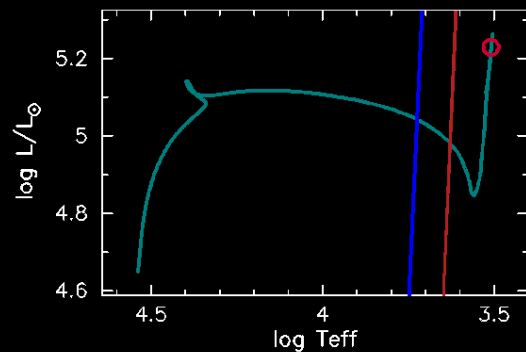
model_number	320	star_mass	4.9867119	log_cntr_T	7.5547301	log_Lnuc	3.0566234
log_star_age	7.9194255	log_abs_mdot	-9.8037155	log_cntr_Rho	2.1296857	log_Lneu	1.8898499
log_dt	3.4552184	he_core_mass	0.5255276	log_center_P	17.4838885	log_LH	3.0566234
log_L	3.0240377	co_core_mass	0	center h1	2.841E-05	log_LHe	-24.3128072
log_Teff	4.1833877	cz_bot_mass	4.9867119	center he4	0.9863897	log_LZ	-12.1661386
log_R	0.6678966	cz_top_mass	4.9867119	center c12	1.326E-04	num_zones	975
log_g	3.8000887	cz_bot_radius	4.6137859	center n14	0.0089545	num_retries	0
log_surf_cell_P	3.0333294	cz_top_radius	4.6285848	center o16	1.812E-04		



Sylvia Ekström and Cyril Georgy, Stellar Evolution Through the Red Supergiant Phase, MDPI AG 2025

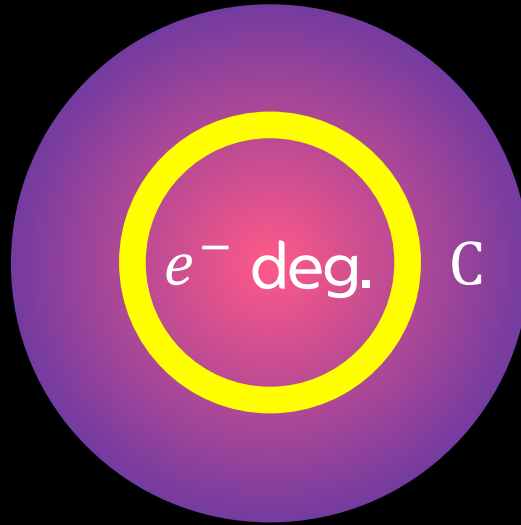
age	1.012177e7 yrs	log_dt	1.7154821	Mass	20.0000000	H_cntr	0	H_rich	11.7941652
model_number	1218	star_age	1.012E+07	lg_Mdot	-99.0000000	He_cntr	6.177E-07	He_core	8.2058348
non_fe_core_infall	4.880E-07	gamma1_min	1.2525179	eta_cntr	-3.4186976	C_cntr	0.3077678	CO_core	6.0844774
fe_core_infall	0	log_max_T	8.5411061	lg_Lnuc	4.2047533	O_cntr	0.6728515	Fe_core	0
		log_cntr_T	8.5411061	lg_Lneu	3.4398854	Ne_cntr	0.0155577	zones	858
		log_cntr_Rho	3.5737612	lg_Lphoto	-55.7310026	Si_cntr	1.071E-08	retries	6

20 M_⊙

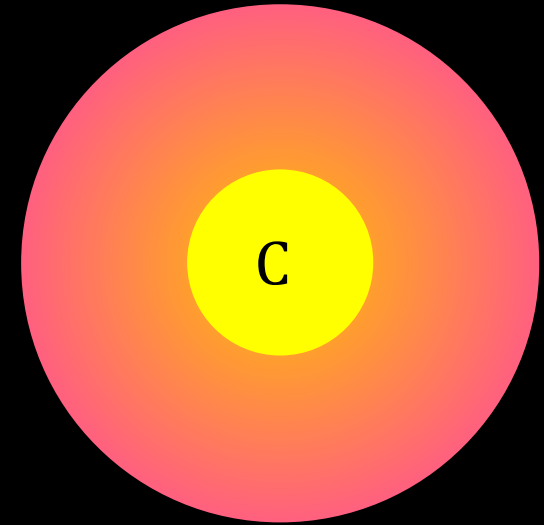




No C burning



Shell C ignition



Core C ignition

Low Intermediate

$0.5 - 2.0 M_{\odot}$ $2.0 - 8.0 M_{\odot}$

Massive

$8.0 M_{\odot} - 10 M_{\odot}$

Massive

$10 M_{\odot} -$

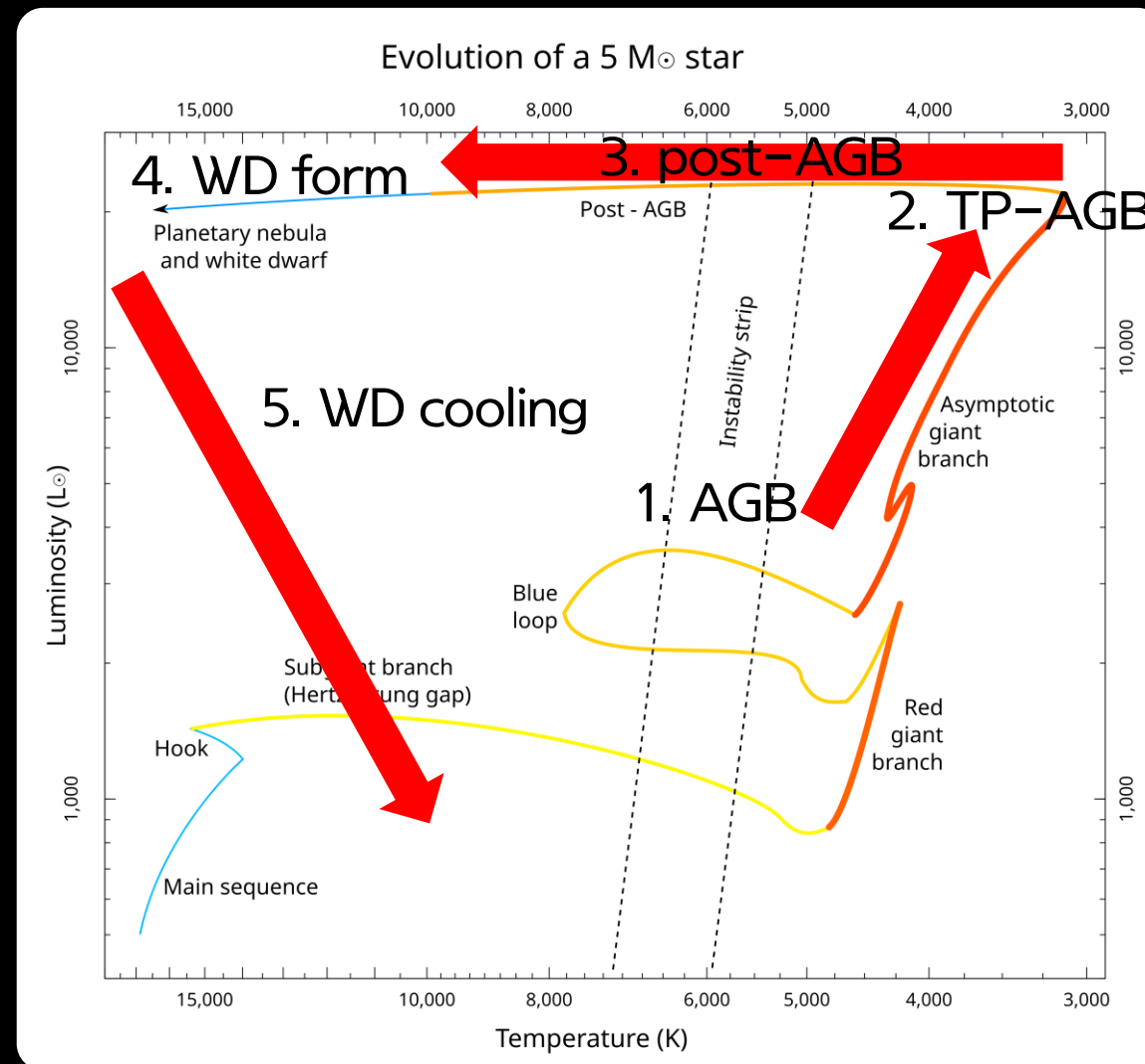


e^- deg.

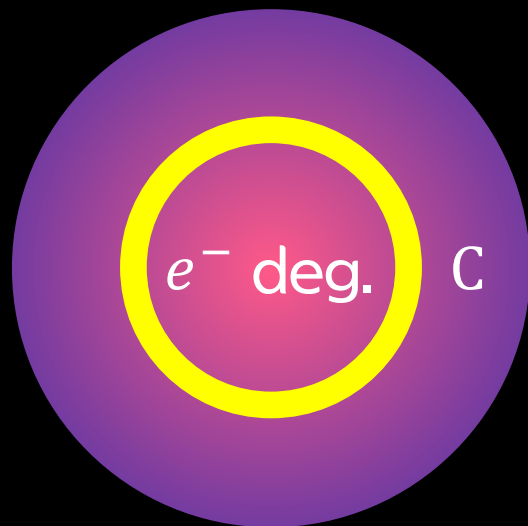
No C burning

Low Intermediate

$0.5 - 2.0 M_{\odot}$ $2.0 - 8.0 M_{\odot}$



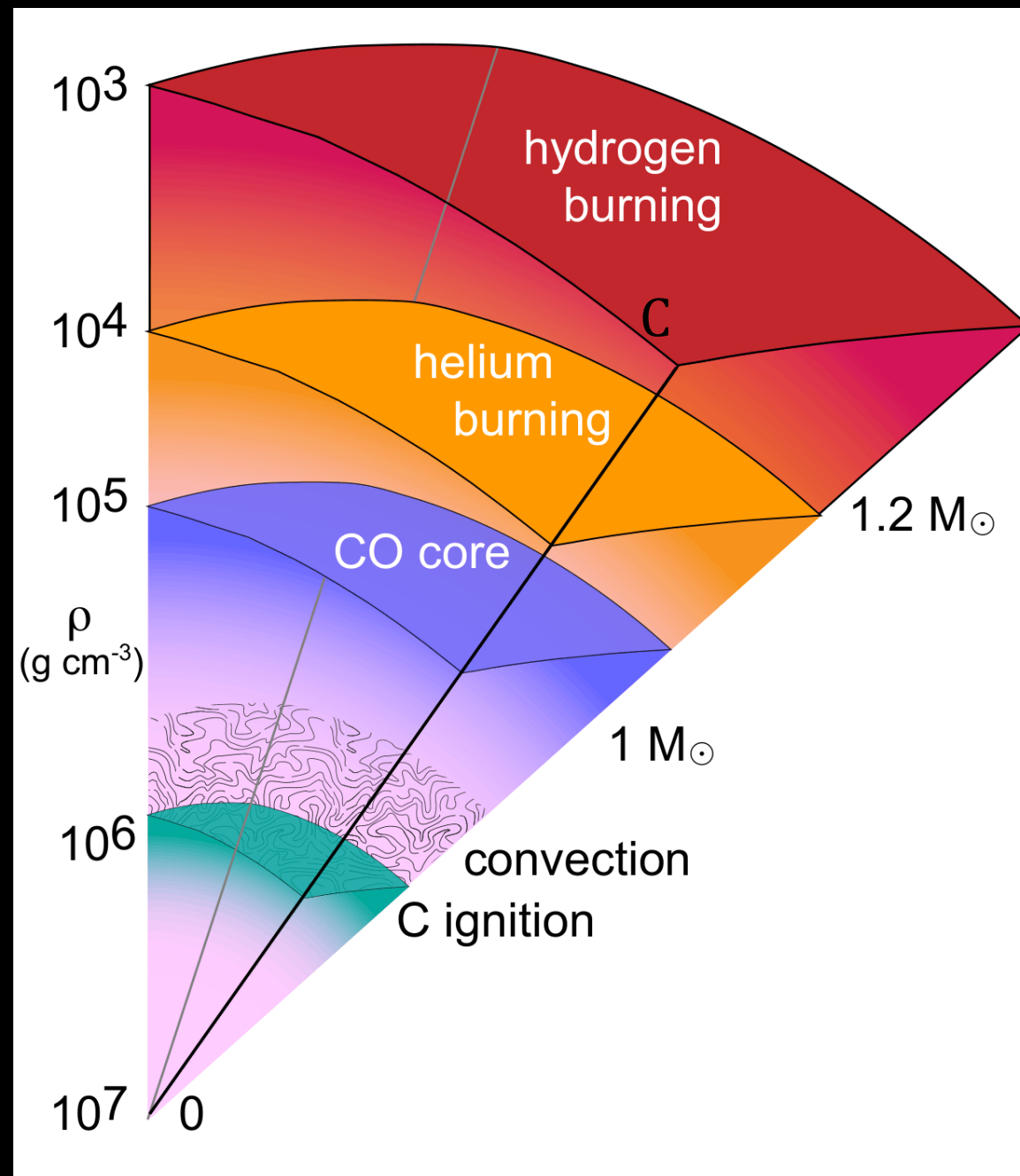
Lithopsian, Evolutionary track 5m, Wikipedia 2016

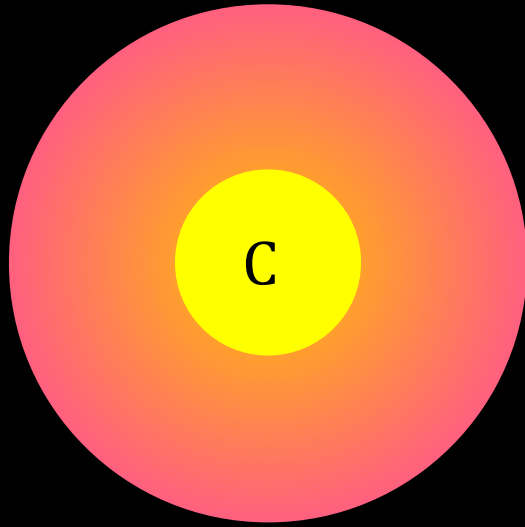


Shell C ignition

Massive

$8.0 M_{\odot} - 10 M_{\odot}$



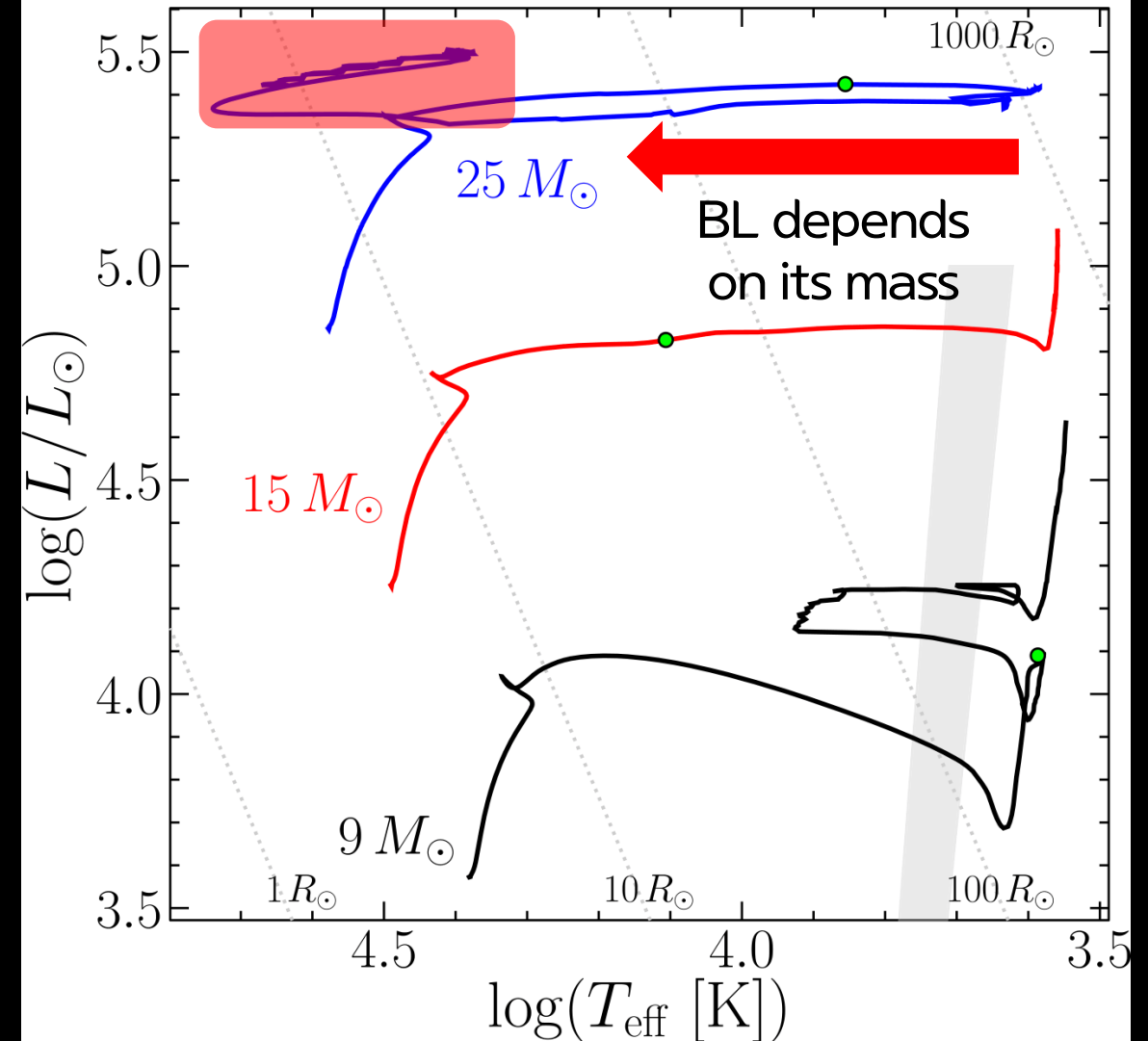


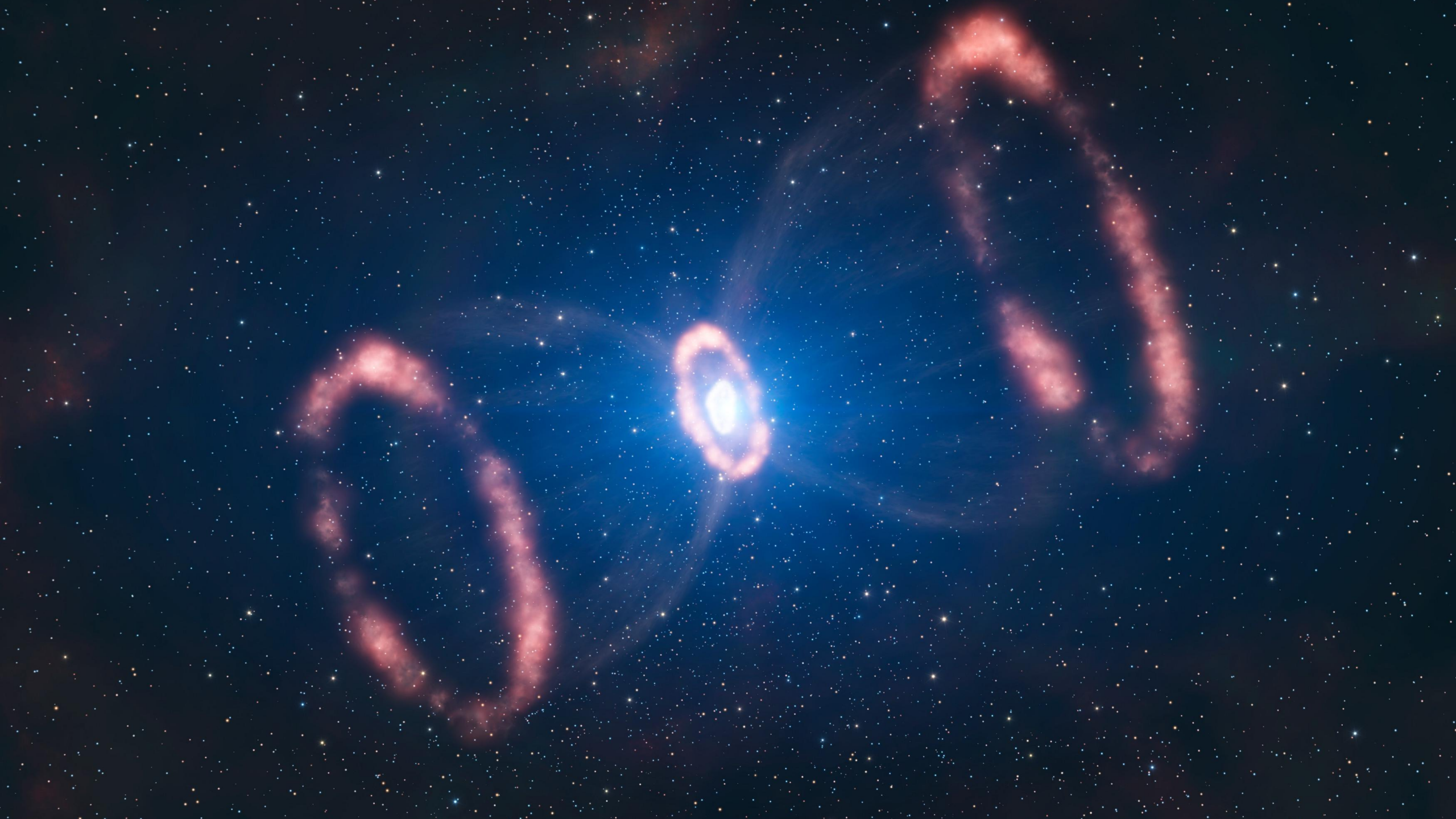
Core C ignition

Massive

$10 M_{\odot} -$

Barely moves during the
advanced fusion





Core–Collapse Supernova

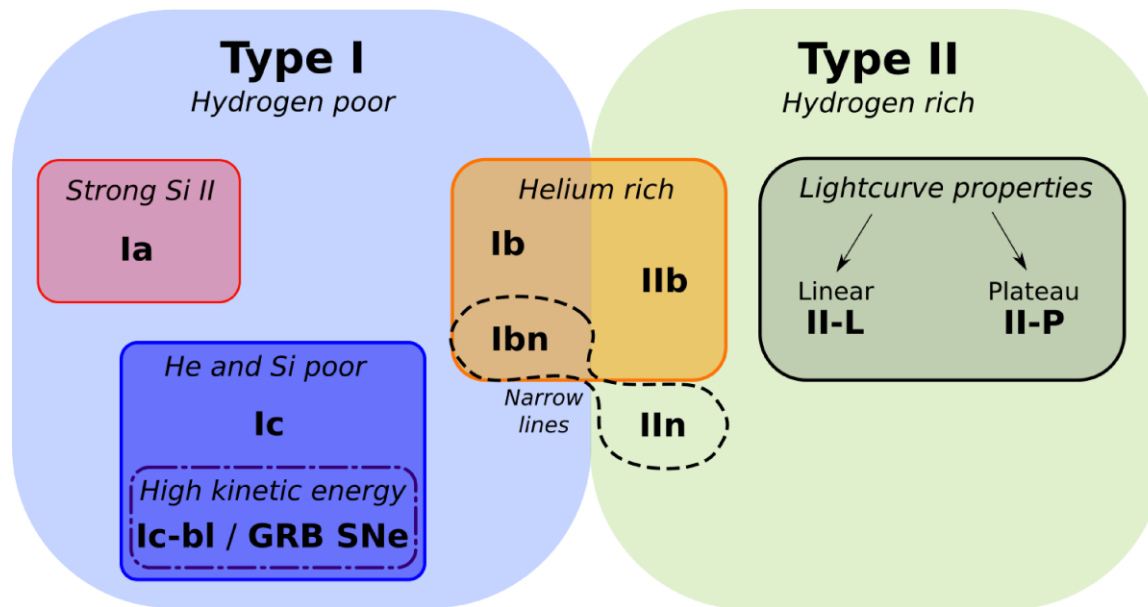
Electron degenerate

Adiabatic exponent $\Gamma_1 \leq \frac{4}{3}$

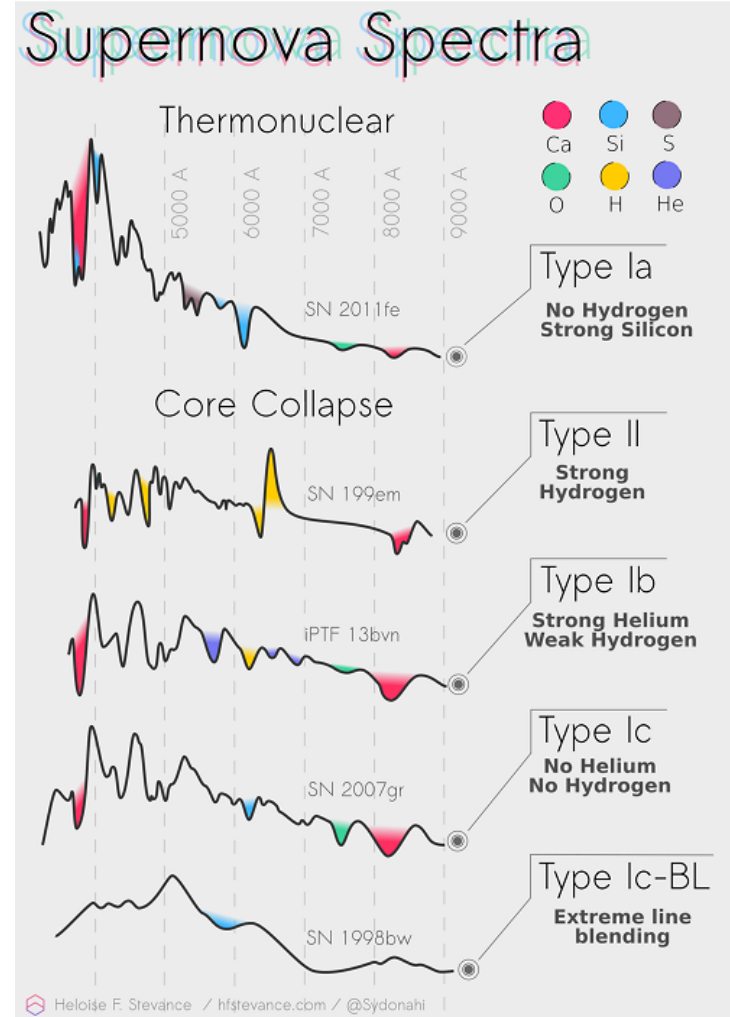


Categorization of Supernovae

Observed Spectra and Light Curves



Heloise F. Stevance, Core-Collapse Supernovae, 2024



Heloise F. Stevance, Supernova Spectra

Explosion Mechanisms



Thermonuclear

Explosion energy source:

Nuclear fusion energy
(Thermal runaway)

Low-mass star
Type Ia



Core-Collapse

Explosion energy source:

Gravitational collapse
(Electron capture)

Massive star
Type Ib, Ic, II



Pair-Instability

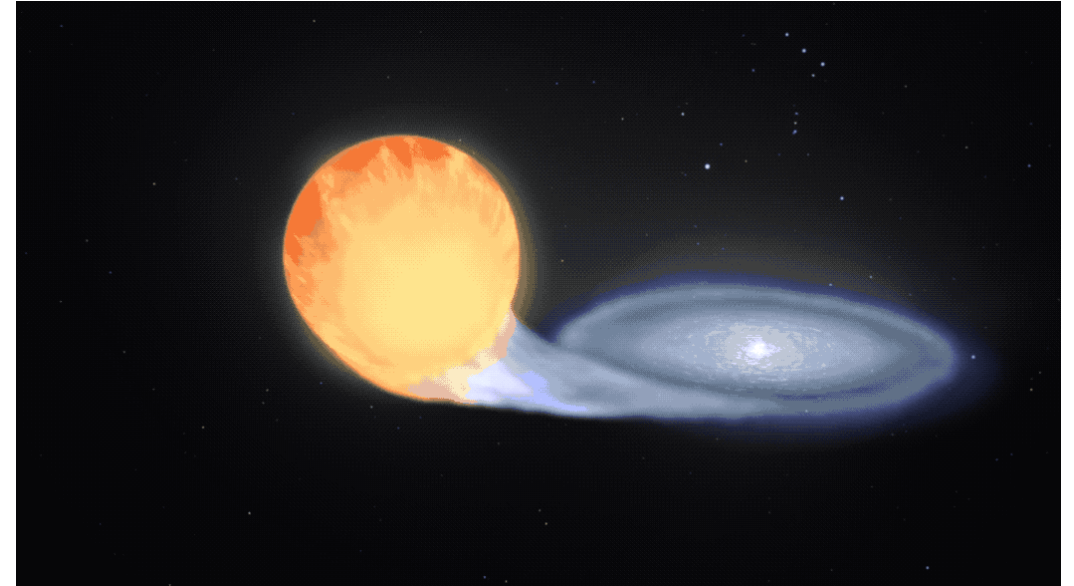
Explosion energy source:

Triggered by pair-production,
driven by nuclear fusion energy

Extremely massive star

Thermonuclear Supernovae

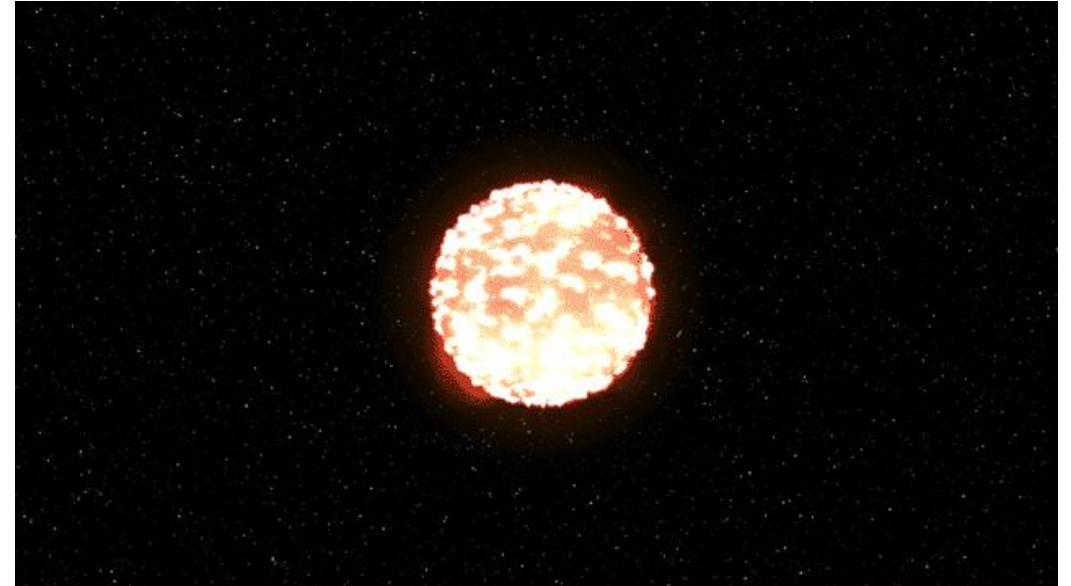
- M_{Ch} of a typical C–O white dwarf is
 $M_{\text{Ch}} \simeq 1.44 M_{\odot}$
- Once the mass exceed M_{Ch} through mass accretion (e.g., in a binary system), gravitational collapse begins, and the temperature increase during the collapse triggers explosive oxygen burning
- Since the exploding mass is nearly fixed, making these events useful as standard candles



NASA's Roman Mission Will Use Exploding Stars to Measure Cosmic Distances,
NASA Goddard's Scientific Visualization Studio, 2021

Core–Collapse Supernovae

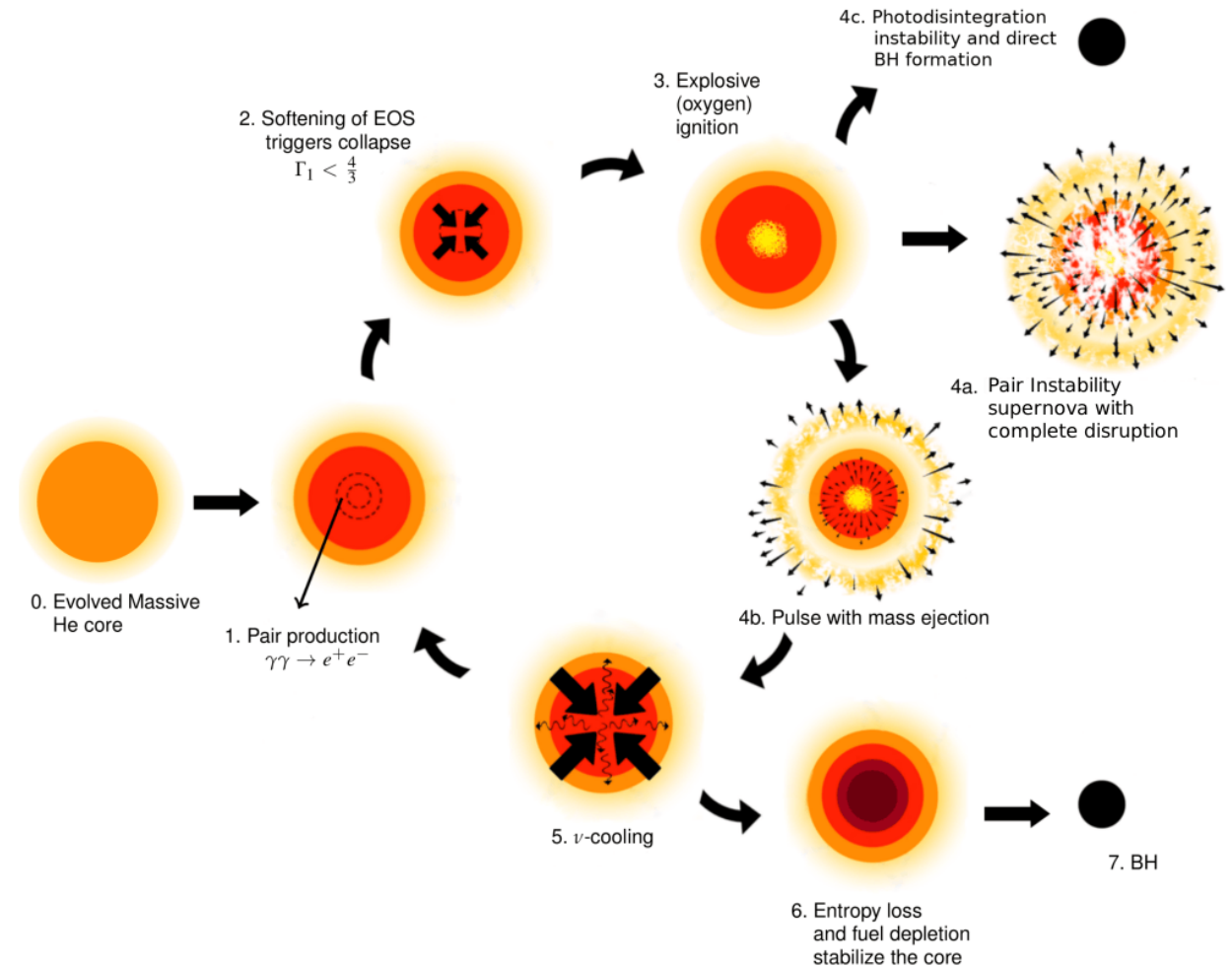
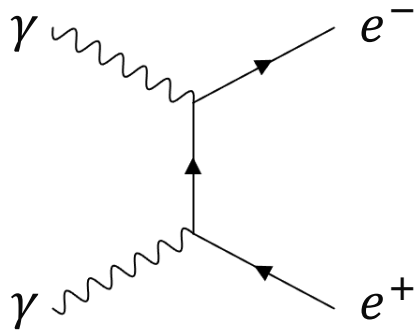
- At the end of stellar evolution, a massive star forms an electron–degenerate iron core
- Further fusion is no longer possible, the core contracts and increases the electron E_F
- As the electron E_F rises, the strongly endothermic electron capture occurs, reducing Y_e
- The decrease in Y_e lowers M_{Ch}



Kepler Beyond Planets: Finding Exploding Stars, NASA/JPL–Caltech, 2018

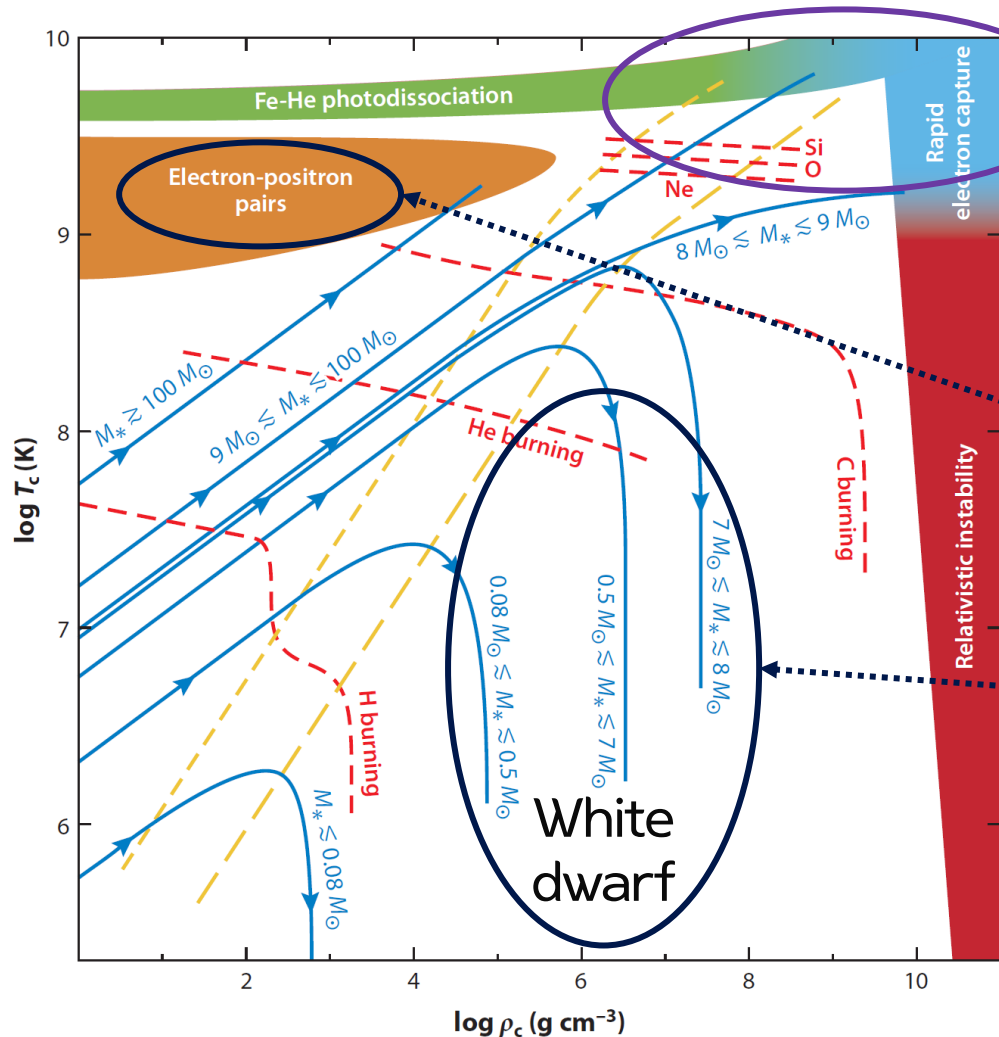
Pair-Instability

- From the Stefan–Boltzmann law, radiation energy density $u \propto T^4$, radiation pressure $P \propto \frac{1}{3}u$
 $\therefore \Gamma_1 \simeq \frac{4}{3} \dots$ dynamically unstable!
- Pair-production reduces Γ_1 , triggering the collapse



M. Renzo et al., Predictions for the hydrogen-free ejecta of pulsational pair-instability supernovae, *Astronomy & Astrophysics* 640 (2020)

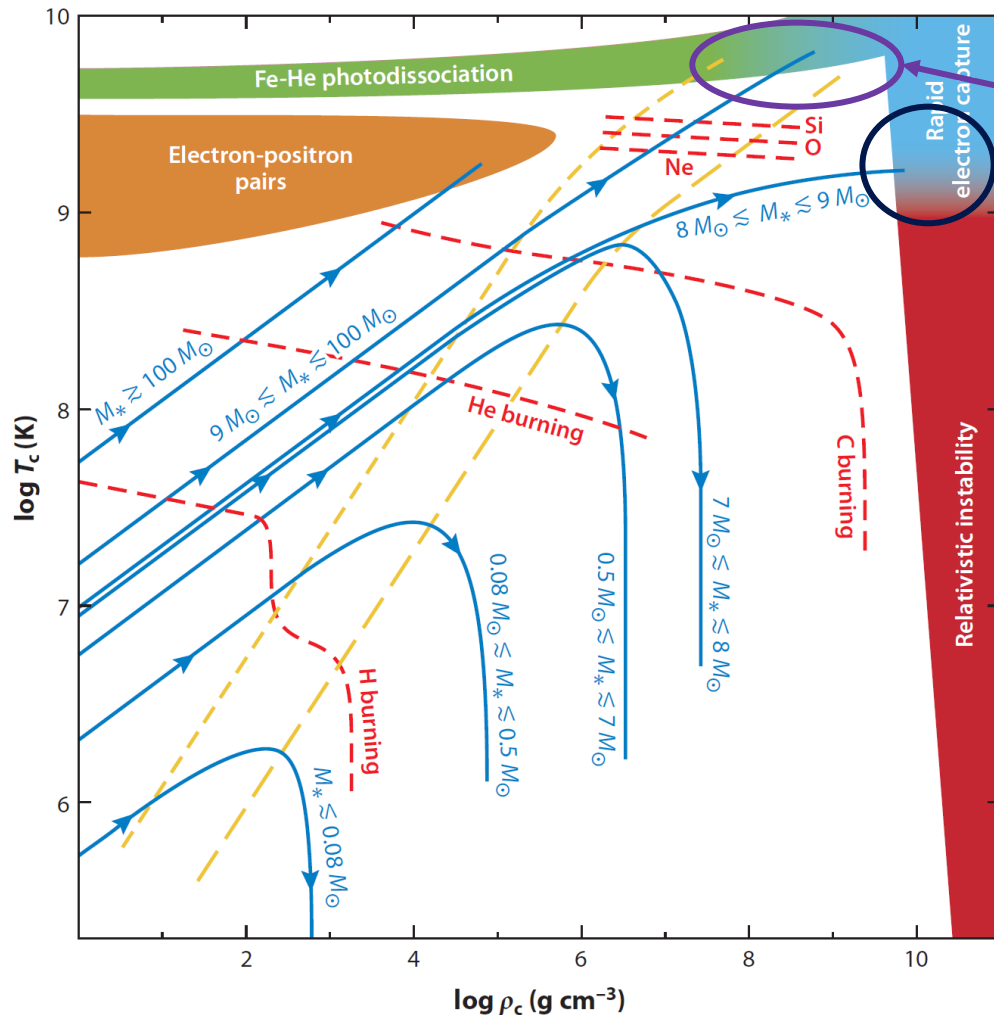
Stellar Evolution Tracks and Explosion Mechanisms



- **Core–Collapse supernovae (CCSNe)**
massive stars ($10 M_\odot \lesssim M_{\text{ZAMS}} \lesssim 70 M_\odot$)
- **Pair–Instability supernovae**
extremely massive stars ($M_{\text{ZAMS}} \gtrsim 70 M_\odot$)
- **Thermonuclear supernovae**
low mass stars ($M_{\text{ZAMS}} \lesssim 8 M_\odot$)

Explosion Mechanism of CCSN

Subtypes of Core–Collapse Supernovae



■ Iron–core supernovae

Collapse occurs in an **Fe core** where electrons are weakly degenerate

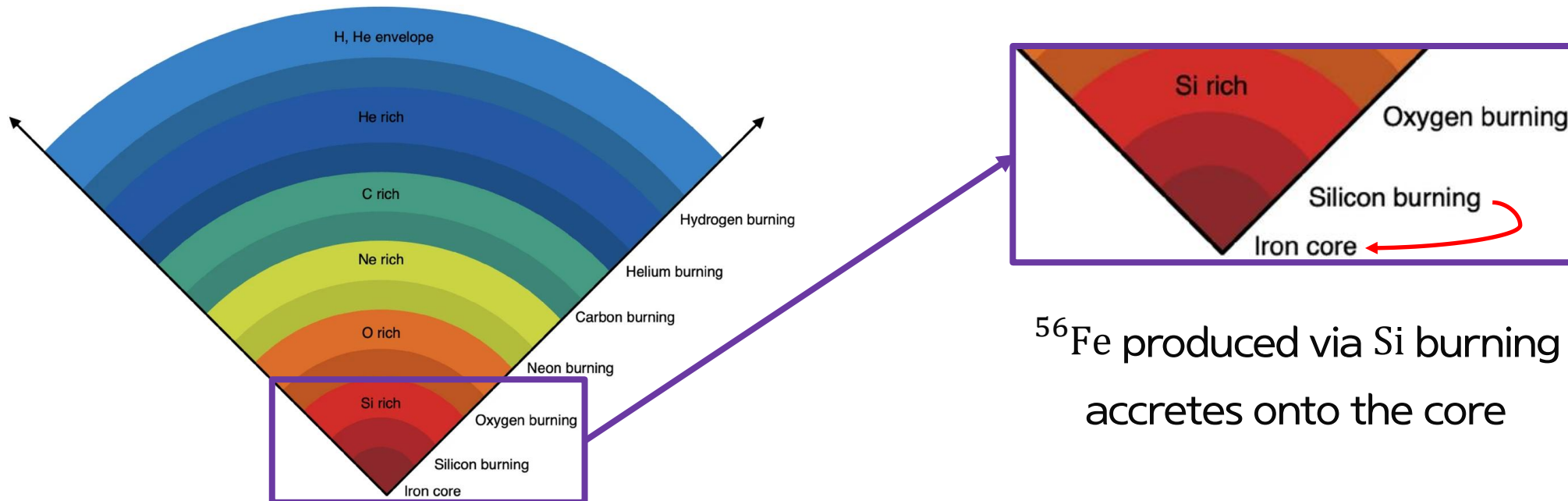
■ Electron–capture supernovae (ECSNe)

Collapse occurs in an **O-Ne-Mg core** where electrons are strongly degenerate

We will mainly focus on the iron–core supernovae

Triggering the Core–Collapse

- During stellar evolution, a star develops an onion-like structure
- Although **mass** produced by shell burning outside the core **accretes onto the core**, its **effect is relatively minor**



^{56}Fe produced via Si burning
accretes onto the core

Triggering the Core–Collapse

- In ECSNe, the core electrons are strongly degenerate, resulting in a high electron p_F
- In iron–core supernovae, once the Fe core is formed, nuclear fusion becomes impossible
The core then contracts, the electrons become degenerate, and the electron p_F rises
- The increased electron p_F triggers the endothermic electron capture process, reducing M_{ch} and ultimately triggering core collapse

Electron capture triggers the core collapse!

Electron Capture

- The Fermi momentum of degenerate electrons is

$$p_F = \left(\frac{3\pi^2}{m_u} \rho Y_e \right)^{1/3}$$

- When degenerate electrons behave relativistically, the electron Fermi energy is

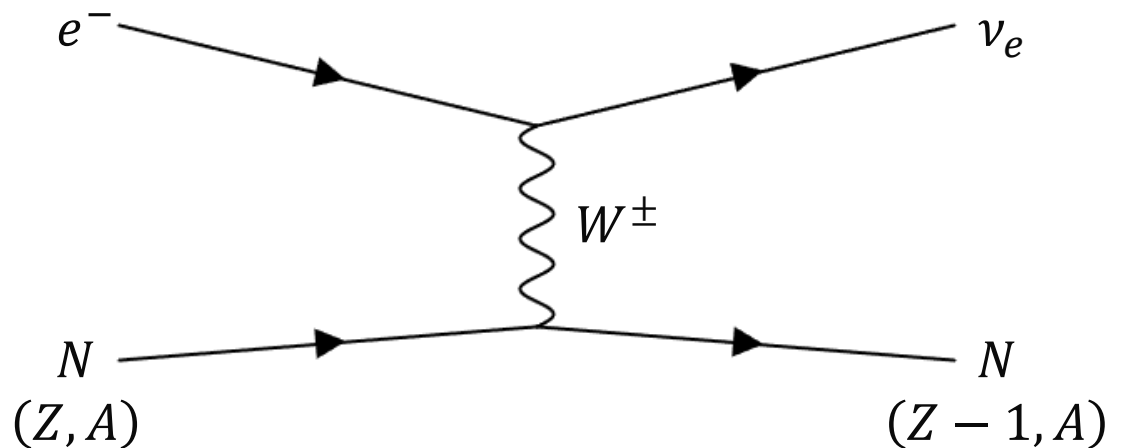
$$E_F \simeq p_F = 11.1 (\rho_{10} Y_e)^{1/3} \text{ MeV}$$

- Electron capture onto nuclei begins

$$e^- + (Z, A) \rightarrow (Z - 1, A) + \nu_e$$

- Electron capture reduces P_e ,
and the produced ν_e cool the core

→ Accelerates the core collapse!

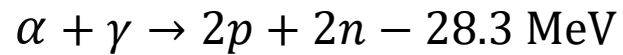
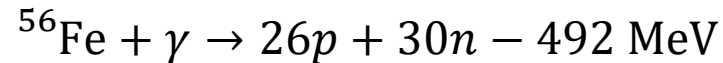
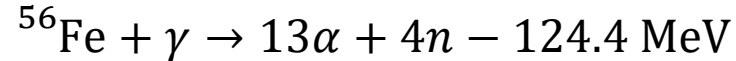


Photodisintegration

- Occurs around $T_c \sim \mathcal{O}(1)$ MeV
- When nuclear statistical equilibrium (NSE) holds, the Helmholtz free energy remains stationary under changes in particle numbers
$$0 = dF = -SdT - PdV + \sum_i \mu_i dN_i = \sum_i \mu_i dN_i$$
and it gives
$$\mu(Z, A) = Z\mu_p + (A - Z)\mu_n$$
- In regions where NSE holds, the nuclear reaction timescale is shorter than the hydrodynamical timescale, so local thermal equilibrium (LTE) can be assumed
- Therefore, in a local region where NES holds, the Helmholtz free energy $F = U - TS$ is minimized

Photodisintegration

- Representative photodisintegration processes are

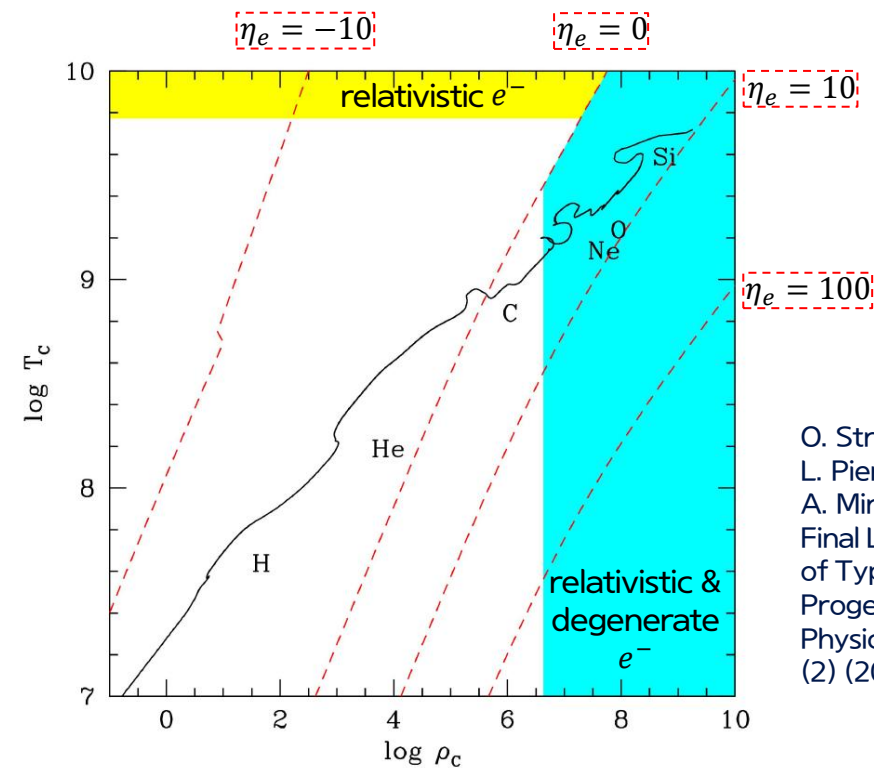


- These processes are strongly endothermic, they increase the internal energy
However, they also produce many light nuclei and therefore increase the entropy
Thus, in high-temperature environments, photodisintegration reduces F
- Photodisintegration is strongly endothermic

→ Accelerates the core collapse!

Progenitor of the Core–Collapse Supernovae

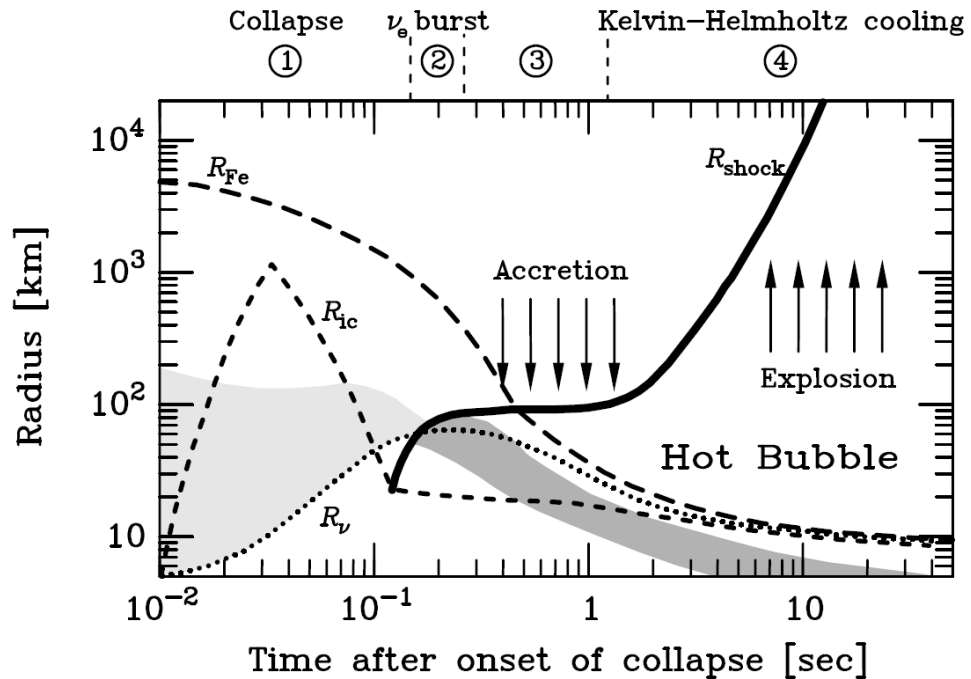
- At the final stage of massive stellar evolution, an Fe core forms and nuclear fusion ceases
- Typical properties of the iron core
 - No energy from nuclear fusion
 - P dominated by degenerated relativistic e^-
 - $\rho_c \sim \mathcal{O}(10^9) \text{ g} \cdot \text{cm}^{-3}$
 - $T_c \sim \mathcal{O}(10^9) \text{ K} (\simeq 10^{2\sim 3} \text{ keV})$
 - $Y_e \sim 0.45$
- Core contraction increases the electron p_F
- $M_c \sim M_{\text{Ch}} = 5.81 Y_e^2 M_\odot$ is reached, core becomes dynamically unstable



O. Straniero, I. Dominguez, L. Piersanti, M. Giannotti, A. Mirizzi, The Initial Mass–Final Luminosity Relation of Type II Supernova Progenitors: Hints of New Physics?, *Astrophys. J.* 881 (2) (2019) 158.

$M_{\text{ZAMS}} = 20 M_\odot, Y_e = 0.464$
 Degeneracy parameter $\eta \equiv \mu/k_B T$

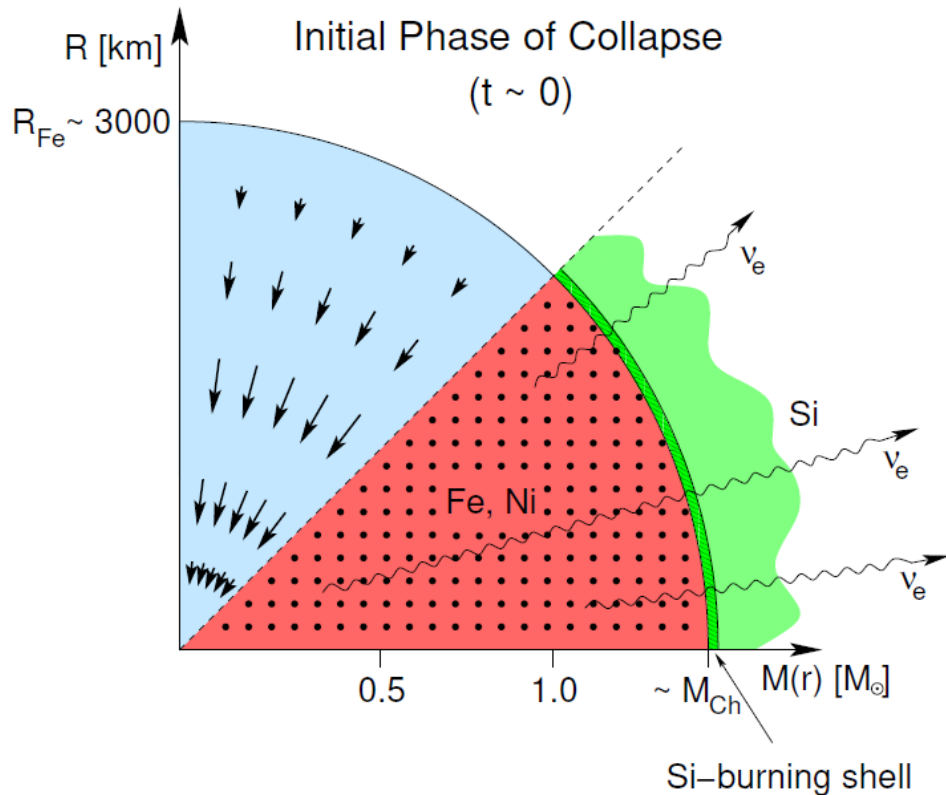
Schematic Overview of a Core-Collapse Supernovae



G. G. Raffelt, Stars as Laboratories for Fundamental Physics, 1996

- Conventional supernova explosion mechanism
→ Neutrino-driven mechanism
- Four phases of the core-collapse supernova based on the neutrino-driven mechanism
 - ①: Collapse
 - ②: Shock propagation and ν_e burst
 - ③: Shock stagnation and ν heating
 - ④: ν -driven explosion and cooling

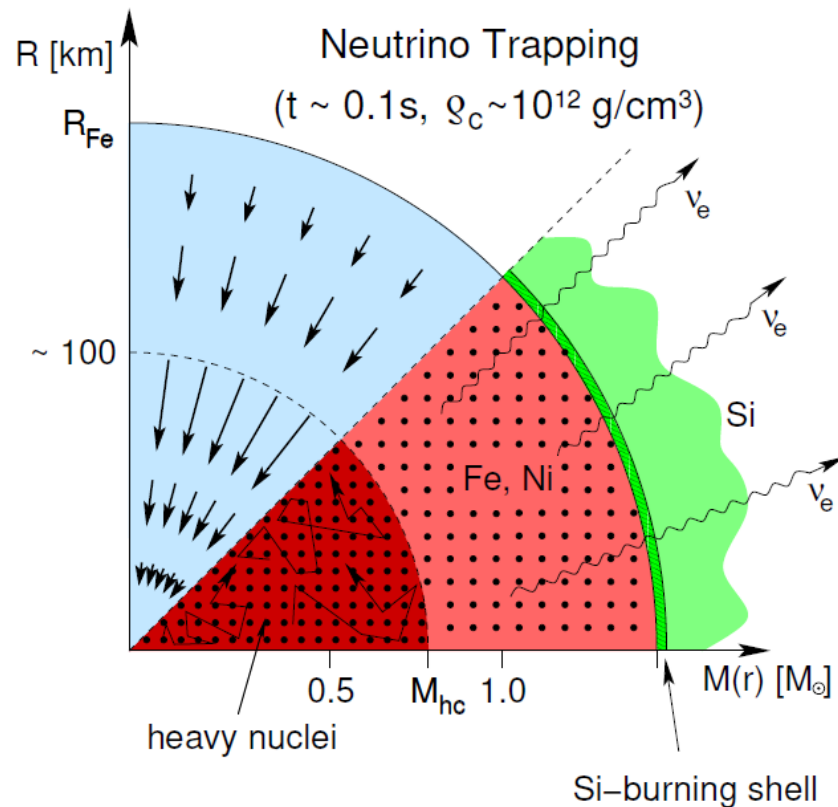
Initial Phase of the Collapse



H. T. Janka et al., Theory of core-collapse supernovae,
Physics Reports 442.1–6 (Apr. 2007)

- Collapse mainly triggered by e^- capture
- e^- capture and photodisintegration accelerate the core collapse
- β decay is less significant in the initial phase due to Pauli blocking

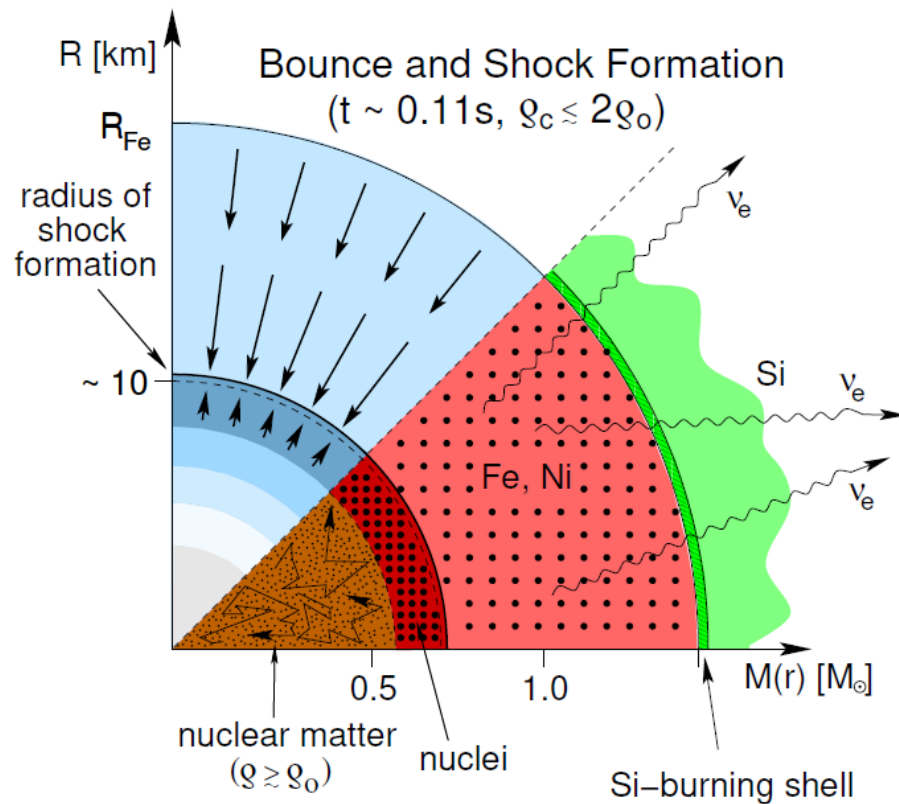
Neutrino Trapping



H. T. Janka et al., Theory of core-collapse supernovae,
Physics Reports 442.1–6 (Apr. 2007)

- $\rho_c \sim 10^{12}\text{ g/cm}^3$ is reached, **neutrinos become trapped in the core** by coherent scatterings on heavy nuclei
- Trapped neutrinos form a degenerate ν_e sea, Y_L approximately conserved
- Nuclear matter incompressibility increases
 - Inner core collapses at subsonic speeds
 - Outer core collapses at supersonic speeds

Core Bounce and Shock Formation



H. T. Janka et al., Theory of core-collapse supernovae, Physics Reports 442.1–6 (Apr. 2007)

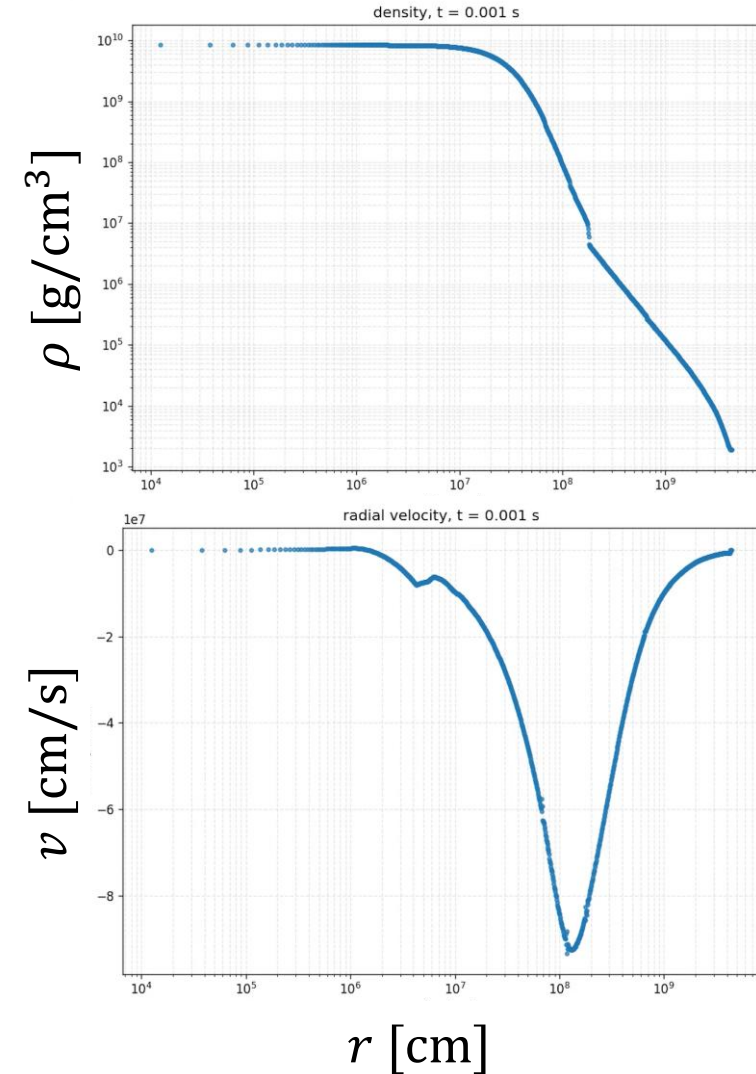
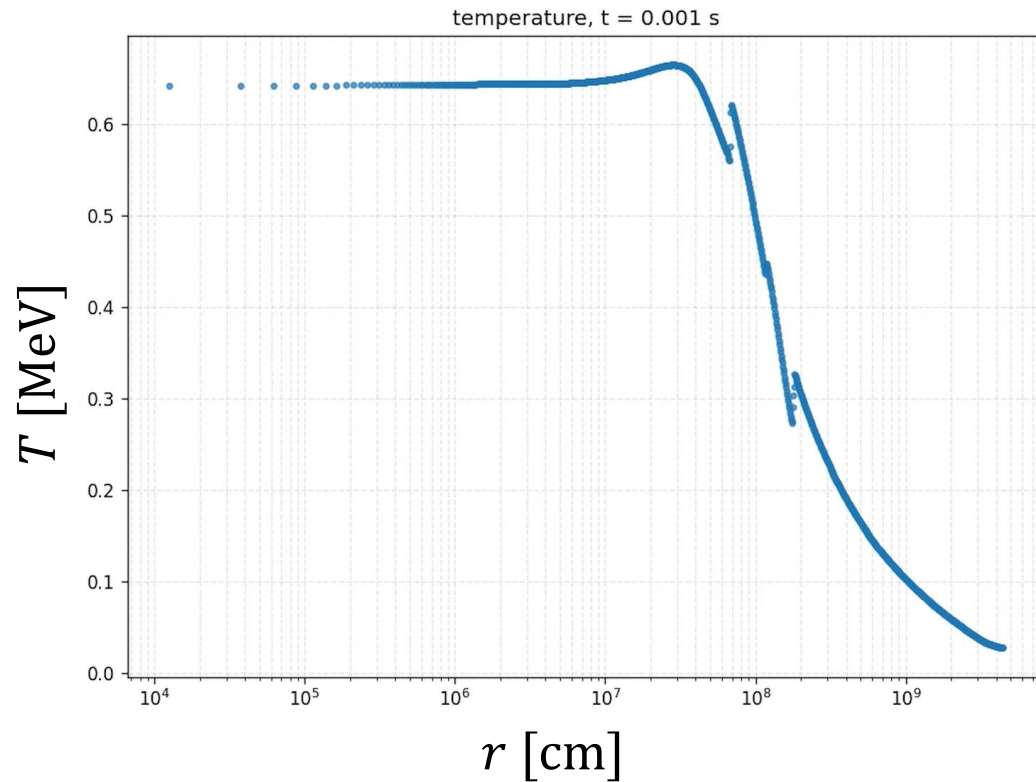
- When the core density reaches $\rho_c \sim 10^{14} \text{ g/cm}^3$, the formation of shock wave begins
- Nuclear matter EOS stiffens due to the degeneracy pressure of the nucleons and repulsive interactions of the strong force
 - Γ_1 increases (EOS stiffens)
 - Infalling matters are decelerated
 - Shock wave forms and propagates

Now the collapse phase turns into the explosion phase

Collapse Phase: Stellar Profiles

GR1D simulation results (successful SN explosion)

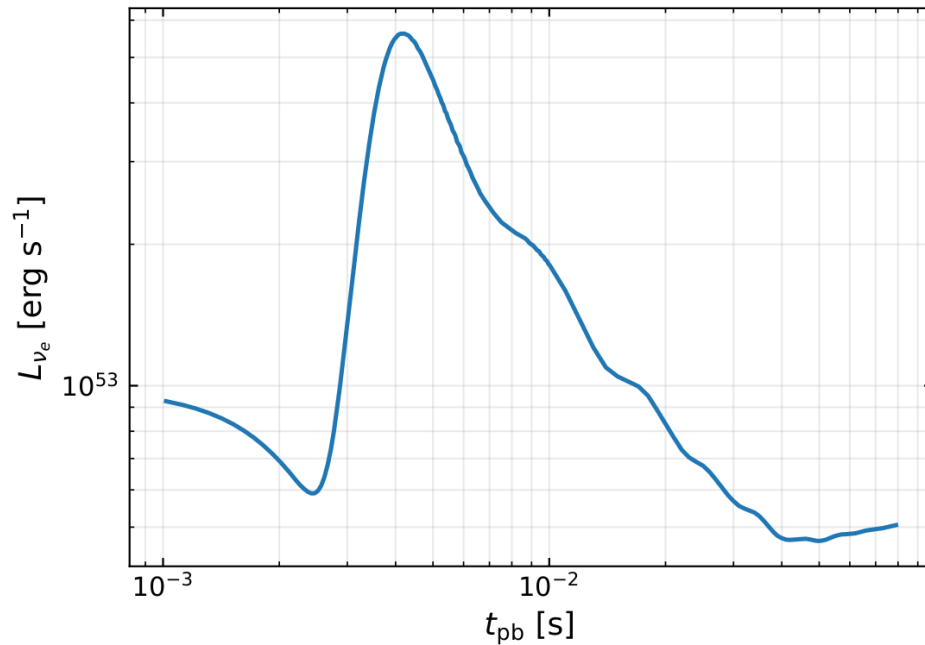
$M_{\text{ZAMS}} = 18 M_{\odot}$, SFHo EOS, $\alpha = 1.48$





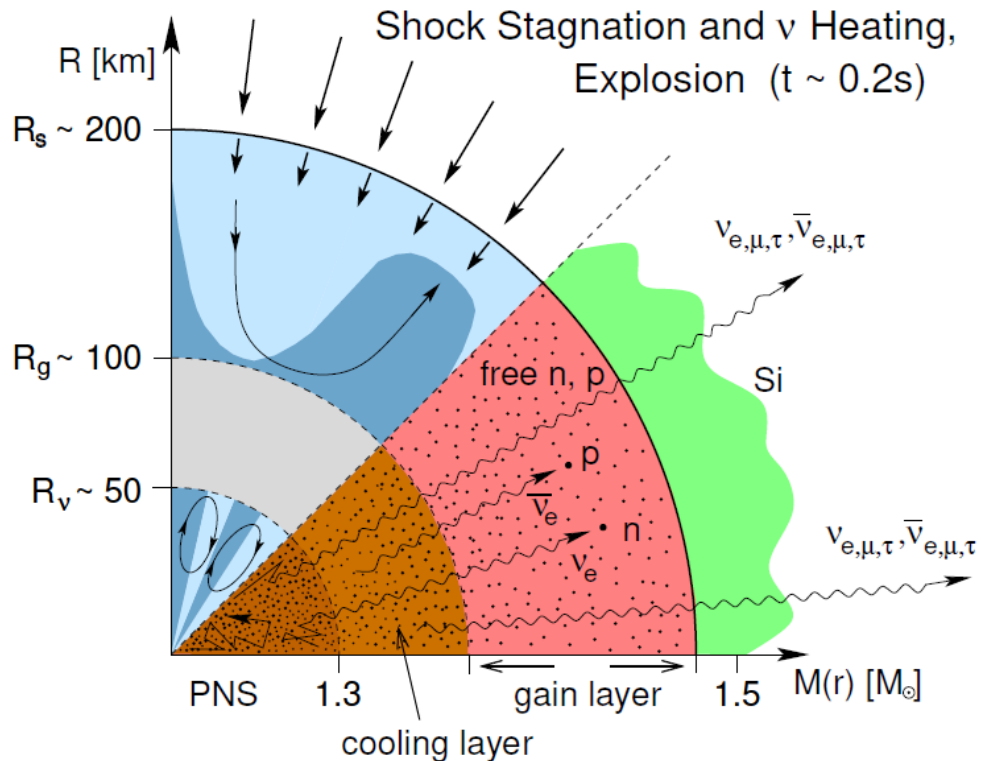
- 61/99

Shock Propagation and ν_e Burst



- The outward-propagating **shock wave** dissociates heavy nuclei
- Increase in free protons enhances the e^- capture rate
- e^- capture further depletes energy, leading to a **ν_e burst** driven by the produced ν_e
- $\mathcal{O}(10^{51})$ erg of energy is released via ν_e burst, $L_{\nu_e} \simeq 10^{53}$ erg/s

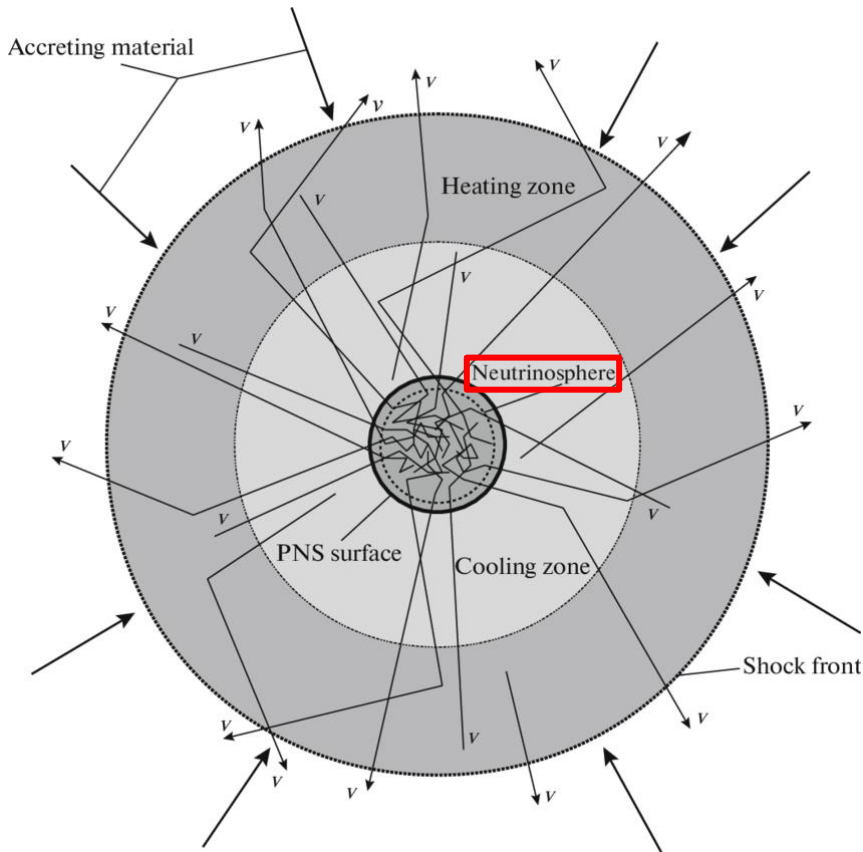
Shock Stagnation and Protoneutron Star Formation



H. T. Janka et al., Theory of core-collapse supernovae,
Physics Reports 442.1–6 (Apr. 2007)

- The shock wave having lost energy, eventually stalls at around $R_s \sim \mathcal{O}(100)$ km
- At the core, a protoneutron star (PNS) forms
- The cooling of the PNS occurs through ν pair production and ν diffusion processes
- Neutrinos emitted from the neutrinosphere can be reabsorbed in the gain layer, may revitalizing the shock wave

Neutrinosphere



Yuri Krivosheyev and Gena Kogan, Monte Carlo Simulations of Radiative and Neutrino Transport under Astrophysical Conditions, Astronomy Reports 62 (May 2018)

- The neutrinosphere is a surface where the neutrinos begin to freely streaming

- The radius of the neutrinosphere (R_ν) is defined at which the neutrino optical depth τ_ν equals 2/3

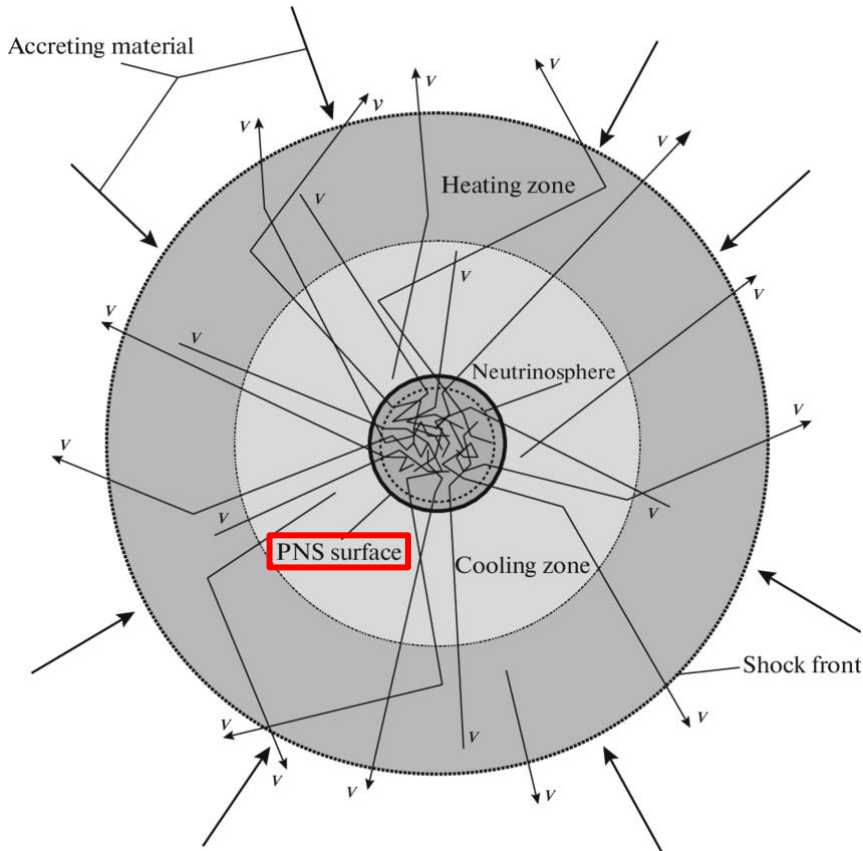
$$\tau_i(r) = \int_r^\infty \Gamma_i(r') dr' = \int_r^\infty \kappa_i(r) \rho_i(r) dr'$$

Γ_i : interaction rate of ν_i

κ_i : opacity of ν_i

τ_i satisfies the relation; $I/I_0 = e^{-\tau}$

Neutrinosphere

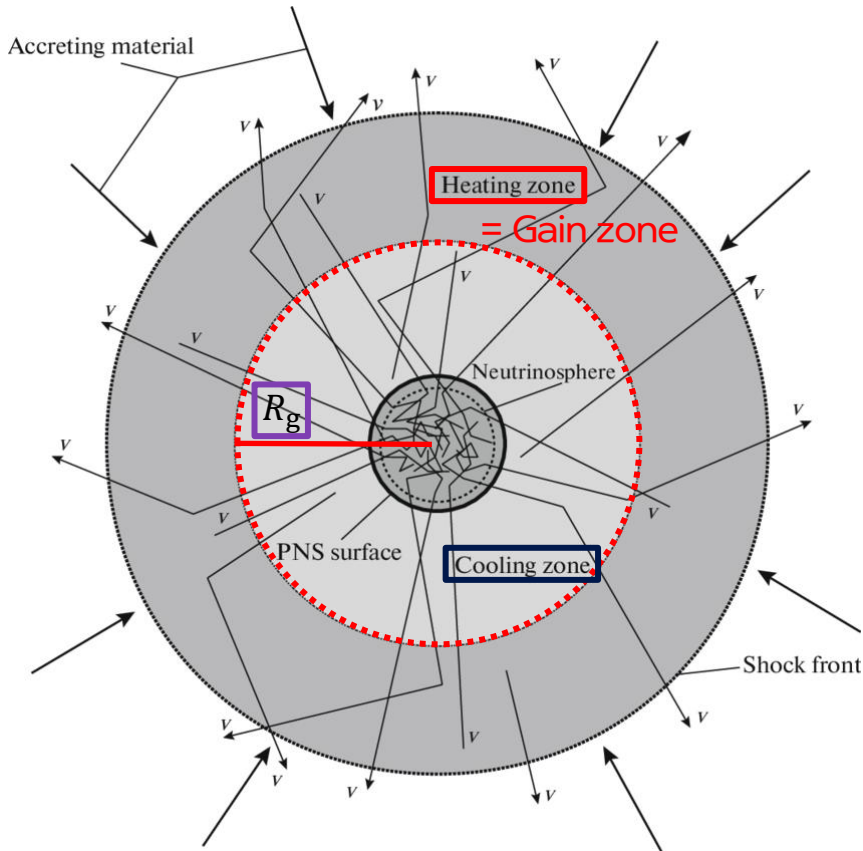


Yuri Krivosheyev and Gena Kogan, Monte Carlo Simulations of Radiative and Neutrino Transport under Astrophysical Conditions, Astronomy Reports 62 (May 2018)

- Inside the PNS, **all neutrino species are produced** via $\mathcal{L} \supset \frac{G_F}{\sqrt{2}} (\bar{\nu} \gamma^\mu (1 - \gamma^5) \nu) (\bar{f} \gamma_\mu (C_V - C_A \gamma^5) f)$
- All neutrinos can be produced through pair processes and neutral-current scattering, (anti) electron neutrinos can also be produced via charge-current scattering
- Therefore, the neutrinosphere radii of the different neutrino species are ordered as

$$R_{\nu_x, \bar{\nu}_x} \lesssim R_{\bar{\nu}_e} \lesssim R_{\nu_e}, \quad x \ni \mu, \tau$$

Gain and Cooling Zones



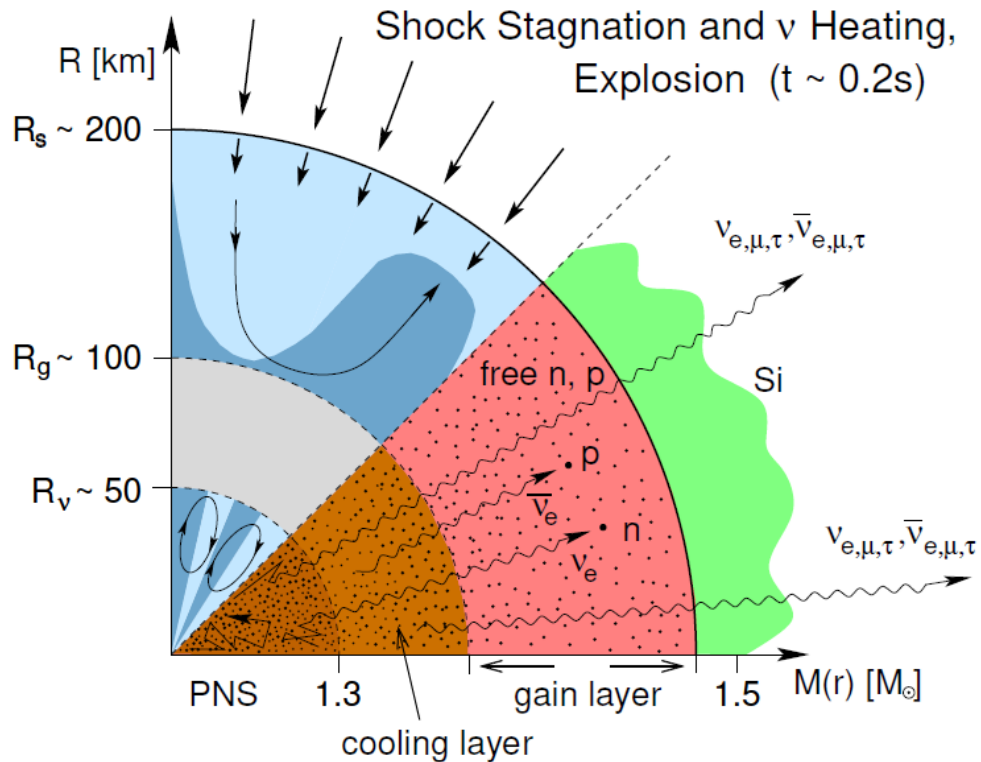
Yuri Krivosheyev and Gena Kogan, Monte Carlo Simulations of Radiative and Neutrino Transport under Astrophysical Conditions, Astronomy Reports 62 (May 2018)

- The gain zone, cooling zone, and the radius of the gain zone (R_g) are determined by the ν -induced heating/cooling rate per nucleon $Q_\nu^\pm$

$$Q_\nu^+ \simeq 110 \left(\frac{L_{\nu e,52} \langle \varepsilon_{\nu e,15}^2 \rangle}{r_7^2 \langle \mu_{\nu e} \rangle} Y_n + \frac{L_{\bar{\nu} e,52} \langle \varepsilon_{\bar{\nu} e,15}^2 \rangle}{r_7^2 \langle \mu_{\bar{\nu} e} \rangle} Y_p \right) \left(\frac{\text{MeV}}{\text{s} \cdot \text{N}} \right)$$

$$Q_\nu^- \simeq 145 \left(\frac{k_B T}{2 \text{ MeV}} \right)^6 \left(\frac{\text{MeV}}{\text{s} \cdot \text{N}} \right)$$

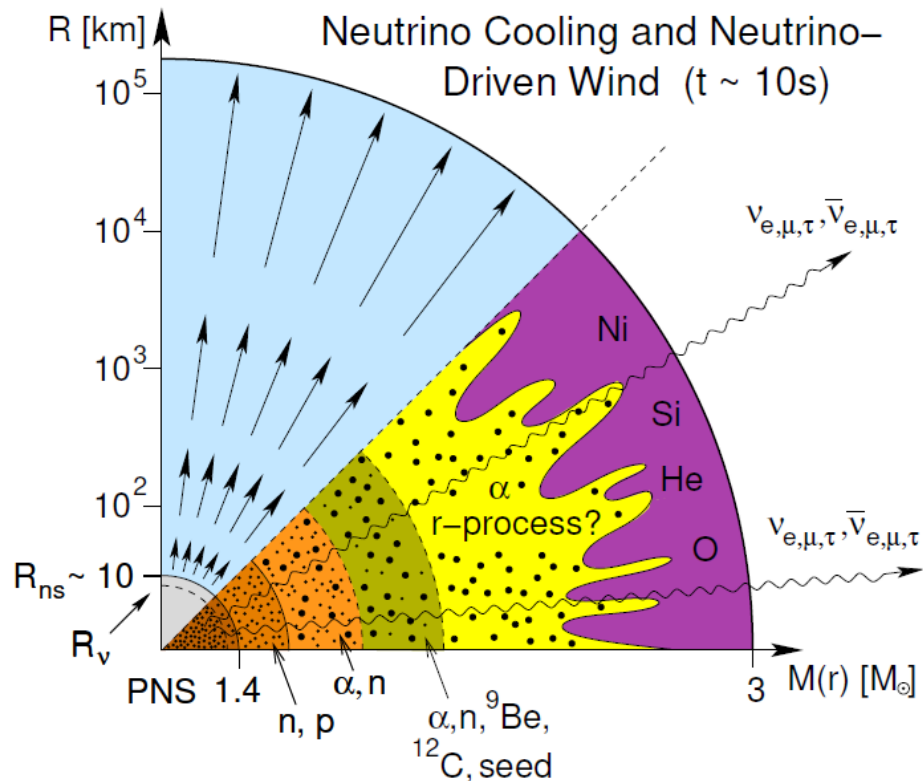
Neutrino Heating and Shock Revitalization



H. T. Janka et al., Theory of core-collapse supernovae, Physics Reports 442.1–6 (Apr. 2007)

- Three possibilities at this stage, depending on the shock revitalization
 1. **Successful shock revitalization**
Shock propagates outward through the star and producing a supernova explosion
The central PNS successfully forms a neutron star
 2. **Shock revitalized but some of the material falls back**
Supernova explosion still occurs, but fallback material accretes onto the PNS, forming a black hole
 3. **Fails to shock revival**
The shock fails to eject the stellar envelope, and the supernova explosion fails
The shock falls back, forming a black hole

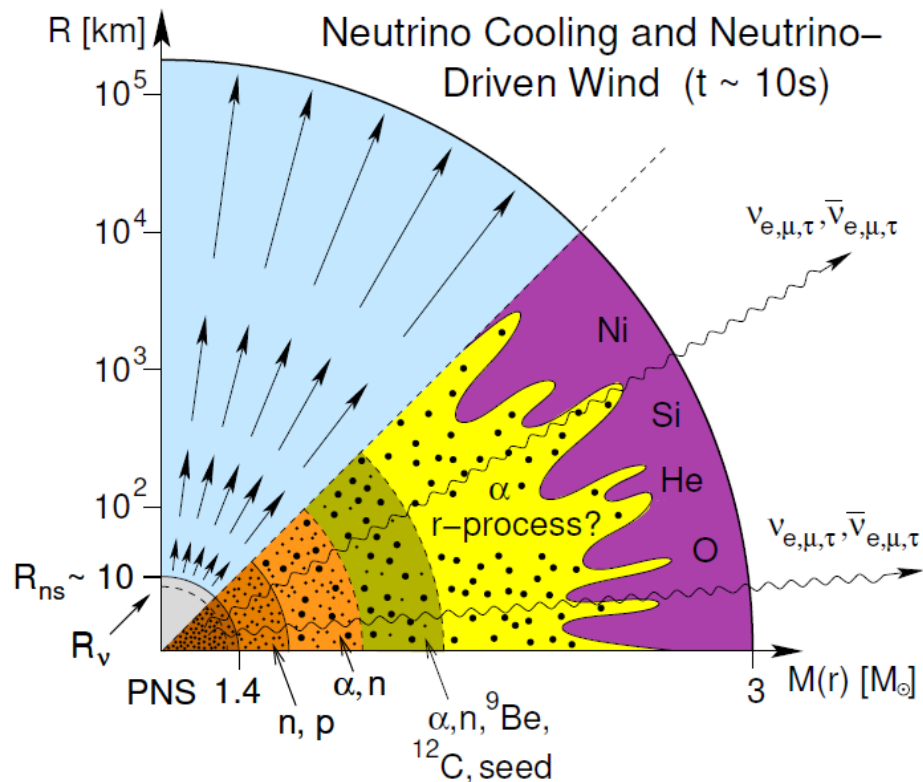
Neutrino Cooling and Neutrino-Driven Wind



H. T. Janka et al., Theory of core-collapse supernovae, Physics Reports 442.1–6 (Apr. 2007)

- Around $t_{pb} \sim \mathcal{O}(0.1)$ s, the shock wave, revitalized by sufficient neutrino heating, ejects the outer layers of the star
- 99% of explosion energy emitted by neutrinos, only 1% is emitted in an optically observable form
- The typical explosion energy of a CCSN is $\mathcal{O}(10^{53})$ erg, while the energy observed in electromagnetic radiation is $\mathcal{O}(10^{51})$ erg

Neutrino Cooling and Neutrino-Driven Wind



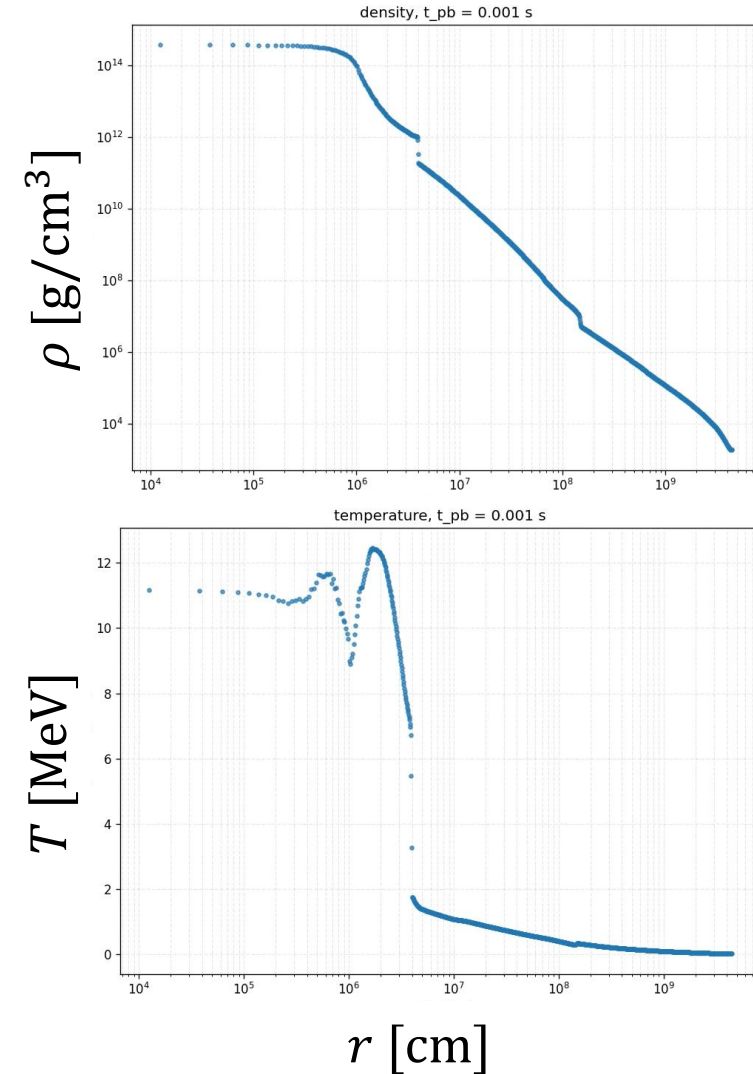
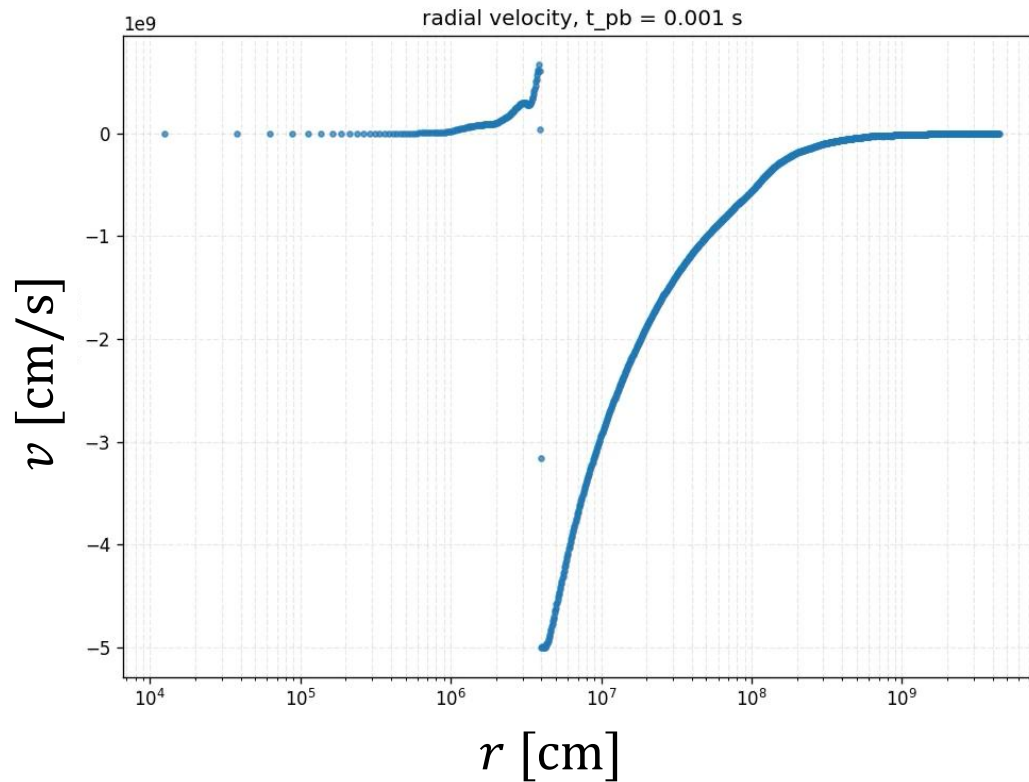
H. T. Janka et al., Theory of core-collapse supernovae,
Physics Reports 442.1–6 (Apr. 2007)

- Most of the energy released by PNS cooling is emitted over $t \sim 5$ to 10 s
- As the PNS gradually becomes transparent to neutrinos, the neutrino luminosity decreases significantly after $t \sim 10$ s

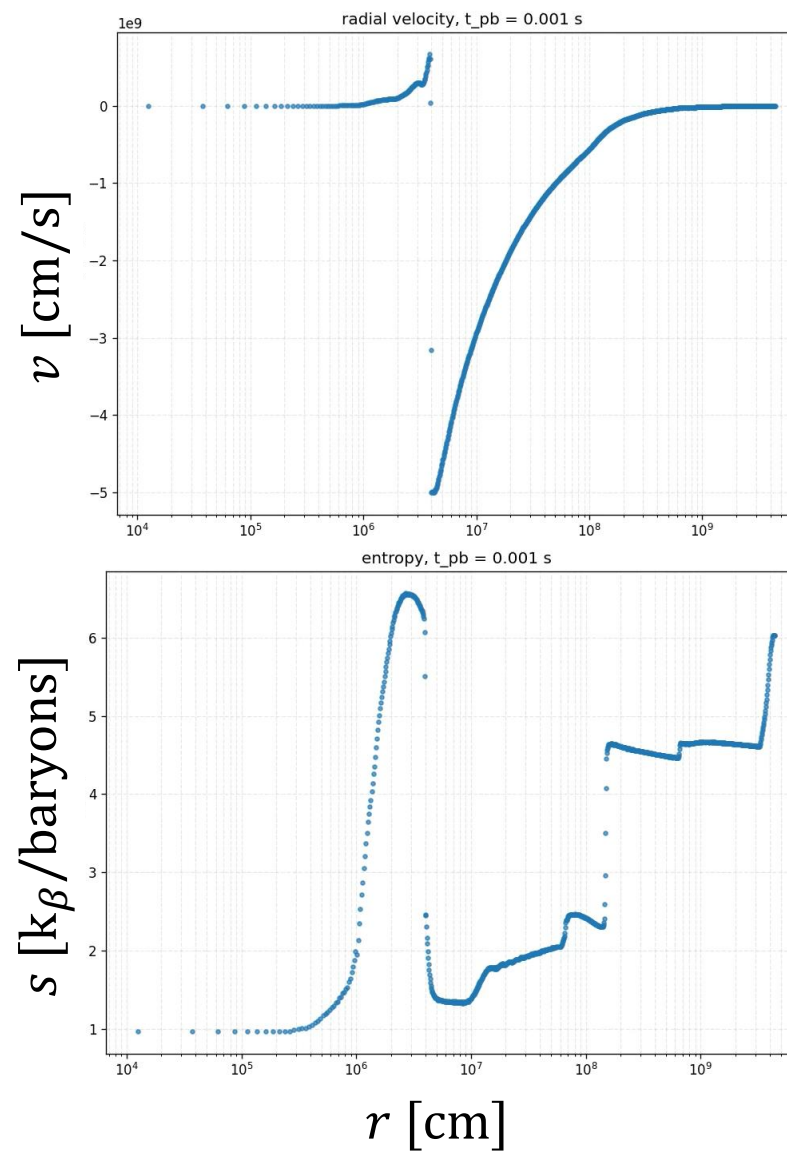
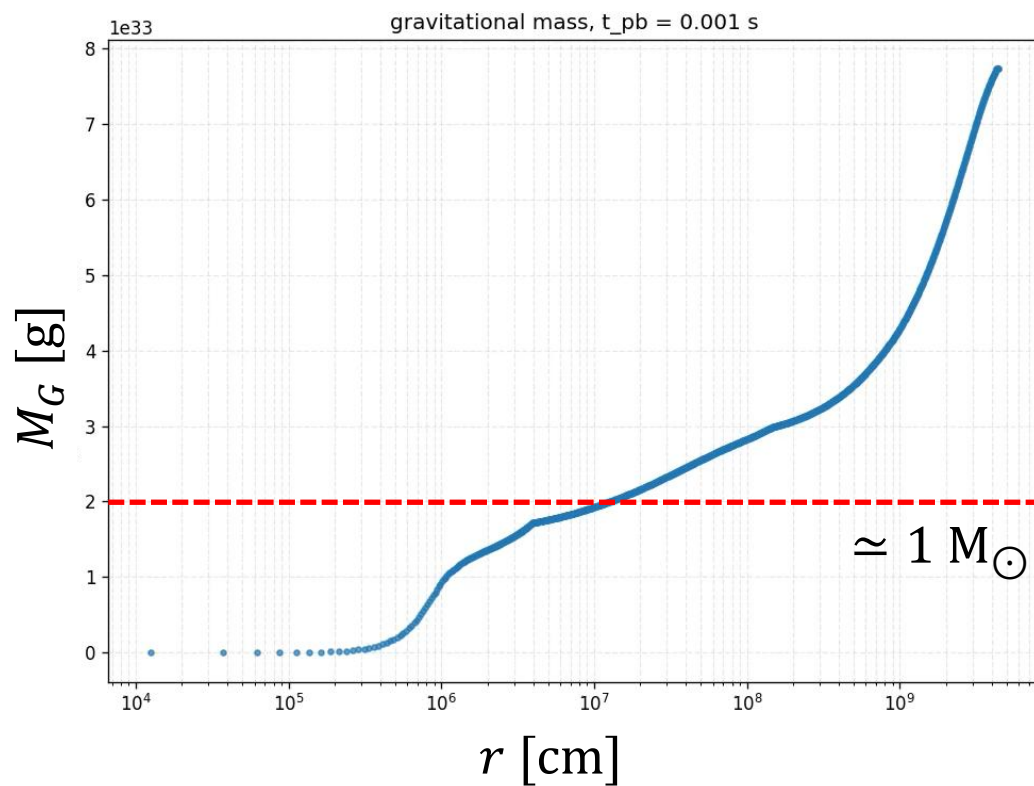
Explosion Phase: Stellar Profiles

GR1D simulation results (successful SN explosion)

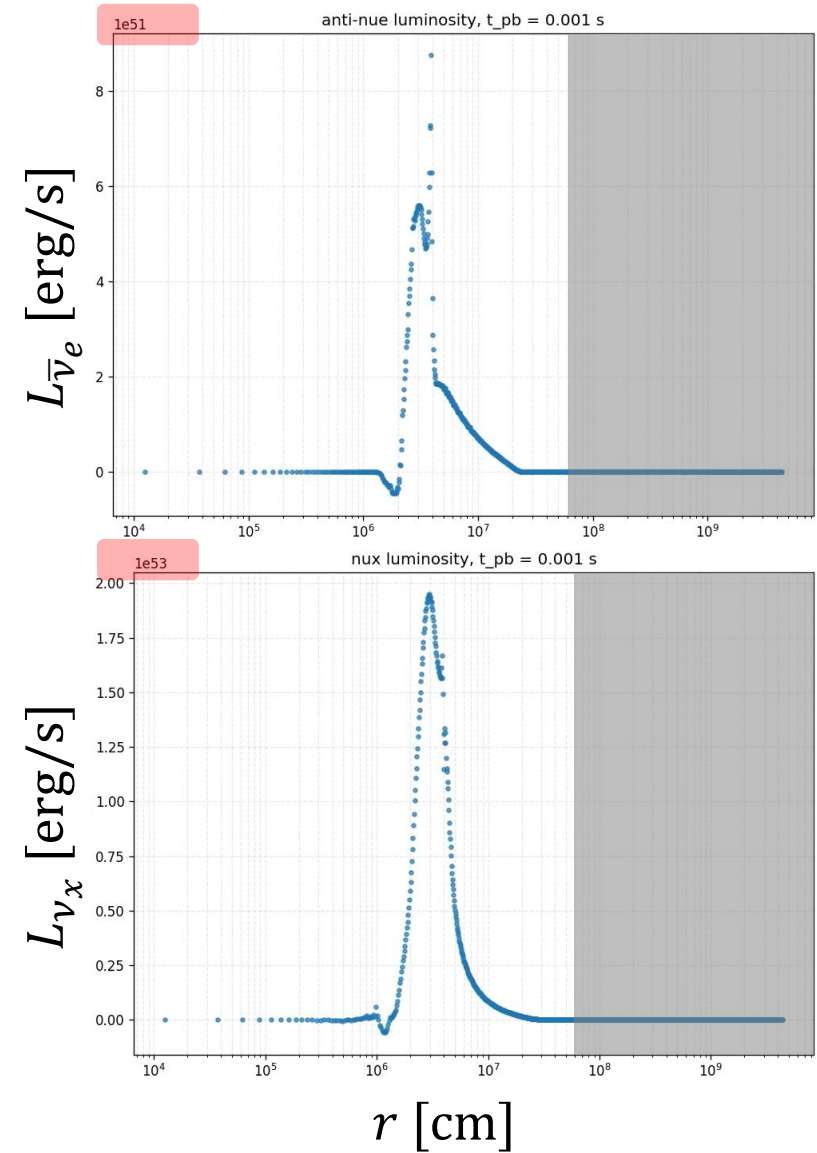
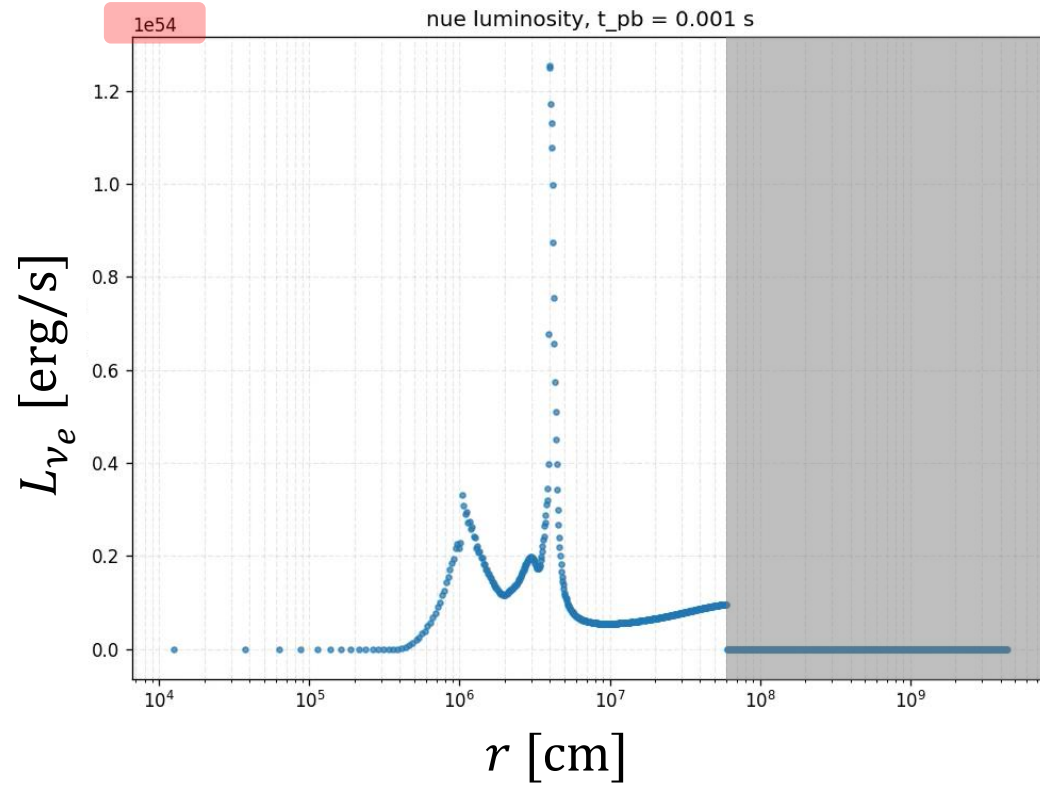
$M_{\text{ZAMS}} = 18 M_{\odot}$, SFHo EOS, $\alpha = 1.48$



Explosion Phase: Stellar Profiles



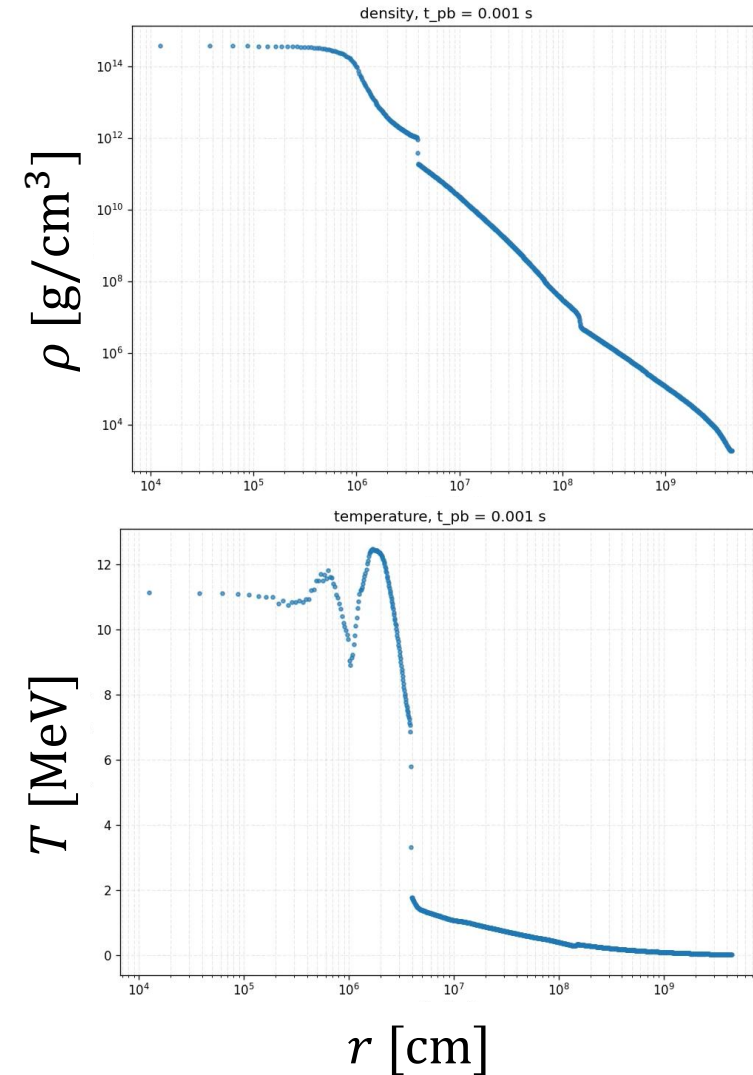
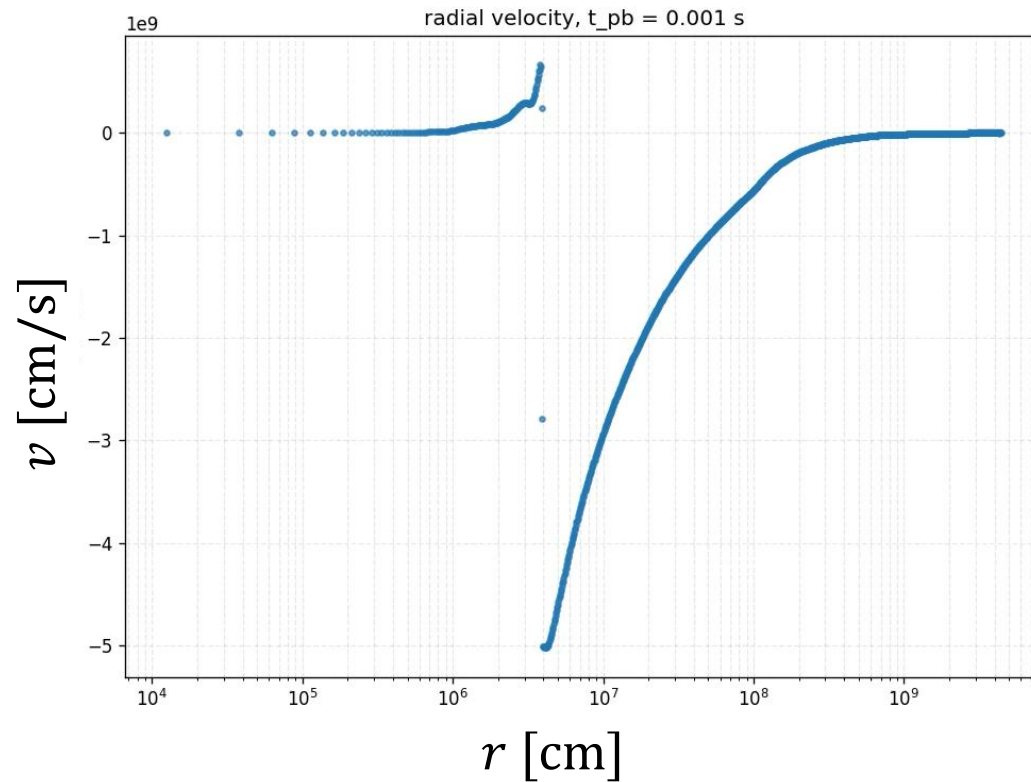
Explosion Phase: Stellar Profiles



Failed Supernova Explosion: Stellar Profiles

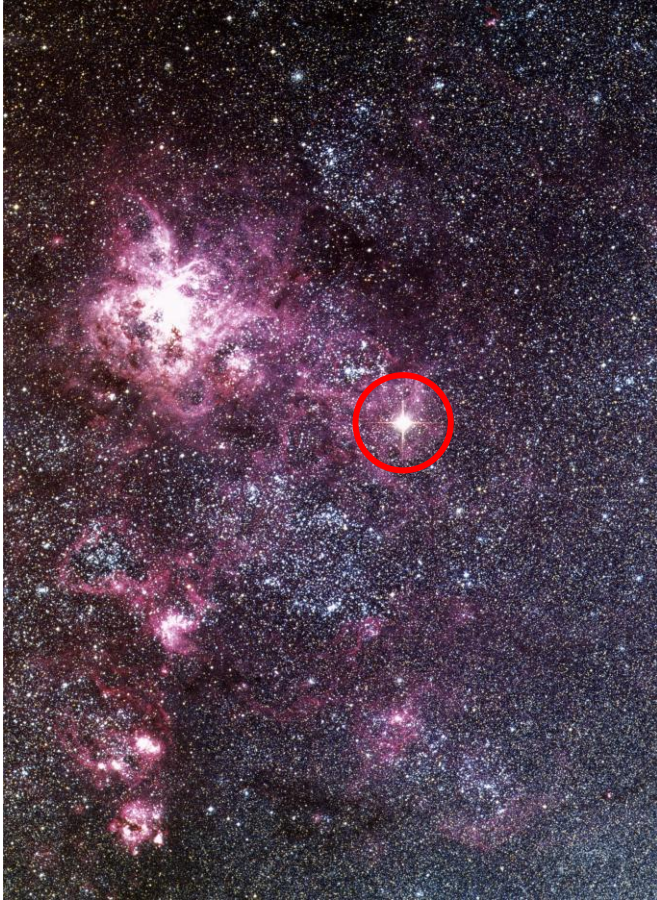
GR1D simulation results (failed SN explosion)

$M_{\text{ZAMS}} = 18 M_{\odot}$, SFHo EOS, $\alpha = 0.25$



Observations of CCSNe

SN 1987A

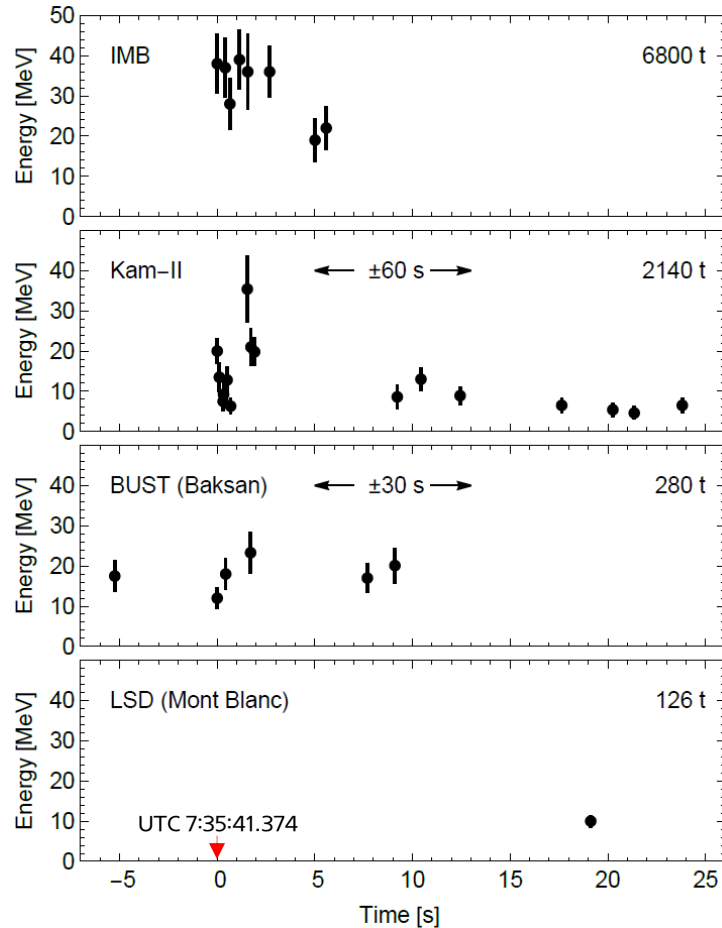


SN1987A in the Large Magellanic Cloud, ESO, 2007

- SN 1987A
 - Type II-pec
 - Date: UTC 7:35 on February 23, 1987
 - Host: Large Magellanic Cloud
 - Distance: 51.4 ± 1.2 kpc
- Progenitor (Sanduleak $-69^\circ 202$)
 - Mass: $\sim 20 M_\odot$
 - Radius: $\sim 41.15 R_\odot$
 - Spectral type: B3 Ia (blue supergiant)
 - Color index: +0.085

The only supernova from which
neutrinos have been directly detected!

Neutrinos from SN 1987A



D. F. G. Fiorillo et al., Supernova simulations confront SN 1987A neutrinos, Physical Review D (Oct 2023)

- $\bar{\nu}_e$ from SN 1987A were detected by Irvine–Michigan–Brookhaven (IMB), Kamiokande–II (Kam–II), and Baksan Underground Scintillator Telescope (BUST)
- Observations were based on detecting Cherenkov radiation produced by positrons generated by $\bar{\nu}_e$

Neutrinos from SN 1987A

IBM

TABLE II. SN 1987A neutrinos in IMB [187]. Time relative to the first event UT 7:35:41.374 $\pm$ 50 ms. The energy refers to the detected $e^\pm$, its angle is relative to the opposite LMC direction (i.e. the scattering angle).

Event No.	Time [sec]	N_{hit}	Energy [MeV]	Angle [Degree]
1	0.000	47	38 ± 7	80 ± 10
2	0.412	61	37 ± 7	44 ± 15
3	0.650	49	28 ± 6	56 ± 20
4	1.141	60	39 ± 7	65 ± 20
5	1.562	52	36 ± 9	33 ± 15
6	2.684	61	36 ± 6	52 ± 10
7	5.010	44	19 ± 5	42 ± 20
8	5.582	45	22 ± 5	104 ± 20

^aNumber of muons following this event according to the trigger numbers in Table III of Bionta et al. [186], with a total of 15 muons.

D. F. G. Fiorillo et al., Supernova simulations confront SN 1987A neutrinos, Physical Review D (Oct 2023)

Kamiokande II

TABLE III. SN 1987A neutrinos in Kamiokande II [189]. Time relative to the first event at UT 7:35:35 $\pm$ 1 min.

Event No.	Time [sec]	N_{hit}	Positron event energy [MeV]	Angle [Degree]
1	0.000	58	20.0 ± 2.9	18 ± 18
2	0.107	36	13.5 ± 3.2	40 ± 27
3	0.303	25	7.5 ± 2.0	108 ± 32
4	0.324	26	9.2 ± 2.7	70 ± 30
5	0.507	39	12.8 ± 2.9	135 ± 23
6 ^a	0.686	16	6.3 ± 1.7	68 ± 77
7	1.541	83	35.4 ± 8.0	32 ± 16
8	1.728	54	21.0 ± 4.2	30 ± 18
9	1.915	51	19.8 ± 3.2	38 ± 22
10	9.219	21	8.6 ± 2.7	122 ± 30
11	10.433	37	13.0 ± 2.6	49 ± 26
12	12.439	24	8.9 ± 1.9	91 ± 39
13 ^{a,b}	17.641		6.5 ± 1.6	103 ± 50
14 ^{a,b}	20.257		5.4 ± 1.4	110 ± 50
15 ^{a,b}	21.355		4.6 ± 1.3	120 ± 50
16 ^{a,b}	23.814		6.5 ± 1.6	112 ± 50

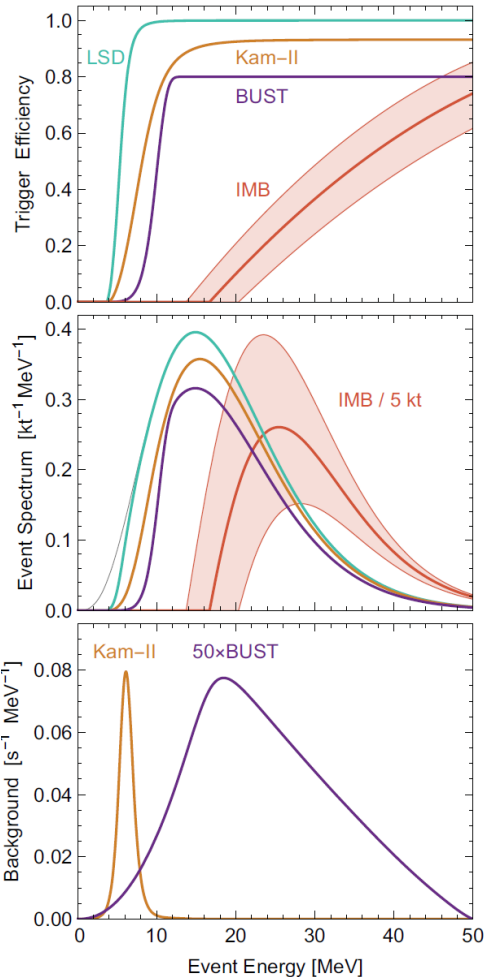
^aUsually attributed to background
^bTimes and energies from Ref. [33], angles from Ref. [190]
D. F. G. Fiorillo et al., Supernova simulations confront SN 1987A neutrinos, Physical Review D (Oct 2023)

BUST

TABLE IV. SN 1987A neutrinos in BUST (Baksan) [191, 192]. Time relative to event No. 1 at UT 7:36:11.818 $\pm$ 2/ $-$ 54 s.

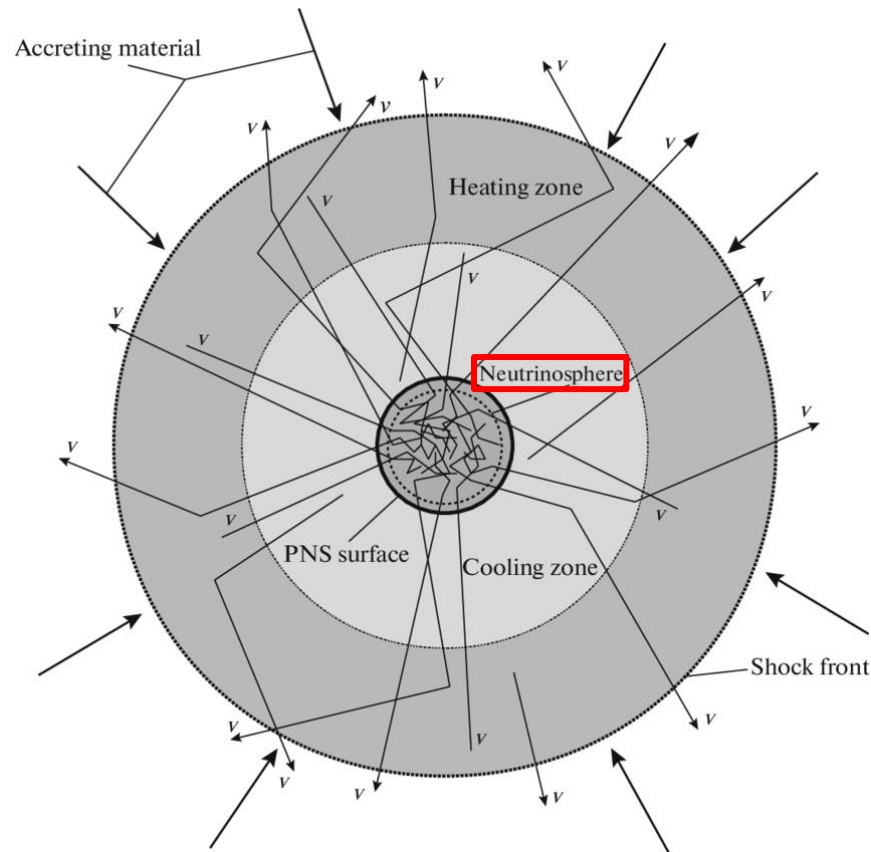
Event No.	Time [sec]	Positron event energy [MeV]
0 ^a	-5.247	17.5 ± 3.5
1	0.000	12.0 ± 2.4
2	0.435	18.0 ± 3.6
3	1.710	23.3 ± 4.7
4	7.687	17.0 ± 3.4
5	9.099	20.1 ± 4.0

^aAttributed to background by Alexeev et al. [191] based on the time structure of the signals in the other detectors and was dropped without mention in subsequent publications.
D. F. G. Fiorillo et al., Supernova simulations confront SN 1987A neutrinos, Physical Review D (Oct 2023)



D. F. G. Fiorillo et al., Supernova simulations confront SN 1987A neutrinos, Physical Review D (Oct 2023)

Neutrino Cooling of SN 1987A



Yuri Krivosheyev and Gena Kogan, Monte Carlo Simulations of Radiative and Neutrino Transport under Astrophysical Conditions, Astronomy Reports 62 (May 2018)

- 99% of the PNS binding energy is released through neutrinos of all flavors
→ Neutrino cooling

$$E_B = \frac{3}{5} \frac{GM_{\text{PNS}}^2}{R_{\text{PNS}}} \simeq 3 \times 10^{53} \left(\frac{M_{\text{PNS}}}{1.4 M_{\odot}} \right)^2 \left(\frac{10 \text{ km}}{R_{\text{PNS}}} \right) \text{ erg}$$

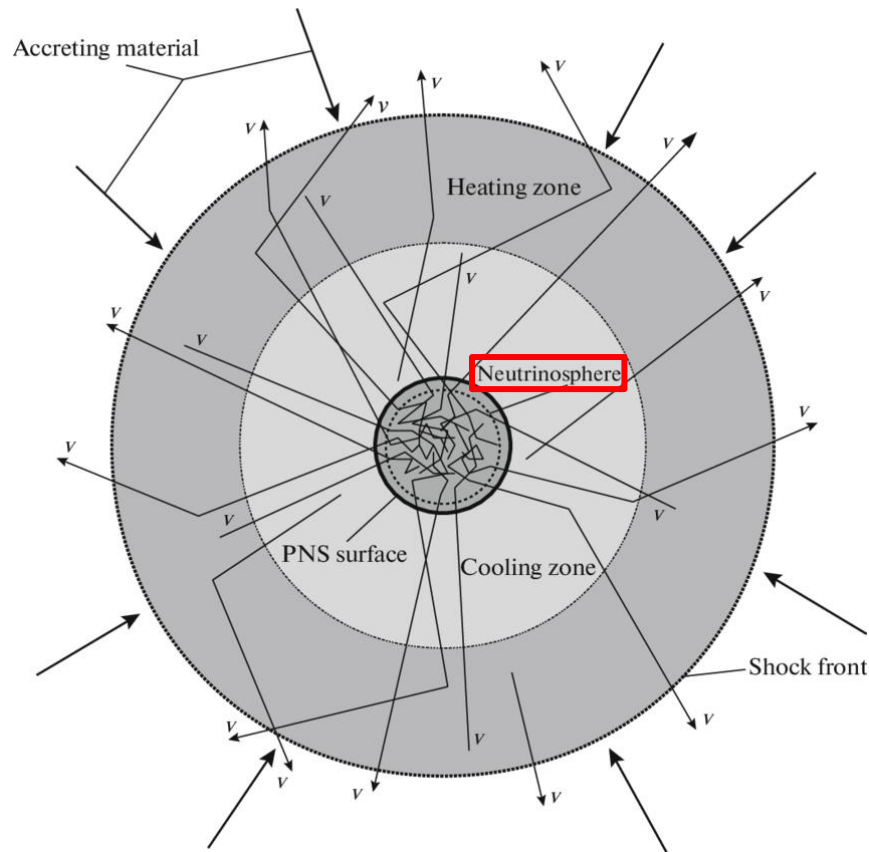
- Duration of neutrino cooling depends on the diffusion time t_{diff}

$$t_{\text{diff}} = \frac{\lambda_{\nu}}{c} \frac{R_{\text{PNS}}^2}{\lambda_{\nu}^2} \sim 10 \text{ s}$$

time interval between successive scattering

number of steps (random walk ... $R \simeq \sqrt{N}\lambda$)

Neutrino Cooling of SN 1987A



Yuri Krivosheyev and Gena Kogan, Monte Carlo Simulations of Radiative and Neutrino Transport under Astrophysical Conditions, Astronomy Reports 62 (May 2018)

$$E_B = \frac{3}{5} \frac{GM_{\text{PNS}}^2}{R_{\text{PNS}}} \simeq 3 \times 10^{53} \left(\frac{M_{\text{PNS}}}{1.4 M_{\odot}} \right)^2 \left(\frac{10 \text{ km}}{R_{\text{PNS}}} \right) \text{ erg}$$

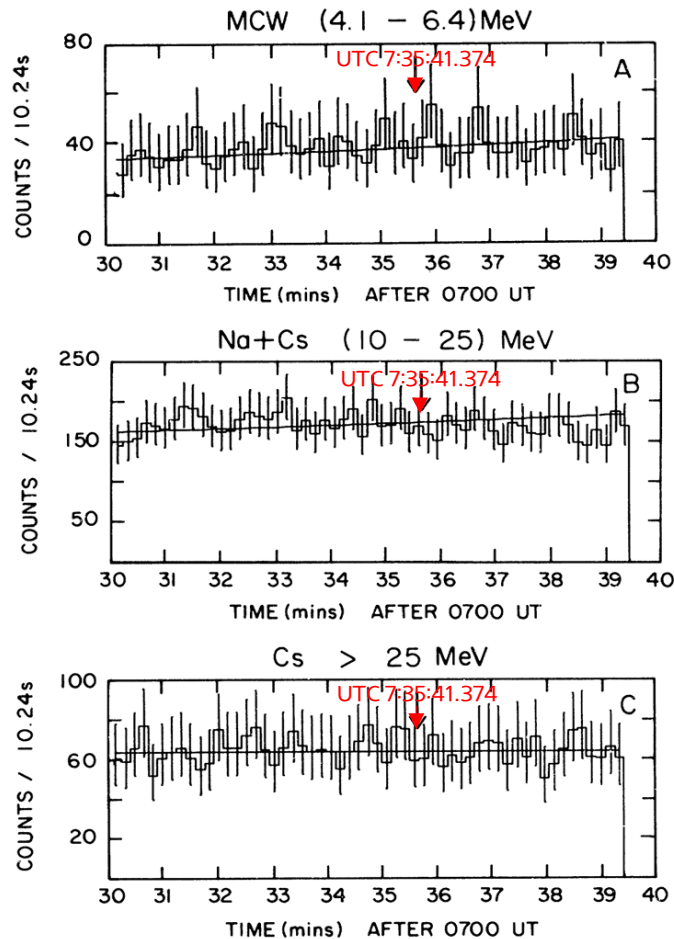
$$t_{\text{diff}} = \frac{\lambda_{\nu}}{c} \frac{R_{\text{PNS}}^2}{\lambda_{\nu}^2} \sim 10 \text{ s}$$

$$\rightarrow \therefore L_{\nu} \sim 10^{52} \text{ erg} \cdot \text{s}^{-1}$$

■ From the virial theorem,

$$2 \langle E_{\nu, \text{kin}} \rangle \simeq \frac{GM_{\text{PNS}} m_N}{R_{\text{PNS}}} \rightarrow \langle E_{\nu, \text{kin}} \rangle \simeq 25 \text{ MeV}$$

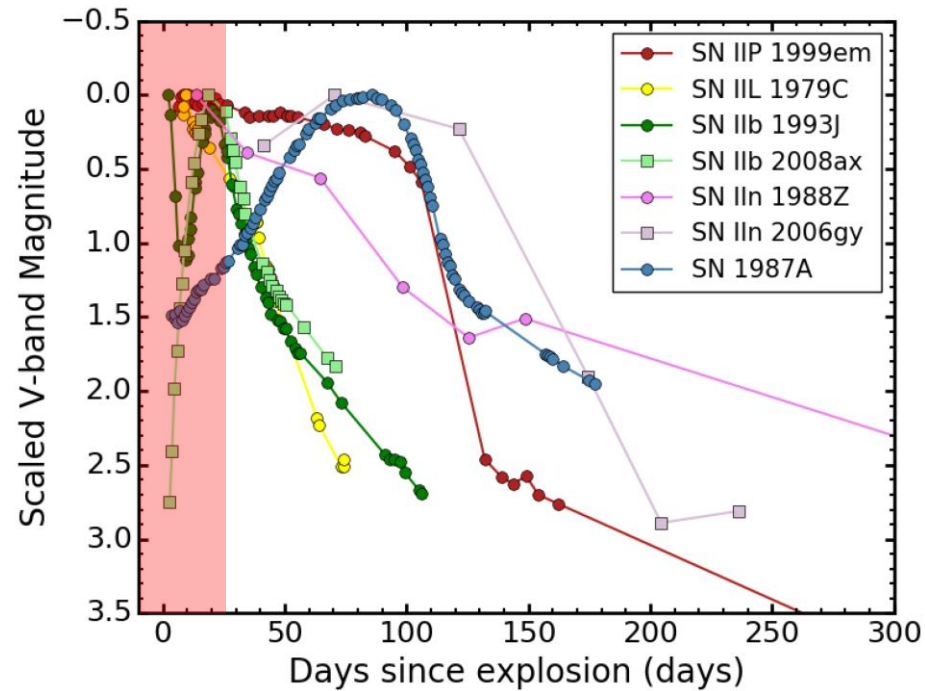
Absence of a Gamma-Ray Burst Signal from SN 1987A



- If a γ -ray burst from SN 1987A had occurred, the Solar Maximum Mission (SMM) satellite could have detected it. However, there was no significant signal that exceeded the background.

E. L. Chupp, W. T. Vestrand, and C. Reppin,
Experimental limits on the radiative decay of SN 1987A
neutrinos, Phys Rev Lett. 1989 Jan

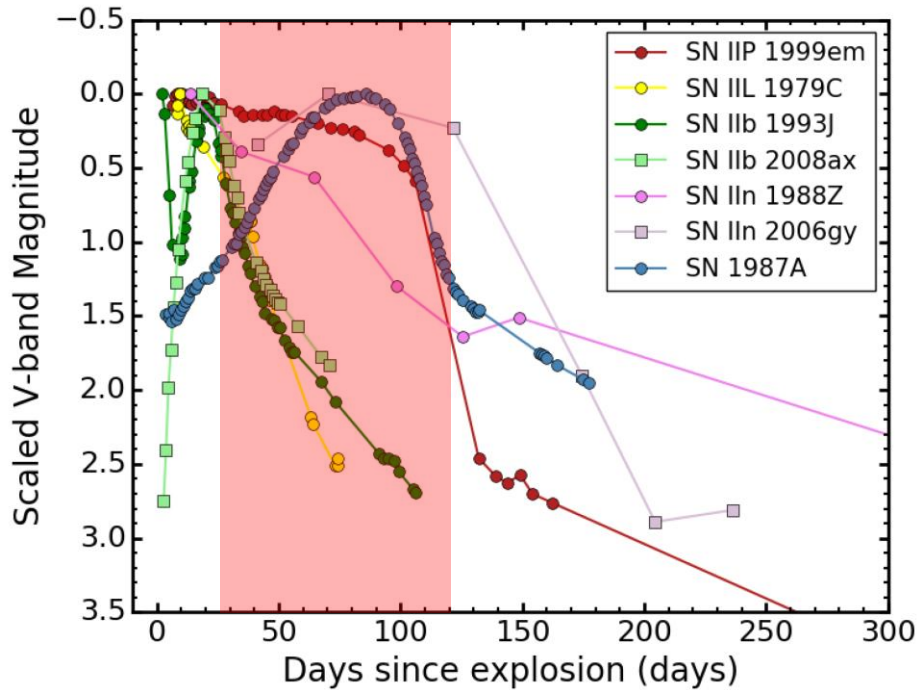
Light Curves of Type II Supernovae



Han Lin, Jujia Zhang, and Xinghan Zhang, Probing Diversity of Type II Supernovae with the Chinese Space Station Telescope, Universe 9.5 (2023)

- Early-time luminosity is dominated by the thermal energy deposited by the shock
- The shock heats the stellar envelope, ionizing hydrogen in the stellar atmosphere
- The resulting increase in the number of free electrons raises the photon opacity, causing the photosphere to expand

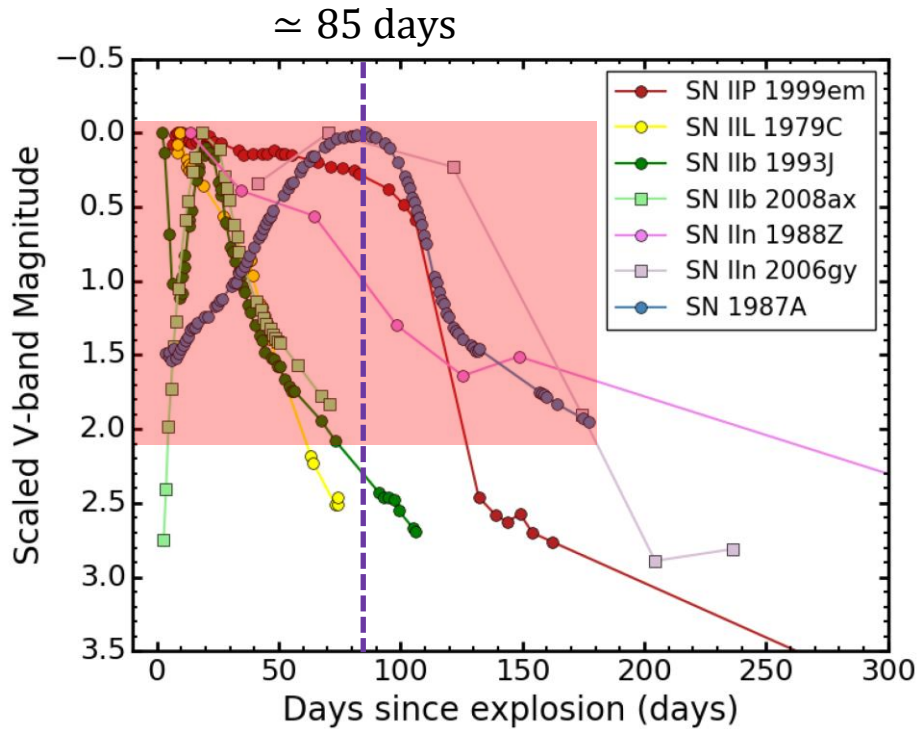
Light Curves of Type II Supernovae



Han Lin, Jujia Zhang, and Xinghan Zhang, Probing Diversity of Type II Supernovae with the Chinese Space Station Telescope, Universe 9.5 (2023)

- Intermediate-time luminosity is dominated by the hydrogen recombination
- As the shock-expanded envelope cools, recombination of ionized hydrogen begins
- The number of free electrons decreases, reducing the photon opacity and causing the photosphere to recede
- Plateau in the light curves of Type II-P SNe lasting approximately 100 days

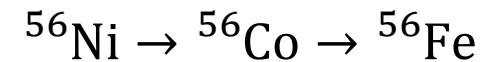
Light Curves of SN 1987A



Han Lin, Jujia Zhang, and Xinghan Zhang, Probing Diversity of Type II Supernovae with the Chinese Space Station Telescope, Universe 9.5 (2023)

- Decay of radioactive isotopes provides an additional contribution to the light curve

- Representative decay chain is

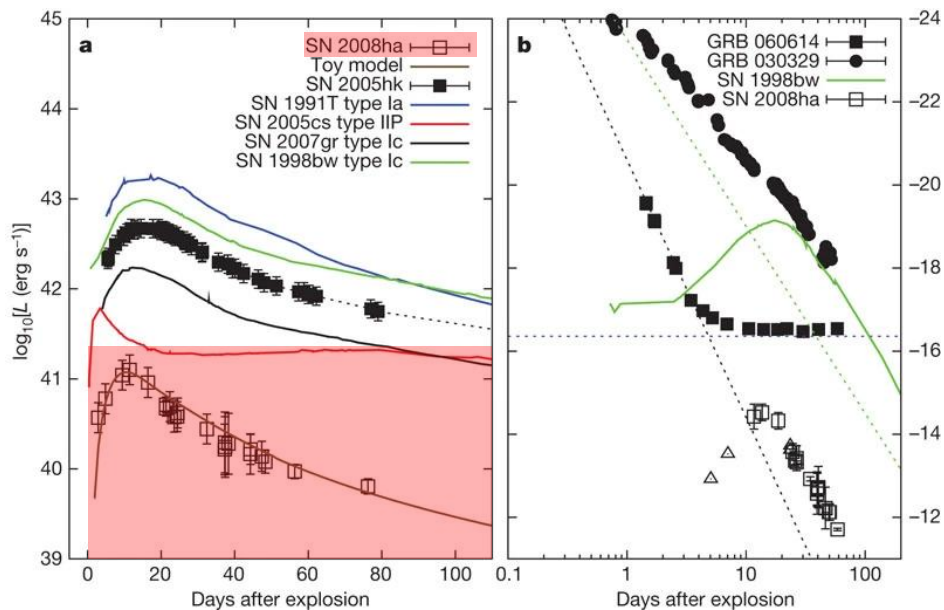


Half-life: 6.1 days

77.3 days

- Effects of radioactive decay are particularly prominent in the light curves of Type Ia SNe
- In the light curve of SN 1987A, the characteristic signature of the decay of ^{56}Co is clearly observed

Low Explosion Energy Supernovae

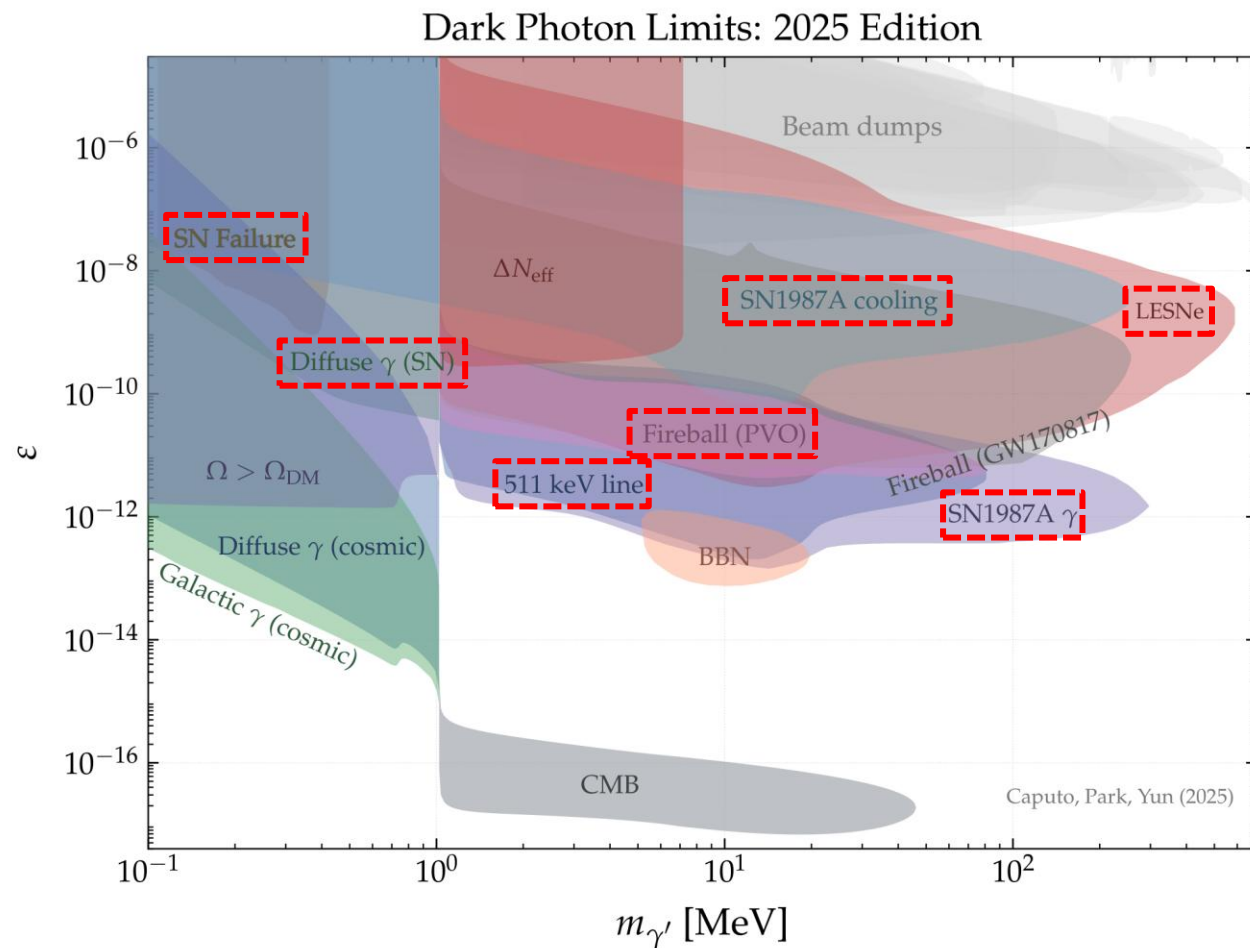


Valenti, S., Pastorello, A., Cappellaro, E. et al., A low-energy core-collapse supernova without a hydrogen envelope, *Nature* 459, 674–677 (2009)

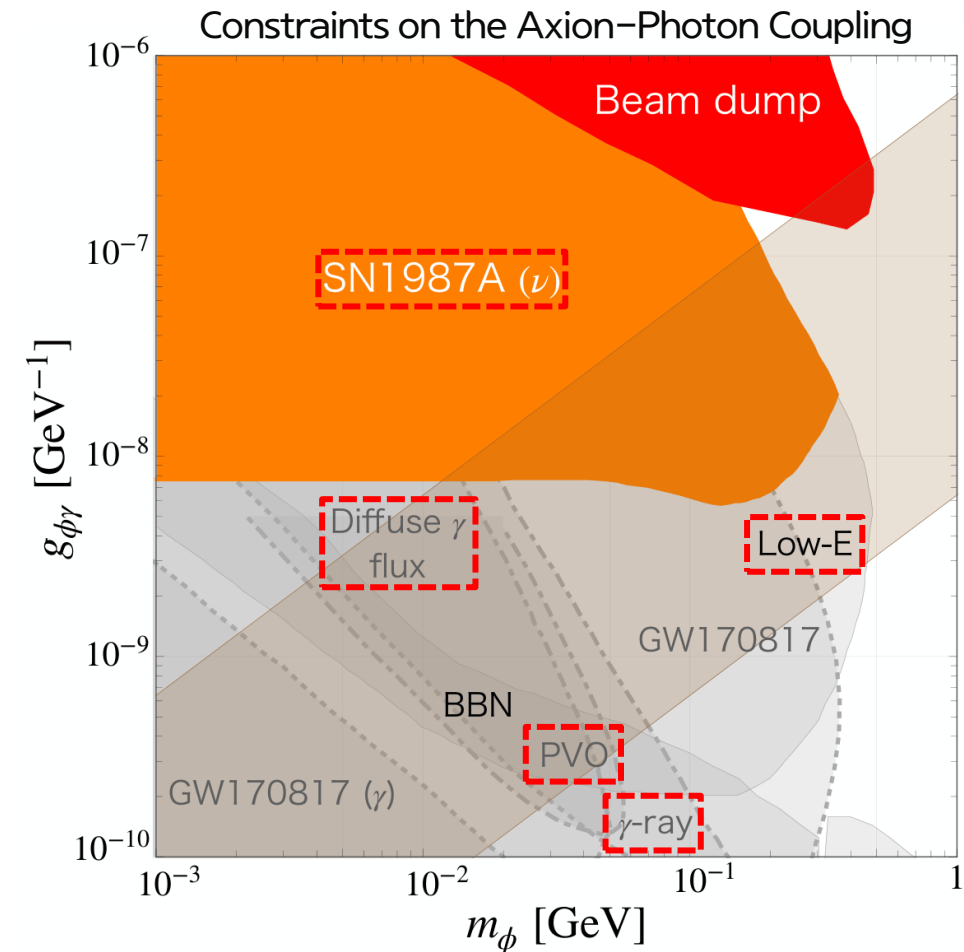
- Low-energy supernovae (LESNe) explode with energies below the canonical CCSN scale, $E_{\text{exp}} \simeq 10^{51}$ erg
Typical values are $\sim 10^{50}$ erg, and extreme faint events can reach $\sim 10^{49}$ erg
- LESNe show low ^{56}Ni yield and slow ejecta velocities
 ^{56}Ni yields are typically a few percent of those in ordinary CCSNe, and ejecta velocities are generally about 10–20% of the canonical CCSN value, $\sim 10^4$ km/s

Strategies to Probe New Particles

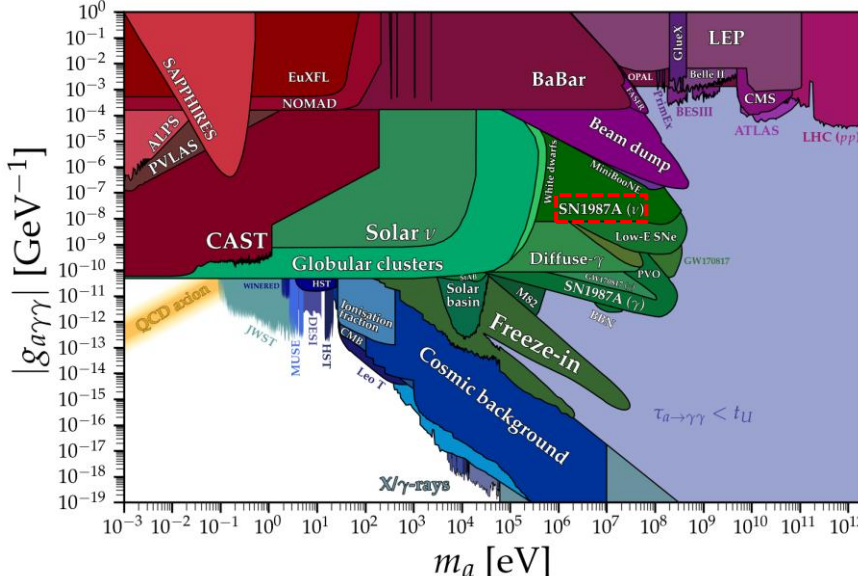
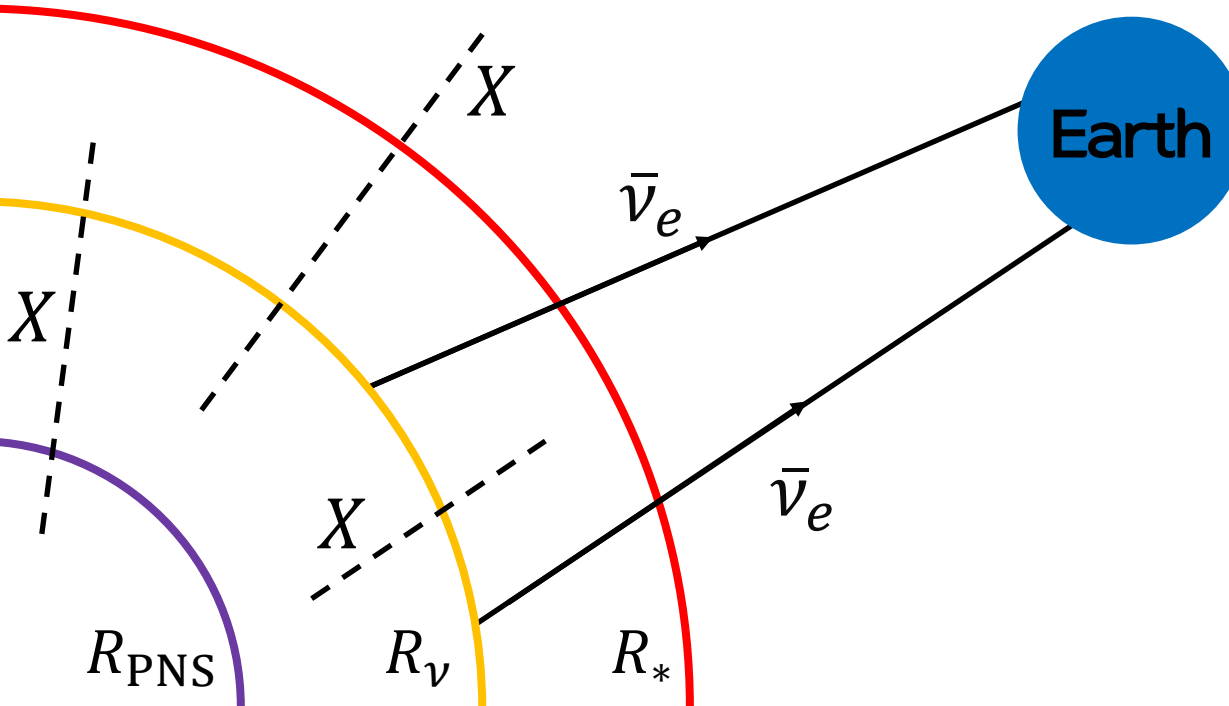
Current Status of Constraints from CCSNe



Andrea Caputo and Jaeyoung Park and Seokhoon Yun, Heavy dark photon handbook: Cosmological and astrophysical bounds, Physical Review D (Apr 2026)



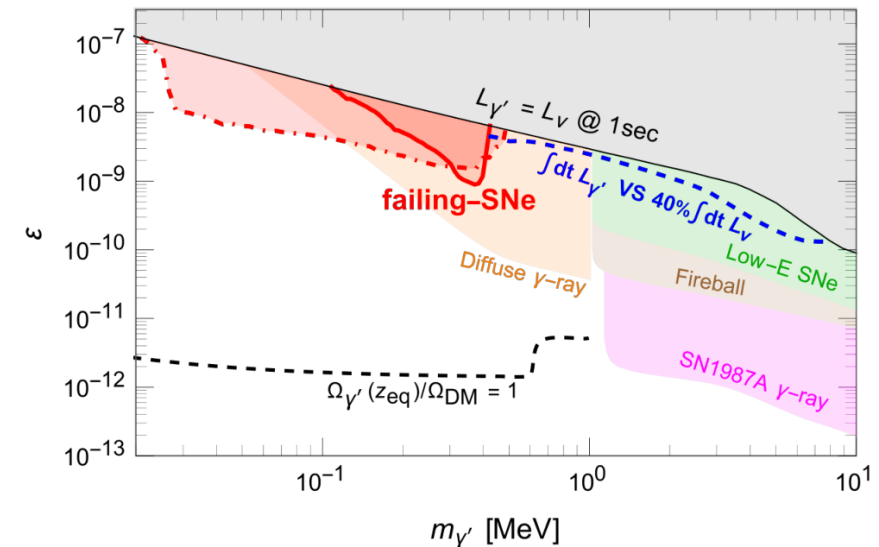
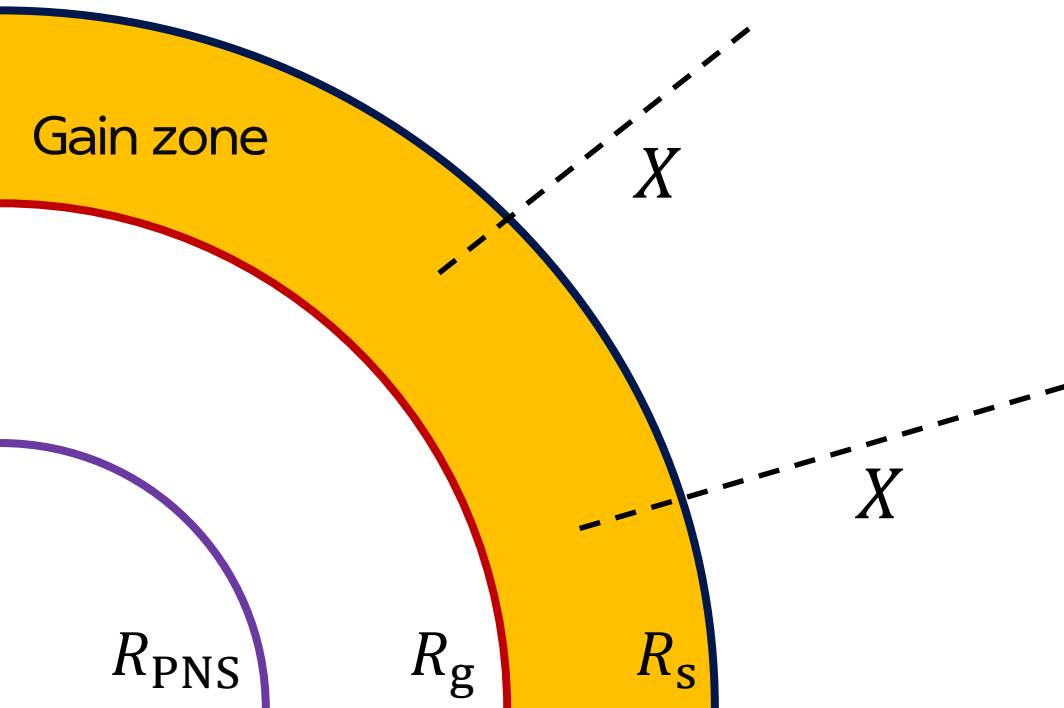
Kwang Sik Jeong and Ju Hyeon Kang and Shota Nakagawa, Cosmological roles of dark photons in axion-induced electroweak baryogenesis, Journal of Cosmology and Astroparticle Physics (Jan 2025)



Ciaran O'Hare, AxionLimits, Zenodo (July 2020)

Supernova Failure Due to Shock Cooling

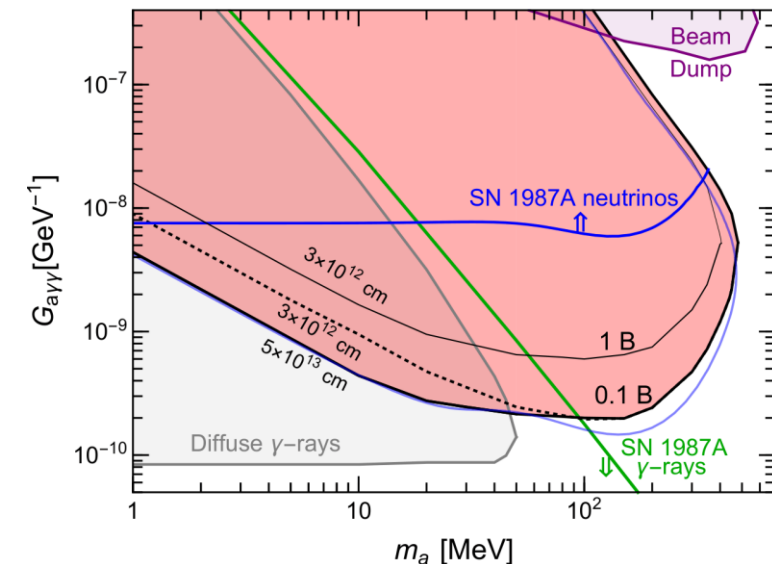
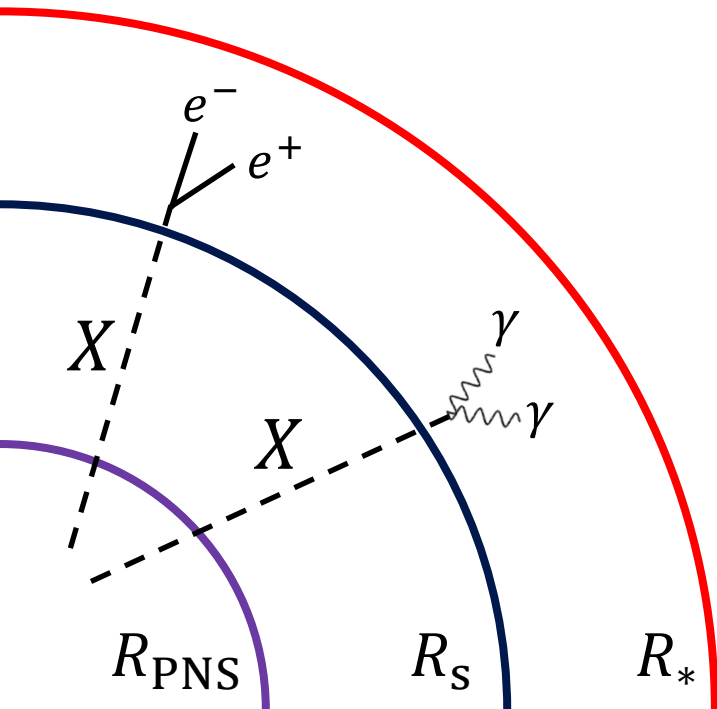
- X particles produced in the gain zone can cool the shock
- Excessive cooling by X can prevent shock revitalization, leading to a failed SN explosion
- This requires $\xi_{X/\nu} \lesssim 0.2$ during the shock revitalization



Andrea Caputo and Hans-Thomas Janka and Georg Raffelt and Seokhoon Yun, Cooling the Shock: New Supernova Constraints on Dark Photons, Physical Review Letters (Apr 2025)

Low-Energy Supernovae

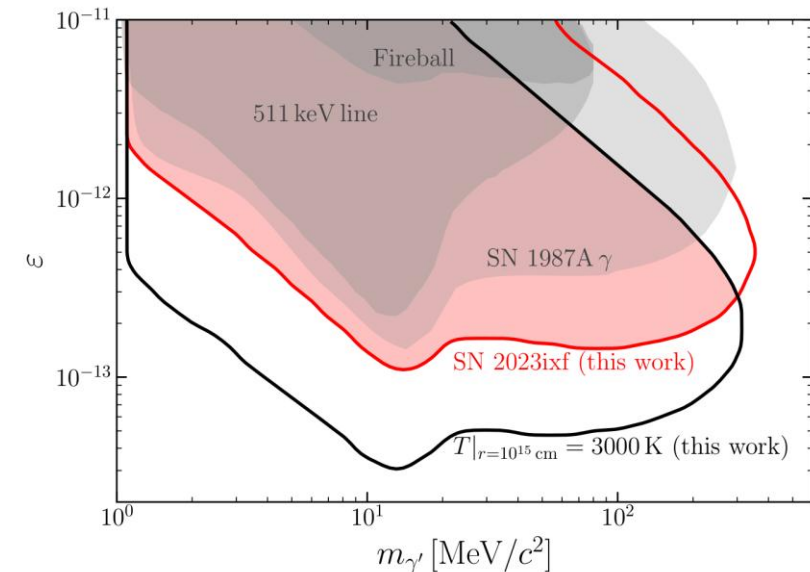
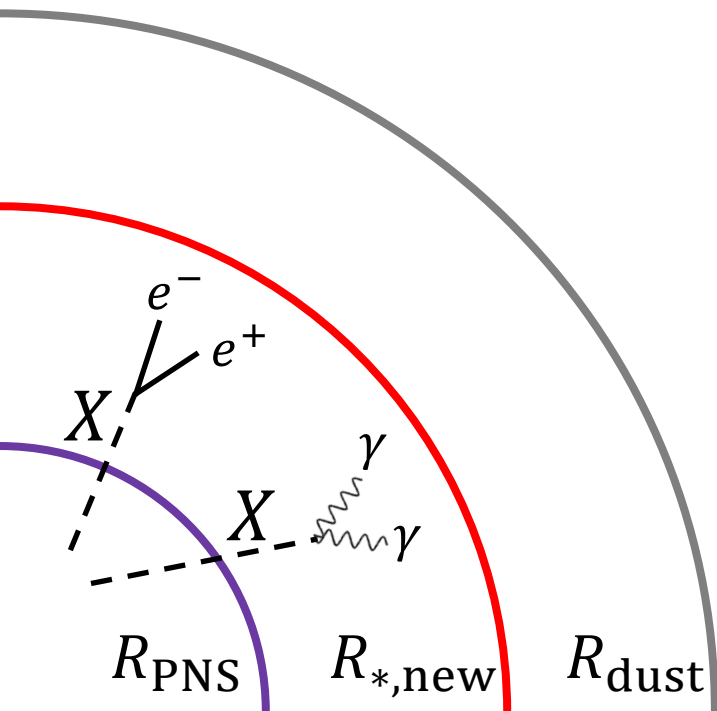
- Explosion energy of observed low-energy supernovae is approximately $\mathcal{O}(10^{50})$ erg
- X produced in PNS can heat the photosphere, thereby enhancing the explosion energy
- This requires energy deposited by X should be $E_X \lesssim 10^{50}$ erg



Andrea Caputo and Hans-Thomas Janka and Georg Raffelt and Edoardo Vitagliano, Low-Energy Supernovae Severely Constrain Radiative Particle Decays, Physical Review Letters (June 2022)

SN 2023ixf Circumstellar Medium Precursor

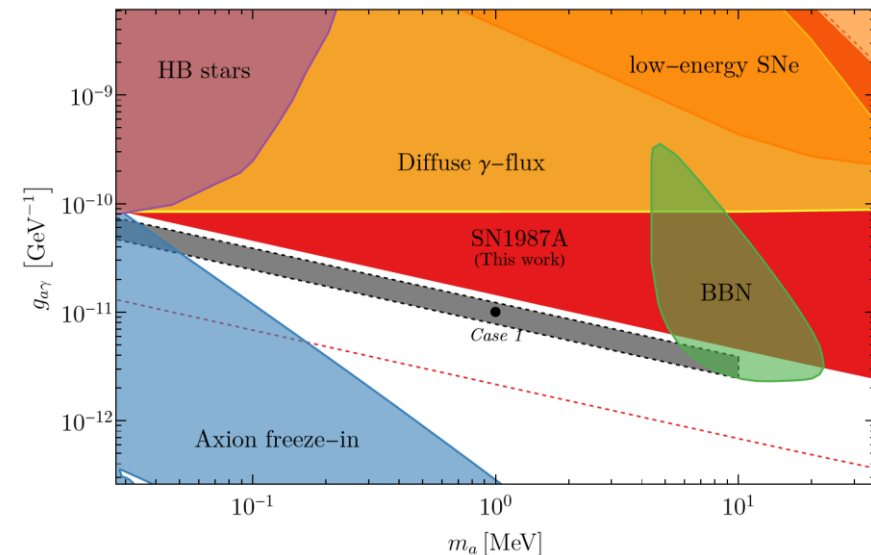
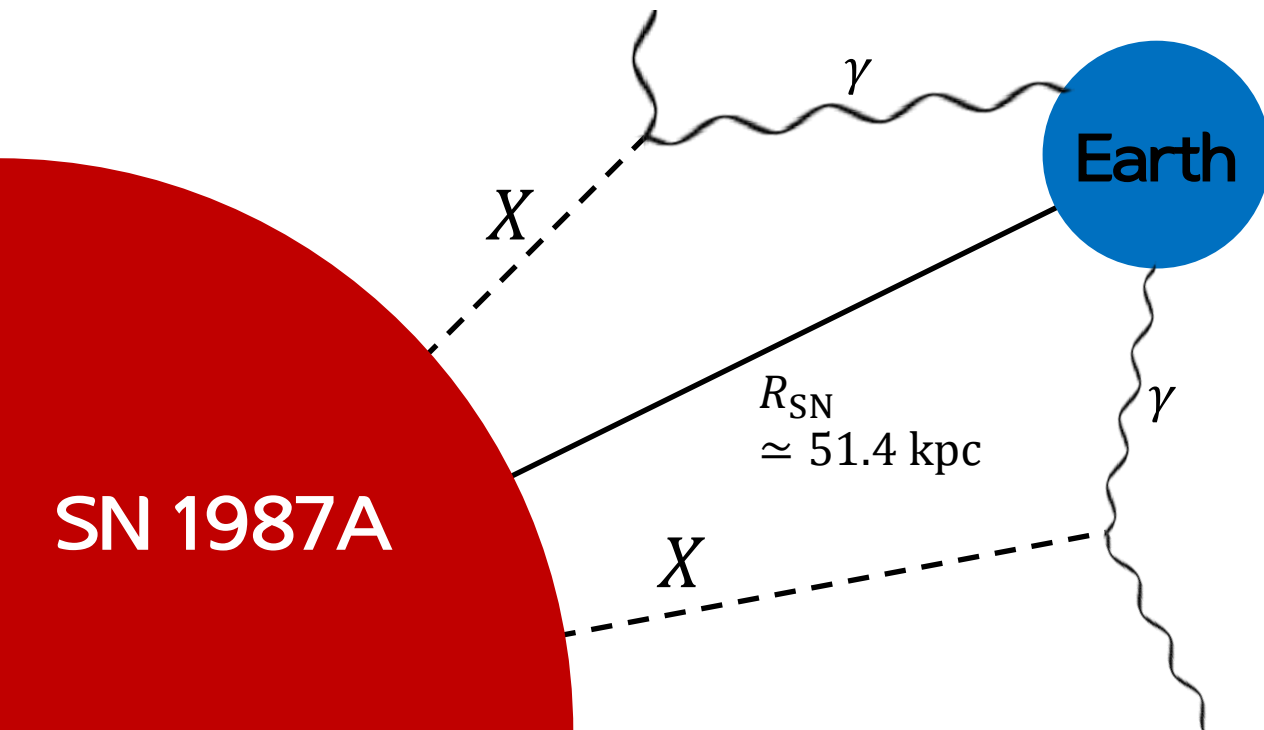
- X particles from the PNS can decay in the dense CSM before shock breakout
- This heating can ionize the CSM and form a new photosphere, producing an optical precursor
- Non-detection of this requires the total deposited energy should be $Q_X \lesssim 2.5 \times 10^{44}$ erg



Yu Cheng and Chui-Fan Kong and Yen-Hsun Lin and Meng-Ru Wu and Seokhoon Yun, Circumstellar Medium of Supernovae as New Probes for Feebly-interacting Particles, arXiv:2603.09615 (Mar 2026)

SN 1987A γ -Ray Burst

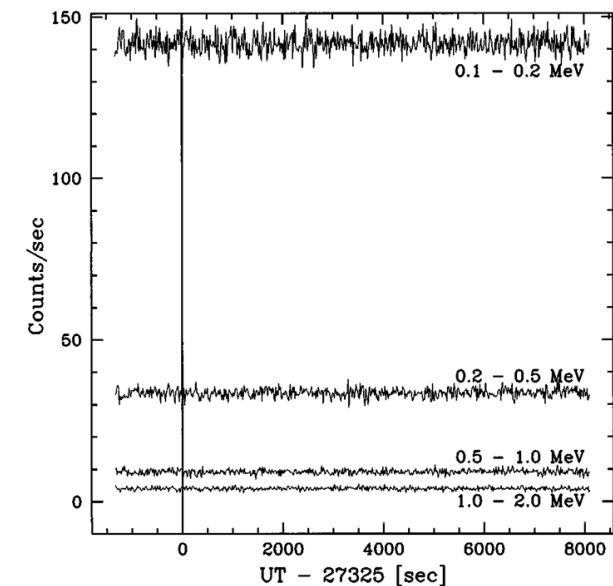
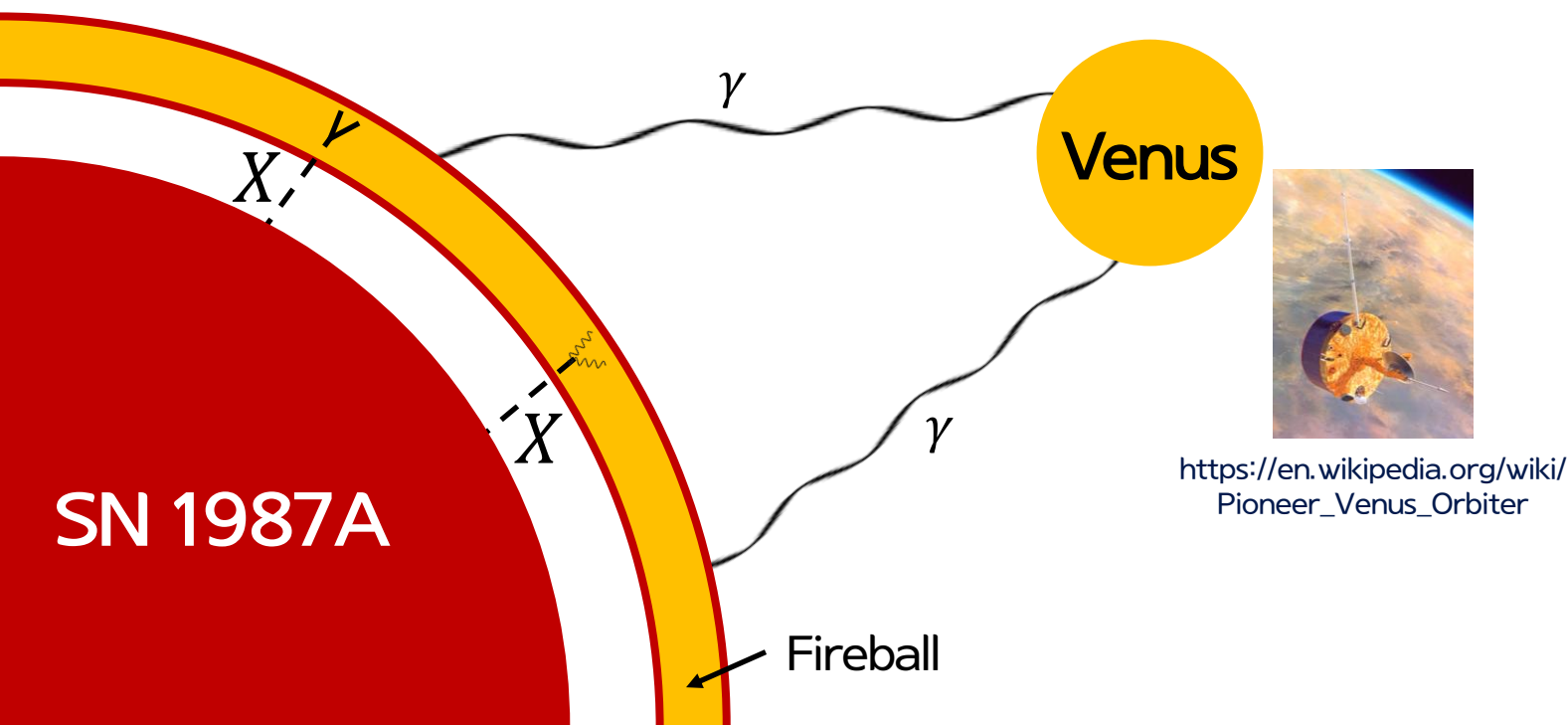
- X produced in a supernova can induce a γ -ray burst if they produce photons
- SMM satellite detected no γ -ray burst in the $\mathcal{O}(1\text{--}10)$ MeV energy range from the SN 1987A
- Absence of a signal places a bound on γ -ray fluence of $\mathcal{F}_\gamma \lesssim \mathcal{O}(1) \text{ cm}^{-2}$, depending on E_γ



E. Müller and F. Calore and P. Carenza and C. Eckner and M.C. David Marsh, Investigating the gamma-ray burst from decaying MeV-scale axion-like particles produced in supernova explosions, *Journal of Cosmology and Astroparticle Physics* (July 2023)

SN 1987A X/γ -Ray from X -Induced Fireball

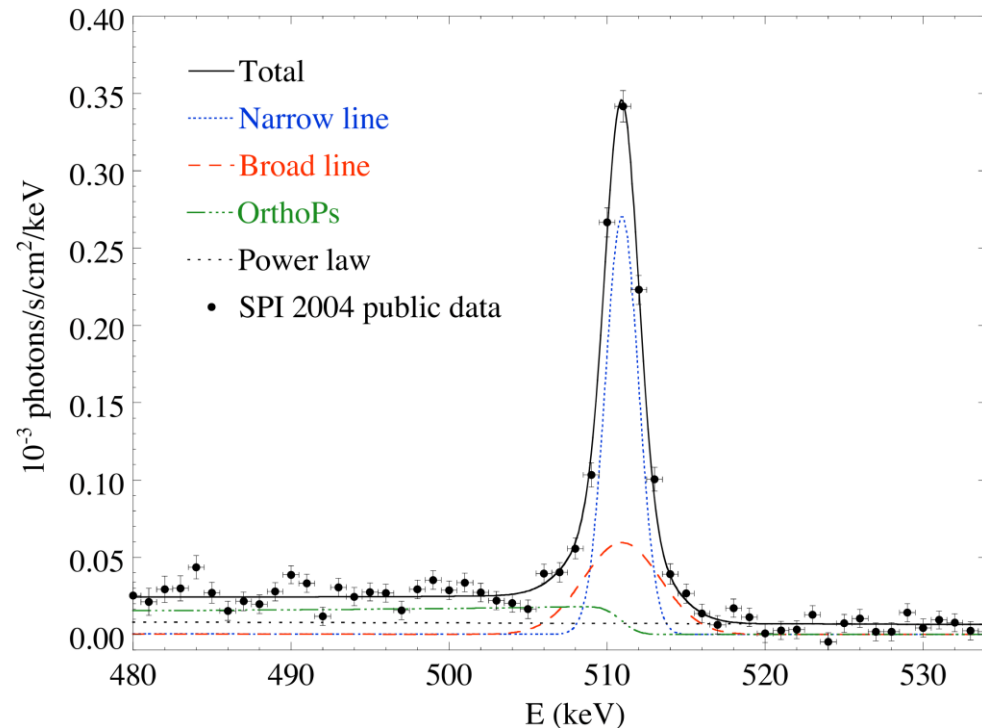
- X produced in a supernova may produce visible particles outside the SN and form a fireball
- For $E_\gamma \sim \mathcal{O}(0.1)$ MeV, the resulting signal would have been detectable by the PVO
- Non-detection of this signal from SN 1987A gives additional constraints on X



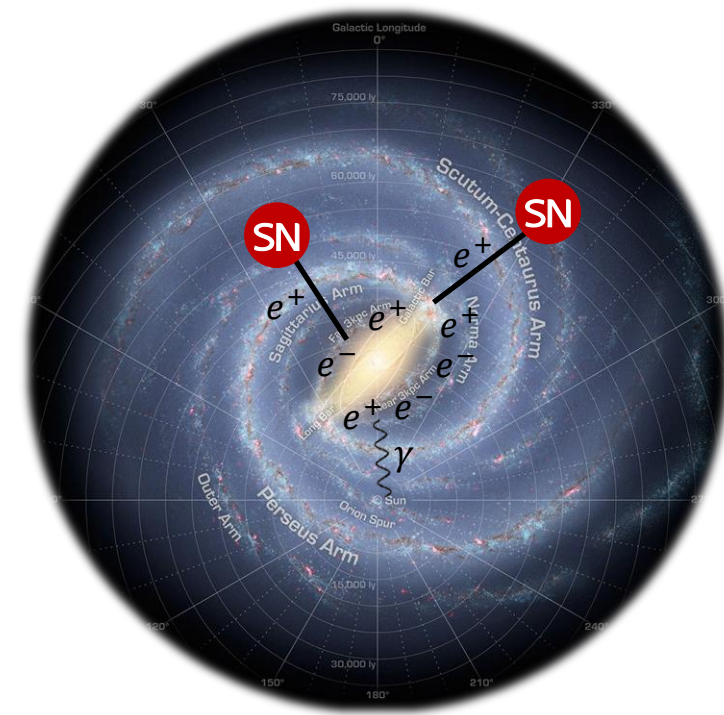
Andrew H. Jaffe and Michael S. Turner, Limits to radiative neutrino decay from SN 1987A, Physical Review D (June 1997)

511 keV Emission from the Galactic Center

- INTEGRAL/SPI has measured the e^-e^+ annihilation spectrum from the Galactic Center
- X produced in supernovae induce $N_{e^+} \gtrsim 10^{53}$, the observed excess cannot be explained



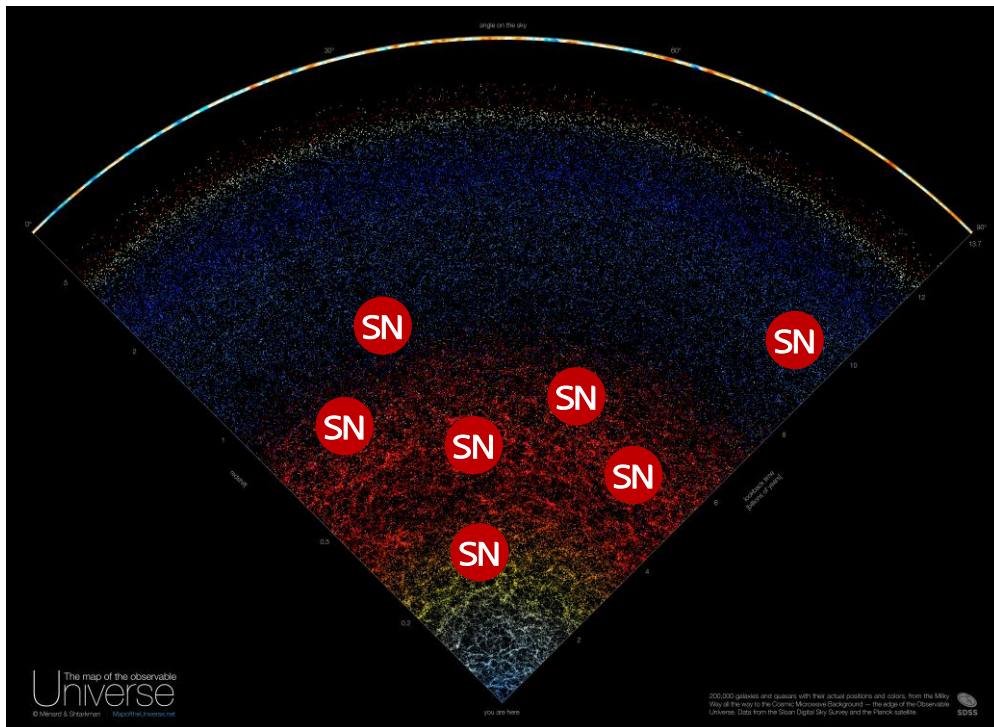
N. Prantzos et al., The 511 keV emission from positron annihilation in the Galaxy, Reviews of Modern Physics (Sep 2011)



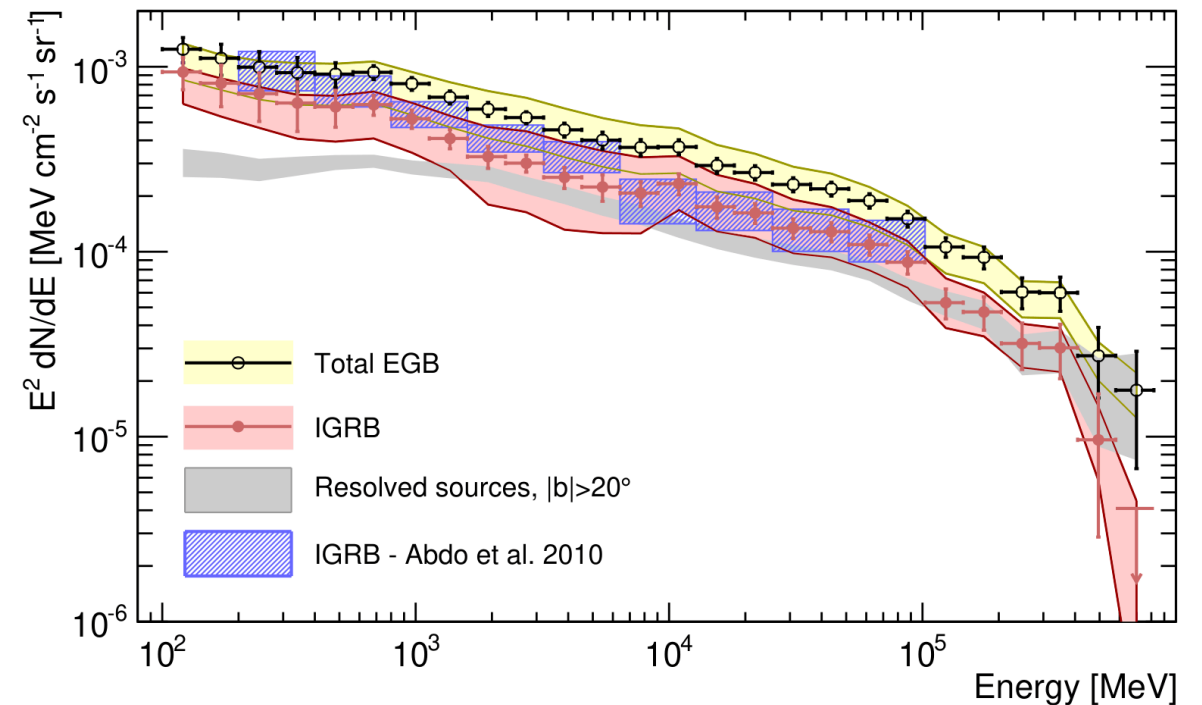
NASA/JPL-Caltech/R. Hurt (SSC/Caltech)

Diffuse γ -Ray Background

- X produced by all past SNe can form the diffuse X background (DXB)
- DXB may contribute to diffuse γ -ray background, but it should not affect the observed data

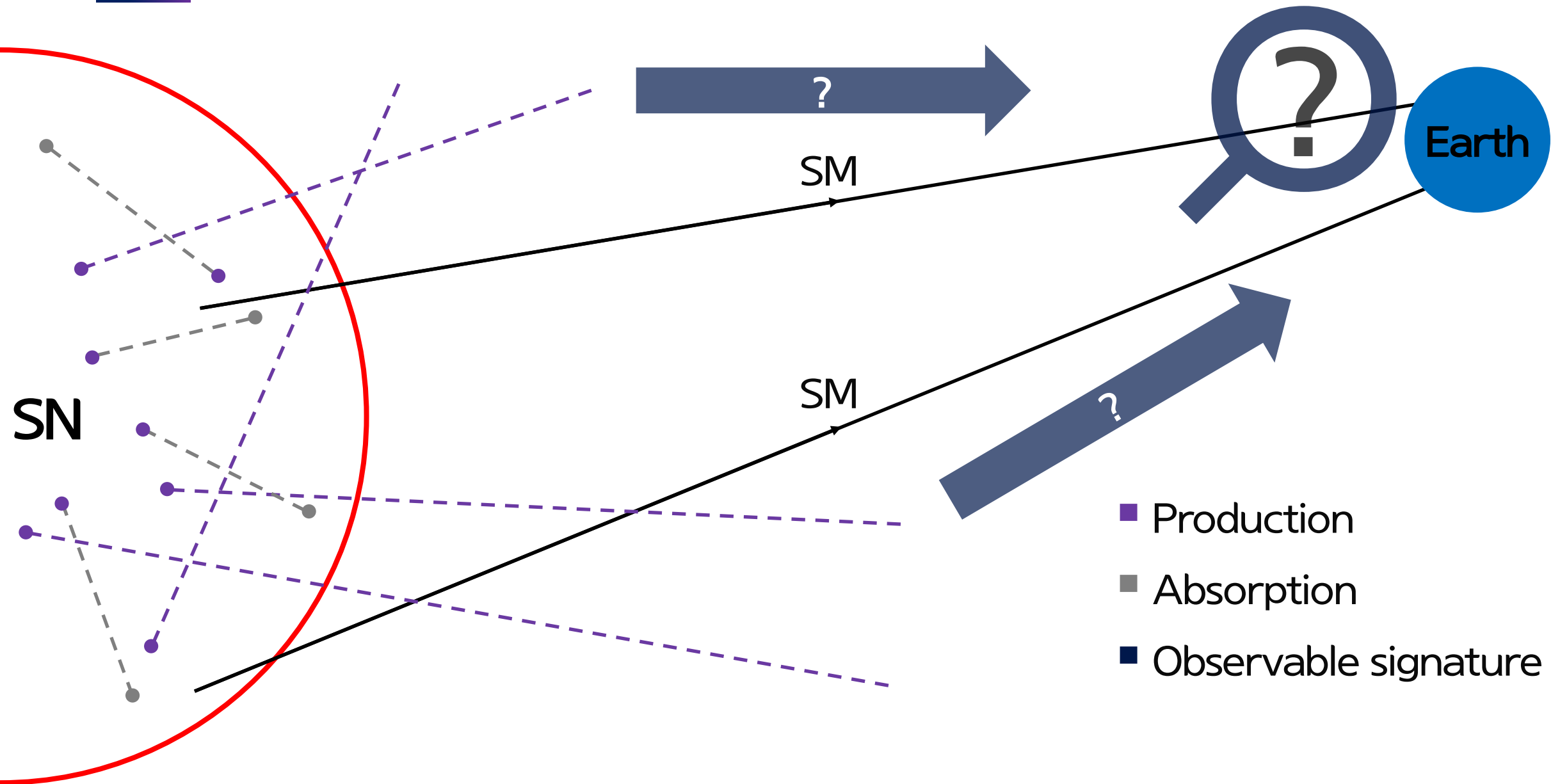


Ménard and Shtarkman, Map of the Universe, 2022



Mattia Fornasa and Miguel A. Sánchez-Conde, The nature of the Diffuse Gamma-Ray Background, Physics Reports (Oct 2015)

Key Ingredients of Supernova Constraints



Example: SN 1987A Cooling Bound

- Observed neutrino signal from SN 1987A requires $L_X \lesssim \mathcal{O}(10^{52}) \text{ erg} \cdot \text{s}^{-1}$ at $t_{\text{pb}} \sim 1 \text{ s}$

$$L_X(g_X, m_X) = 4\pi \int_0^{R_\nu} dr r^2 \int_{m_X/\alpha(r)}^\infty d\omega \alpha^2(r) \frac{dQ(g_X, m_X, \omega, r)}{d\omega} e^{-\tau(g_X, m_X, \omega, r)}$$

↑
gravitational trapping redshift and time dilation

$\frac{dQ}{d\omega}$: Spectral emissivity (production rate)

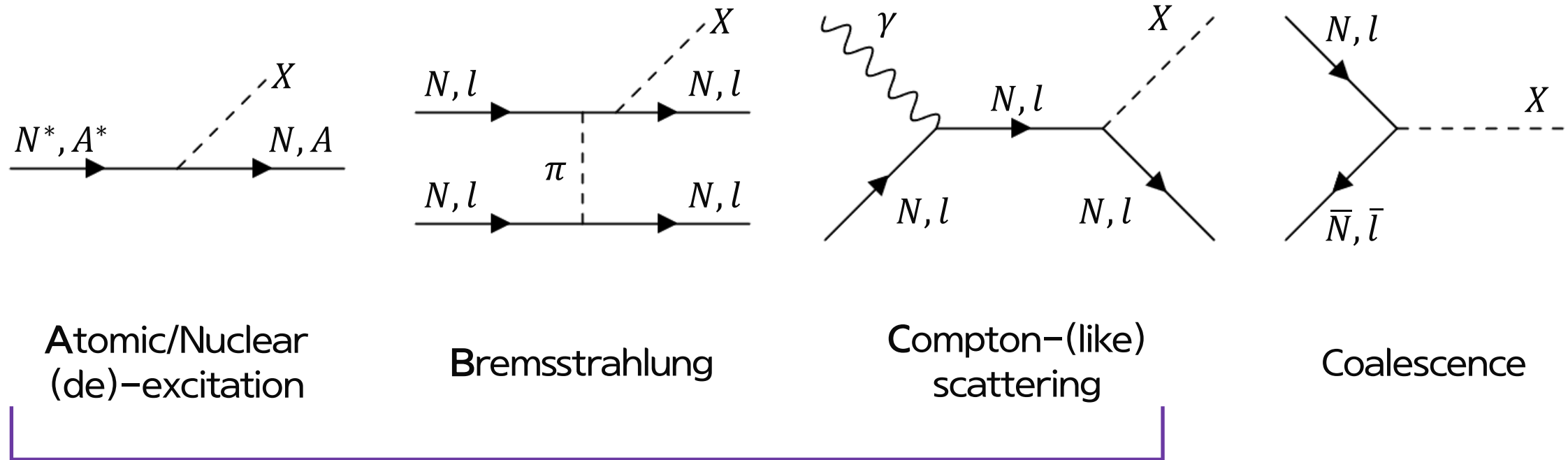
τ : Optical depth (absorption rate)

α : Lapse function (gravitational effect)

- Production
- Absorption
- Observable signature

Particle Interaction Processes in Stellar Medium

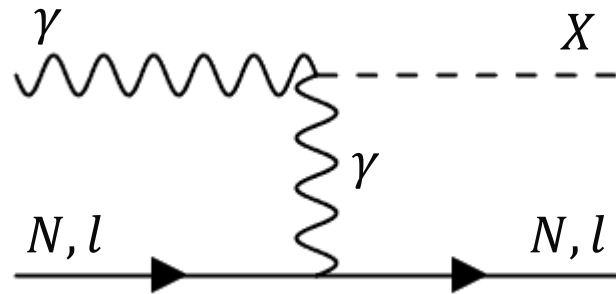
- Coupled to nucleons/leptons



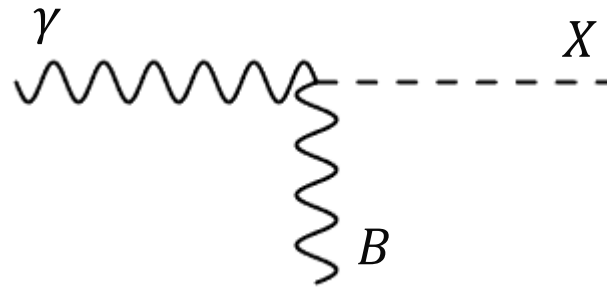
ABC processes

Particle Interaction Processes in Stellar Medium

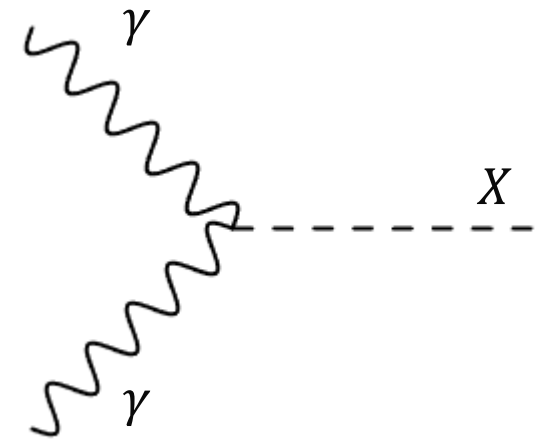
- Coupled to photon



Primakoff



Conversion in external B field



Coalescence

Medium Effects

- Multiple scattering effect

$$\frac{\omega^2}{\omega^2 + \Gamma^2/4}$$

- Photon self energy

$$\omega^2 + k^2 = \pi_{T,L} \approx \omega_{\text{pl}}^2 \quad (\text{for transverse photon})$$

$$\omega_{\text{pl}} = \sqrt{\frac{4\pi\alpha n_e}{m_e}}$$

- Effective nucleon mass

$$m_N^* = m + \Sigma_S$$

